Class 9
Physics

Table of Content

1.	Vectors	1 - 15
2.	Motion in a Plane	16 – 32
3.	Force and Laws of Motion	33 – 62
4.	Work, Energy And Power	63 – 79
5.	Circular Motion	80 – 113
6.	Gravitation	114 – 146
7.	Heat	147 – 179

1.Vectors

SOLUTIONS:

TOPIC-1: Introduction to Vectors and Vector Operations

LEVEL-I

1. Answer: B) Scalars

Explanation: Physical quantities that have only magnitude and no direction are called scalars (e.g., mass, time).

2. Answer: C) Vector

Explanation: A vector is a physical quantity that has both magnitude and direction.

3. Answer: D) Vectors

Explanation: Vectors cannot be added or multiplied using ordinary algebra because their directions must also be considered.

4. Answer: A) Scalar

Explanation: The magnitude (length) of a vector has only size and no direction, so it is a scalar quantity.

5. Answer: B) Mass

Explanation: Mass has only magnitude and no direction, whereas force, velocity, and displacement are vectors**.**

6. Answer: C) Null vector

Explanation: A vector with zero magnitude is called a null (zero) vector.

7. Answer: A) $|\vec{A}| \neq 0$

Explanation: A proper vector is any vector with a non-zero magnitude.

8. Answer: C

Explanation: Negative vectors have the same magnitude but point in exactly opposite directions. Diagram C represents this correctly.

9. Answer: D) 40 m

Explanation: "40 m" represents only distance (a scalar), while the other options include direction and represent displacement vectors.

10. Answer: C) 13

Explanation: The magnitude of the vector is:

$\sqrt{3^2 + (-12)^2 + 4^2} = \sqrt{9 + 144 + 16} = \sqrt{169} = 13$

11. Answer: D) $\sqrt{0.11}$

Explanation: For a unit vector:

$0.5^2 + (-0.8)^2 + P^2 = 1$

$0.25 + 0.64 + P^2 = 1$

$P^2 = 0.11$

$P = \sqrt{0.11}$

12. Answer: A) $10\sqrt{2}$

Explanation: The magnitude is:

$\sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$

13. Answer: B) $\sqrt{186}$

Explanation: First find the vector:

$\overrightarrow{PQ} = (6-2)\vec{\imath} + (6-(-5))\vec{\jmath} + (4-(-3))\vec{k} = 4\vec{\imath} + 11\vec{\jmath} + 7\vec{k}$

Its length is:

$\sqrt{4^2 + 11^2 + 7^2} = \sqrt{16 + 121 + 49} = \sqrt{186}$

14. Answer: A) 6

Explanation: Given the magnitude is 7:

$\sqrt{2^2 + 3^2 + C^2} = 7$

$4 + 9 + C^2 = 49$

$C^2 = 36$

$C = 6$

15. Answer: C) $-\frac{3}{2}$

Explanation: For parallel vectors:

$\frac{4}{8} = \frac{-y}{3}$

$\frac{1}{2} = \frac{-y}{3}$

$y = -\frac{3}{2}$

16. Answer: D) $8\vec{\imath} + 6\vec{\jmath} - 5\vec{k}$

Explanation: Add corresponding components:

$(2 + 6)\vec{\imath} + (3 + 3)\vec{\jmath} + (-4 - 1)\vec{k} = 8\vec{\imath} + 6\vec{\jmath} - 5\vec{k}$

17. Answer: C) $\frac{\vec{\imath}+\vec{\jmath}}{\sqrt{2}}$

Explanation: Divide the vector by its magnitude:

$|\ \vec{\imath} + \vec{\jmath}\ | = \sqrt{1^2 + 1^2} = \sqrt{2}$

Hence the unit vector is:

$\frac{\vec{\imath} + \vec{\jmath}}{\sqrt{2}}$

18. Answer: B) $\vec{\jmath} - \vec{k}$

Explanation: Since:

$\vec{A} + \vec{B} = \vec{\imath}$

$\vec{B} = \vec{\imath} - \vec{A} = \vec{\imath} - (\vec{\imath} - \vec{\jmath} + \vec{k}) = \vec{\jmath} - \vec{k}$

19. Answer: A) $\frac{15\vec{\imath}+25\vec{\jmath}-5\vec{k}}{\sqrt{35}}$

Explanation: The magnitude of $\vec{A} = 3\vec{\imath} + 5\vec{\jmath} - \vec{k}$is:

$\sqrt{3^2 + 5^2 + (-1)^2} = \sqrt{35}$

A vector of magnitude 5 in the same direction is:

$5\left(\frac{\vec{A}}{|\ \vec{A}\ |}\right) = \frac{15\vec{\imath} + 25\vec{\jmath} - 5\vec{k}}{\sqrt{35}}$

20. Answer: B) 45°

Explanation: Since vectors $\vec{B}$and $\vec{C}$are perpendicular (90°) and $\vec{A}$makes 45°with $\vec{C}$, the angle between $\vec{A}$and $\vec{B}$is:

$90° - 45° = 45°$

21. Answer: A) 75°

Explanation: One vector is 30°above the x-axis and the other is 45°below it. Therefore, the angle between them is:

$30° + 45° = 75°$

22. Answer: C) 5

Explanation: The projection on the yz-plane is:

$3\vec{\jmath} + 4\vec{k}$

Its magnitude is:

$\sqrt{3^2 + 4^2} = \sqrt{25} = 5$

23. Answer: A) $2\vec{\imath} - 4\vec{\jmath} + 6\vec{k}$, $\sqrt{56}$

Explanation: The vector is:

$\overrightarrow{AB} = (3-1)\vec{\imath} + (-2-2)\vec{\jmath} + (4-(-2))\vec{k} = 2\vec{\imath} - 4\vec{\jmath} + 6\vec{k}$

Its magnitude is:

$\sqrt{2^2 + (-4)^2 + 6^2} = \sqrt{4 + 16 + 36} = \sqrt{56}$

24. Answer: C) $4\vec{\imath} + 6\vec{\jmath} + 16\vec{k}$

Explanation: Multiplying the vector $2\vec{\imath} + 3\vec{\jmath} + 8\vec{k}$by 2 gives:

$4\vec{\imath} + 6\vec{\jmath} + 16\vec{k}$

Multiplying by a scalar changes the magnitude but keeps the direction the same, so the new vector remains parallel to the original.

LEVEL-II

Single Option Correct Type

1. Answer: A

Explanation: Density (mass/volume)has only magnitude and no direction. Hence, it is a scalar **quantity.**

2. Answer: C

Explanation: Vectors lying in different planes are called non-coplanar vectors.

3. Answer: B

Explanation: A vector whose magnitude is equal to 1 is called a unit vector.

4. Answer: A

Explanation: The magnitude of $3\vec{\imath} - 4\vec{\jmath}$is

$\sqrt{3^2 + (-4)^2} = 5.$

A vector of the same magnitude but opposite direction is

$-(3\vec{\imath} - 4\vec{\jmath}) = -3\vec{\imath} + 4\vec{\jmath}.$

5. Answer: D

Explanation: Since the vector is a unit vector,

$\sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + a^2} = 1.$

Therefore,

$\frac{1}{4} + \frac{1}{4} + a^2 = 1$

$a^2 = \frac{1}{2}$

$a = \pm\frac{1}{\sqrt{2}}.$

Option D contains the correct value.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

6. Correct Options: A, B, C

Explanation:

Length, Time, and Temperature require only magnitude to be completely described, so they are scalar quantities. Displacement requires both magnitude and direction, making it a vector quantity.

7. Correct Options: A, B, C

Explanation:

- kg (kilogram) is the unit of mass, which is a scalar quantity.
- s (second) is the unit of time, which is a scalar quantity.
- kg m^{-3} is the unit of density, which is also a scalar quantity.
- N (newton) is the unit of force, which is a vector quantity.

8. Correct Options: B, C, D

Explanation:

Two vectors are equal only if they have the same magnitude and the same direction. They are unequal if they differ in magnitude, direction, or both.

9. Correct Options: A, C, D

Explanation:

A null (zero) vector has zero magnitude.

- A: A body moving with uniform velocity has zero acceleration, so the acceleration vector is a null vector.
- C: The position vector of the origin with respect to itself is $(0, 0, 0)$, which is a null vector.
- D: A stationary body has zero displacement, so its displacement vector is a null vector.
- B: A body in circular motion has a non-zero velocity that continuously changes direction, so it is not a null vector.

10. Correct Options: A, B, C, D

Explanation:

Force, Torque, Acceleration, and Velocity all have both magnitude and direction and obey the laws of vector addition. Therefore, all of them are vector quantities.

Passage-I

11. Answer: A

Explanation: For a unit vector,

$\sqrt{(0.3)^2 + (0.4)^2 + a^2} = 1.$

Hence,

$0.09 + 0.16 + a^2 = 1$

$0.25 + a^2 = 1$

$a^2 = 0.75$

$a = \sqrt{0.75}.$

12. Answer: B

Explanation: The magnitude of

$\vec{A} = 2\vec{\imath} + 3\vec{\jmath} + \vec{k}$

is

$|\vec{A}| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14}.$

Therefore, the unit vector is

$\frac{2\vec{\imath} + 3\vec{\jmath} + \vec{k}}{\sqrt{14}}.$

13. Answer: A

Explanation: First,

$\vec{A} + \vec{B} = (2 + 3)\vec{\imath} + (-1 - 2)\vec{\jmath} + (3 - 2)\vec{k} = 5\vec{\imath} - 3\vec{\jmath} + \vec{k}.$

Its magnitude is
$\sqrt{5^2 + (-3)^2 + 1^2} = \sqrt{35}$.

Hence, the unit vector is
$\frac{5\vec{\imath} - 3\vec{\jmath} + \vec{k}}{\sqrt{35}}$.

Passage-II

14. Answer: C

Explanation: The displacement vector is
$\vec{r}_B - \vec{r}_A = (3-2)\vec{\imath} + (2-4)\vec{\jmath} + (-3-6)\vec{k} = \vec{\imath} - 2\vec{\jmath} - 9\vec{k}$.

15. Answer: C

Explanation: The displacement vector is
$(5-1)\vec{\imath} + (4-2)\vec{\jmath} + (2-3)\vec{k} = 4\vec{\imath} + 2\vec{\jmath} - \vec{k}$.

16. Answer: D

Explanation: The displacement vector is
$(\vec{\imath} + \vec{\jmath} + 3\vec{k}) - (2\vec{\imath} + 3\vec{\jmath} + 5\vec{k}) = -\vec{\imath} - 2\vec{\jmath} - 2\vec{k}$.
Its magnitude is
$\sqrt{(-1)^2 + (-2)^2 + (-2)^2} = \sqrt{9} = 3$ m.

Statement Type

17. Answer: A

Explanation: Sliding a vector parallel to itself does not change its magnitude or direction, so Statement I is true. A vector with non-zero magnitude is called a proper vector, so Statement II is also true.

18. Answer: D

Explanation: A vector of magnitude 6 along the negative y-axis is
$-6\vec{\jmath}$,
not
$+6\vec{\jmath}$,
so Statement I is false. Statement II correctly gives the magnitude of a vector joining
$P(x_1, y_1, z_1)$
and
$Q(x_2, y_2, z_2)$,
hence it is true.

Matrix Matching Type

19. Answer:

- **a → Q, R**
- **b → Q, R**
- **c → P, R, S**
- **d → P, R, S**

Explanation: Density and work done are scalar quantities, so they have only magnitude. Force, displacement, and velocity are vector quantities, so they have both magnitude and direction.

20. Answer:

- **a → T**
- **b → S**
- **c → Q**
- **d → R**

Explanation:

- x–y plane: $\sqrt{x^2 + y^2} = \sqrt{2^2 + 3^2}$→ T
- x–z plane: $\sqrt{x^2 + z^2} = \sqrt{2^2 + 5^2}$→ S
- y–z plane: $\sqrt{y^2 + z^2} = \sqrt{3^2 + 5^2}$→ Q
- Three-dimensional magnitude: $\sqrt{x^2 + y^2 + z^2} = \sqrt{2^2 + 3^2 + 5^2}$→ R

Integer Type

21. Answer: 1

Explanation: The angle between the two vectors is 45°. Therefore,
$45° = 1 \times 45°$.

22. Answer: 1

Explanation: The magnitude of every unit vector is always
$|\vec{A}| = 1$.

TOPIC-2:
Resolution of Vectors & Laws of Vector Addition
LEVEL-I

1. Correct Option: C

By definition, the sum of the squares of the direction cosines of any vector is always equal to 1:
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

2. Correct Option: B

The component of a vector $\vec{A}$in the direction of another vector $\vec{B}$is given by $A\cos\theta$. Since the vectors are perpendicular, $\theta = 90°$and $\cos 90° = 0$. Hence, the component is 0.

3. Correct Option: C

Using the identity $\sin^2\theta = 1 - \cos^2\theta$:
$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = (1 - \cos^2\alpha) + (1 - \cos^2\beta) + (1 - \cos^2\gamma) = 3 - 1 = 2$

4. Correct Option: A

For rectangular components,
$P_x^2 + P_y^2 = P^2$
Given $P_x = \frac{3P}{5}$,

$$P_y = \sqrt{P^2 - \left(\frac{3P}{5}\right)^2} = \frac{4P}{5}$$

5. Correct Option: D

If the horizontal component ($F\sin\theta$)and vertical component ($F\cos\theta$)are equal, then
$\tan\theta = 1$
Hence,
$\theta = 45°$

6. Correct Option: B

Let the total force be F. The component making 60°with the force is

$F\cos\ 60° = 25$

so

$F = 50$ N

The other perpendicular component is

$F\sin\ 60° = 50 \times \frac{\sqrt{3}}{2} = 25\sqrt{3}$ N

7. Correct Option: C

The rectangular components are:

x-component $= 50\cos\ 30° = 25\sqrt{3}$ N

y-component $= 50\sin\ 30° = 25$ N

8. Correct Option: A

The vector is

$\vec{A} = 0\hat{\imath} + 1\hat{\jmath} + 1\hat{k}$

Its magnitude is

$|\ \vec{A}\ | = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$

Therefore, the direction cosines are

$\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

9. Correct Option: A

Resolving the 10 cm vector:

$\overrightarrow{OA} = (10\cos\ 30°)\hat{\imath} + (10\sin\ 30°)\hat{\jmath} = 5\sqrt{3}\,\hat{\imath} + 5\,\hat{\jmath}$

10. Correct Option: B

The components of the 20 N force are:

$20\cos\ 30° = 10\sqrt{3}$ N

$20\sin\ 30° = 10$ N

11. Correct Option: C

Two forces of equal magnitude can keep a body in equilibrium only if they act in exactly opposite directions (angle 180°). Hence, the minimum number of forces required is 2.

12. Correct Option: D

Electric current has both magnitude and direction, but it follows the algebraic rules of addition rather than vector addition. Therefore, it is treated as a scalar quantity.

13. Correct Option: A

According to the parallelogram law of vector addition, the magnitude of the resultant is

$R = \sqrt{A^2 + B^2 + 2AB\cos\ \theta}$

14. Correct Option: B

Using

$R^2 = F^2 + F^2 + 2F^2\cos\ \theta$

and $R = F$,

$F^2 = 2F^2(1 + \cos\ \theta)$

$\cos\ \theta = -\frac{1}{2}$

Therefore,

$\theta = 120°$

15. Correct Option: D

To obtain a zero resultant using vectors in different (non-coplanar) planes, a minimum of 4 vectors is required.

16. Correct Option: C

For two unit vectors whose sum is also a unit vector,

$\theta = 120°$

The magnitude of their difference is

$|\vec{u}_1 - \vec{u}_2| = \sqrt{1^2 + 1^2 - 2(1)(1)\cos 120°} = \sqrt{3}$

17. Correct Option: B

Using

$R^2 = P^2 + Q^2 + 2PQ\cos\theta$

and

$S^2 = P^2 + Q^2 - 2PQ\cos\theta$

Adding the equations gives

$R^2 + S^2 = 2(P^2 + Q^2)$

18. Correct Option: B

The relation

$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$

is possible only when the vectors act in the same direction. Hence,

$\theta = 0°$

19. Correct Option: D

When three equal-magnitude vectors satisfy

$\vec{A} + \vec{B} = \vec{C}$

the angle between $\vec{A}$and $\vec{B}$is

120°

20. Correct Option: B

The vectors must be placed tail-to-tail to find the included angle. Thus, the angle is

$180° - 60° = 120°$

Two equal forces at 120°produce a resultant equal to the magnitude of either force.

21. Correct Option: A

Maximum resultant:

$F_1 + F_2 = 10$

Minimum resultant:

$F_1 - F_2 = 6$

Solving,

$F_1 = 8\text{ N}, F_2 = 2\text{ N}$

After increasing each by 3 N:

$11\text{ N}, 5\text{ N}$

At 90°,

$R = \sqrt{11^2 + 5^2} = \sqrt{146}\text{ N}$

22. Correct Option: A

Given

$\vec{P} = 7\hat{\imath} + 6\hat{\jmath}$

and

$\vec{P} + \vec{Q} = 11\hat{\imath} + 9\hat{\jmath}$

Then

$\vec{Q} = 4\hat{\imath} + 3\hat{\jmath}$

Its magnitude is

$|\vec{Q}| = \sqrt{4^2 + 3^2} = 5$

23. Correct Option: C

Using the resultant formula:

$(\sqrt{10}F)^2 = (2F)^2 + (\sqrt{2}F)^2 + 2(2F)(\sqrt{2}F)\cos\theta$

$10 = 6 + 4\sqrt{2}\cos\theta$

$\cos\theta = \frac{1}{\sqrt{2}}$

Hence,

$\theta = 45°$

24. Correct Option: A

Let the forces be $3x$and $5x$.

$28^2 = (3x)^2 + (5x)^2 + 2(3x)(5x)\cos 60°$

$784 = 49x^2$

$x = 4$

Therefore, the forces are

12 N and 20 N

25. Correct Option: B

According to the triangle law of vector addition (used in Lami's theorem), three forces keeping a point in equilibrium are proportional to the sides of the closed triangle representing them:

$\frac{P}{AB} = \frac{Q}{BC} = \frac{R}{CA}$ = constant

26. Answer: A

According to the Law of Sines (Sine Rule), for any triangle, the ratio of the length of a side to the sine of its opposite angle remains constant.

Since:

- Side P is opposite angle α
- Side Q is opposite angle β
- Side R is opposite angle γ

The correct relation is:

$\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$

27. Answer: C

For horizontal equilibrium, the horizontal components of the tensions must balance:

$T_1\cos 30° = T_2\cos 60°$

Therefore,

$\frac{T_1}{T_2} = \frac{\cos 60°}{\cos 30°} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$

28. Answer: D

Using horizontal equilibrium:

$T_1\cos 60° = T_2\cos 30°$

$\frac{T_1}{2} = \frac{\sqrt{3}}{2}T_2$

$T_1 = \sqrt{3}\,T_2$

Using vertical equilibrium:

$T_1 \sin 60^\circ + T_2 \sin 30^\circ = 200$ N

Substituting $T_1 = \sqrt{3}T_2$:

$(\sqrt{3}T_2)\left(\frac{\sqrt{3}}{2}\right) + T_2\left(\frac{1}{2}\right) = 200$

$\frac{3}{2}T_2 + \frac{1}{2}T_2 = 200$

$2T_2 = 200$

$T_2 = 100$ N

Hence,

$T_1 = \sqrt{3} \times 100 = 100\sqrt{3}$ N

29. Answer: A

The vertical component of T_1 balances the suspended weight:

$T_1 \sin 60^\circ = 10$ kgwt

$T_1\left(\frac{\sqrt{3}}{2}\right) = 10$

$T_1 = \frac{20}{\sqrt{3}}$ kgwt

The horizontal component gives:

$T_2 = T_1 \cos 60^\circ$

$T_2 = \frac{20}{\sqrt{3}} \times \frac{1}{2} = \frac{10}{\sqrt{3}}$ kgwt

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LEVEL-II
SINGLE CORRECT OPTION TYPE

1. Answer: B

The maximum resultant is $P + Q$ and the minimum resultant is $P - Q$ (assuming $P > Q$).

Given,

$\frac{P+Q}{P-Q} = 3$

$P + Q = 3(P - Q)$

$P + Q = 3P - 3Q$

$4Q = 2P$

$P = 2Q$

2. Answer: C

Twelve equal forces placed at intervals of 30° form a complete closed polygon, giving a resultant of zero.

Since only 11 forces are present, the system is equivalent to a complete 12-force system with one force missing.

Therefore, the resultant of the 11 forces is equal in magnitude to the missing force.

Hence, the resultant is 5 N.

3. Answer: D

From the given maximum and minimum resultants:

$F_1 + F_2 = 7$ N

$F_1 - F_2 = 3$ N

Adding the equations:

$2F_1 = 10$

$F_1 = 5$ N

$F_2 = 2$ N

After increasing each force by 3 N:

$F_1' = 8$ N, $F_2' = 5$ N

Using the law of cosines for an angle of 60°:

$R = \sqrt{8^2 + 5^2 + 2(8)(5)\cos\ 60°}$

$R = \sqrt{64 + 25 + 40}$

$R = \sqrt{129}$ N

4. Answer: B

The magnitude of the difference of two vectors is

$\sqrt{a^2 + b^2 - 2ab\cos\ \theta}$

Given that it equals $a + b$:

$a^2 + b^2 - 2ab\cos\ \theta = (a + b)^2$

$a^2 + b^2 - 2ab\cos\ \theta = a^2 + b^2 + 2ab$

$-2ab\cos\ \theta = 2ab$

$\cos\ \theta = -1$

$\theta = 180°$

5. Answer: C

For two forces of magnitudes $A$and B, the resultant always lies between

$|\ A - B\ | \leq R \leq A + B$

For forces of 3 N and 8 N:

- Minimum resultant = $8 - 3 = 5$ N
- Maximum resultant = $8 + 3 = 11$ N

Since 4 N is less than the minimum possible resultant of 5 N, it cannot be obtained.

Hence, the combination 3 N, 8 N, and resultant 4 N is impossible.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

6. Answer: B and D

Explanation: Since $(\vec{A}+\vec{B})$is perpendicular to $(\vec{A}-\vec{B})$, their dot product is zero:

$(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 0$

Expanding,

$|\ \vec{A}\ |^2 - \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} - |\ \vec{B}\ |^2 = 0$

Since $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$, the middle terms cancel:

$|\ \vec{A}\ |^2 - |\ \vec{B}\ |^2 = 0$

$|\ \vec{A}\ | = |\ \vec{B}\ |$

Thus, the magnitudes of the two vectors are equal, irrespective of the angle between them.

7. Answer: A

Explanation: The vectors $\overrightarrow{OA}$and $\overrightarrow{OC}$are perpendicular and each has magnitude R. Their resultant is

$\sqrt{R^2 + R^2} = R\sqrt{2}$

This resultant lies along the angle bisector (45°), which is the same direction as $\overrightarrow{OB}$. Hence, the magnitudes can be added directly:

$R_{\text{total}} = R\sqrt{2} + R = R(1 + \sqrt{2})$

8. Answer: A and C

Explanation: Given

$|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$
Squaring both sides,
$A^2+B^2+2AB\cos\theta = A^2+B^2-2AB\cos\theta$
$4AB\cos\theta = 0$
$\cos\theta = 0$
Therefore,
$\theta = 90° = \frac{\pi}{2}$

9. Answer: A and B

Explanation: Applying equilibrium conditions:
Horizontal equilibrium:
$T_1\cos 60° = T_2\cos 30°$
$T_1 = T_2\sqrt{3}$
Vertical equilibrium:
$T_1\sin 60° + T_2\sin 30° = 80$
Substituting $T_1 = T_2\sqrt{3}$,
$(T_2\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) + T_2\left(\frac{1}{2}\right) = 80$
$\frac{3T_2}{2} + \frac{T_2}{2} = 80$
$2T_2 = 80$
$T_2 = 40$ kg wt
Hence,
$T_1 = 40\sqrt{3}$ kg wt

10. Answer: B

Explanation: Since the resultant $\vec{R}$is perpendicular to $\vec{A}$, the vectors form a right triangle with $\vec{B}$as the hypotenuse:
$A^2 + R^2 = B^2$
Given
$R = \frac{B}{2}$
$A^2 + \left(\frac{B}{2}\right)^2 = B^2$
$A^2 = \frac{3B^2}{4}$
$A = \frac{\sqrt{3}}{2}B$
Using the condition for a perpendicular resultant,
$\cos\theta = -\frac{A}{B}$
$\cos\theta = -\frac{\sqrt{3}}{2}$
$\theta = 150°$
COMPREHENSION TYPE
Passage-i

11. Correct Option: D

For rectangular components, the components satisfy the Pythagorean identity:
$F^2 = x^2 + y^2$
Therefore,

$y = \sqrt{F^2 - x^2}$

12. Correct Option: B

If the vector magnitude is V, its vertical component is given by

$y = V\sin\,\theta$

Rearranging,

$V = \frac{y}{\sin\theta} = y\csc\,\theta$

13. Correct Option: D

Since the angle $\theta$is measured with the vertical axis, the horizontal component is

$H = V\sin\,\theta$

Hence,

$V = \frac{H}{\sin\theta}$

The vertical component is

$V_y = V\cos\,\theta = \left(\frac{H}{\sin\theta}\right)\cos\,\theta = H\cot\,\theta$

14. Correct Option: C

When two forces act in the same direction, the angle between them is 0°, so their magnitudes add directly:

$R = 5\,\text{N} + 3\,\text{N} = 8\,\text{N}$

15. Correct Option: A

For two equal forces F, the resultant is

$R = \sqrt{F^2 + F^2 + 2F^2\cos\,\theta}$

$= \sqrt{2F^2(1 + \cos\,\theta)}$

Using the identity $1 + \cos\,\theta = 2\cos^2\frac{\theta}{2}$,

$R = \sqrt{4F^2\cos^2\frac{\theta}{2}} = 2F\cos\,\frac{\theta}{2}$

16. Correct Option: B

Using the formula from Question 15,

$R = 2F\cos\,\frac{\theta}{2}$

Substituting $F = 10$and $\theta = 60°$,

$R = 2(10)\cos\,30° = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3}$ units

17. Correct Option: A

Statement I is true because the parallelogram law is a fundamental law of vector addition.

Statement II is also true because the magnitude of the resultant equals the sum of the magnitudes of the two vectors only when both vectors act in the same direction ($\theta = 0°$).

18. Correct Option: C

Statement I is true because, unless a direction is specified, a vector component refers to its algebraic scalar value.

Statement II is false because the vector

$4\hat{\imath} + 5\hat{\jmath}$

has a y-component equal to 5, not 4.

19.

a) → R (0)

The magnitude of the resultant vector is exactly equal to the algebraic difference of their individual magnitudes ($C = A - B$). This condition occurs when the two vectors are in the same direction, i.e., the angle between them is $0°(\theta = 0)$.

b) → P (π)

The resultant vector has the minimum possible magnitude, equal to the difference of the two vector magnitudes ($C = A - B$). This happens when the vectors are in opposite directions, i.e., the angle between them is $180°(\theta = \pi)$.

c) → S, T ($\frac{\pi}{2}$ or 90°)

The magnitudes satisfy the Pythagorean relation ($A^2 + B^2 = C^2$), indicating that the two vectors are perpendicular to each other. Hence, the angle between them is $90°\left(\theta = \frac{\pi}{2}\right)$.

d) → Q ($\frac{2\pi}{3}$ or 120°)

When two equal vectors combine to produce a resultant of the same magnitude ($A = B = C$), they form an equilateral triangle configuration. Therefore, the angle between the vectors is $120°\left(\theta = \frac{2\pi}{3}\right)$.

20.

- a → P: Horizontal component = $P\cos\theta$
- b → Q: Vertical component = $P\sin\theta$
- c → R, S: The angle between P_x and P_y is 90°(or $\frac{\pi}{2}$).
- d → T: Since

$\tan\theta = \frac{P_y}{P_x}$,

$\theta = \tan^{-1}\left(\frac{P_y}{P_x}\right)$

21. Answer: 6

Let O be the centre of the regular hexagon.

The position vectors satisfy

$\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} + \vec{F} = \vec{0}$.

Now,

$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = (\vec{B} - \vec{A}) + (\vec{C} - \vec{A}) + (\vec{D} - \vec{A}) + (\vec{E} - \vec{A}) + (\vec{F} - \vec{A})$

$= (\vec{B} + \vec{C} + \vec{D} + \vec{E} + \vec{F}) - 5\vec{A}$.

Since

$\vec{B} + \vec{C} + \vec{D} + \vec{E} + \vec{F} = -\vec{A}$,

we get

Sum $= -\vec{A} - 5\vec{A} = -6\vec{A}$.

Because

$\overrightarrow{AO} = -\vec{A} = \overrightarrow{OD}$,

$-6\vec{A} = 6\overrightarrow{OD}$.

22. Answer: 1

Given

$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$,

squaring both sides,

$A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB\cos\theta$.

Therefore,

$4AB\cos\theta = 0$

$\cos\theta = 0$

$\theta = 90°$.
Given
$9 \times 10^x = 90$,
$10^x = 10$,
$x = 1$.

2. Motion in a Plane

TOPIC-1: Motion in two dimensions

LEVEL – I

1. Answer: B) Parabola

Explanation: The horizontal displacement is linear with time ($x = v_x t$), while the vertical displacement is quadratic $\left(y = v_y t - \frac{1}{2} g t^2\right)$. Eliminating time gives a quadratic equation representing a parabola.

2. Answer: B) Vertical component of velocity is zero

Explanation: At the highest point, the vertical component of velocity becomes zero $(v_y = 0)$, while the horizontal component remains constant.

3. Answer: C) Perpendicular to each other

Explanation: At the highest point, the velocity is horizontal while the acceleration due to gravity is vertically downward. Hence, they are perpendicular.

4. Answer: C) $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

Explanation: Using the trajectory equation and substituting the range expression $R = \frac{2u^2 \sin\theta \cos\theta}{g}$, the equation simplifies to this form.

5. Answer: C) $\tan^2\theta$

Explanation: Since

$H = \frac{u^2 \sin^2\theta}{2g}$,

for complementary angles,

$\frac{H_1}{H_2} = \frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta$.

6. Answer: A) $\tan\theta$

Explanation: Time of flight is

$T = \frac{2u\sin\theta}{g}$.

Therefore,

$\frac{T_1}{T_2} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

7. Answer: D) $2u\sin\theta$

Explanation: The horizontal velocity remains unchanged, while the vertical velocity changes from $+u\sin\theta$ to $-u\sin\theta$. Hence,

$\Delta v = 2u\sin\theta$.

8. Answer: A) $\cot^2\theta$

Explanation: At the highest point,

$KE = \frac{1}{2}mu^2\cos^2\theta, PE = \frac{1}{2}mu^2\sin^2\theta$.

Therefore,

$\frac{KE}{PE} = \cot^2\theta$.

9. Answer: D) 39.2 m

Explanation: Using

$t = \frac{u\sin 45^\circ}{9.8}$,

we get $u = 39.2$ m/s. Then,

$H = \frac{u^2 \sin^2 45^\circ}{2g} = 39.2$ m.

10. Answer: B) 3 km

Explanation: Since

$R = \frac{u^2 \sin 2\theta}{g}$,

using the first condition gives

$\frac{u^2}{g} = 3$.

At 45°,

$R = \frac{u^2}{g} = 3$ km.

11. Answer: A) 1 s

Explanation:

$T = \frac{2u \sin 30^\circ}{g} = \frac{2 \times 10 \times 0.5}{10} = 1$ s.

12. Answer: C) 1: 1

Explanation: Complementary angles produce the same horizontal range for the same initial speed.

13. Answer: D) $\frac{8u^2}{17g}$

Explanation: Equating range and height gives

$\tan\theta = 4$.

Substituting into the maximum height formula gives

$H = \frac{8u^2}{17g}$.

14. Answer: D) $\frac{R}{4}$

Explanation: Maximum range occurs at 45°. Since

$R = \frac{u^2}{g}$,

the corresponding height is

$H = \frac{u^2}{4g} = \frac{R}{4}$.

15. Answer: B) 1

Explanation: Maximum range occurs at

$\theta = 45^\circ$.

Hence,

$\tan 45^\circ = 1$.

16. Answer: D) 30°

Explanation: Using

$\tan\theta = \frac{4H}{R}$,

we get

$\tan\theta = \frac{1}{\sqrt{3}}$,

so

$\theta = 30^\circ$.

17. Answer: A) $19.6\ \mathbf{m/s}$

Explanation:

$$T = \frac{2u\sin 30^\circ}{9.8},$$

which gives

$u = 19.6\ \text{m/s}$.

18. Answer: C) 0°

Explanation: At $t = 1.5$ s,

$v_y = u\sin 30^\circ - gt = 15 - 15 = 0$.

Hence, the velocity is purely horizontal.

19. Answer: B) $\frac{k}{2}$

Explanation:

$KE_{\text{top}} = k\cos^2 45^\circ = \frac{k}{2}$.

20. Answer: D) $\frac{k}{4}$

Explanation:

$KE_{\text{top}} = k\cos^2 60^\circ = \frac{k}{4}$.

LEVEL – II

1. Answer: B) $9.6\ \mathbf{m}$

Explanation: Here,

$u_x = 6, u_y = 8$.

Therefore,

$$R = \frac{2u_x u_y}{g} = \frac{2 \times 6 \times 8}{10} = 9.6\ \text{m}.$$

2. Answer: C) 30°

Explanation: Using

$$y = u\sin\theta\, t - \frac{1}{2}gt^2,$$

$2 = 14\sin\theta - 5$.

Hence,

$$14\sin\theta = 7, \sin\theta = \frac{1}{2},$$

so

$\theta = 30^\circ$.

3. Answer: B) $\sqrt{\frac{8h}{g}}$

Explanation: Since

$$h = \frac{u^2\sin^2\theta}{2g},$$

we get $u\sin\theta = \sqrt{2gh}$.

Substituting into $T = \frac{2u\sin\theta}{g}$

Gives $T = \sqrt{\frac{8h}{g}}$.

4. Answer: D) $\sqrt{2g}$

Explanation: Using $R = \frac{u^2 \sin 2\theta}{g}$,

$2 = \frac{u^2}{g}$,

So $u = \sqrt{2g}$.

5. Answer: C) 60°

Explanation: Since

$PE_{\max} = \frac{1}{2}mu^2 \sin^2\theta$,

and

$KE_{\text{initial}} = \frac{1}{2}mu^2$,

$\sin^2\theta = \frac{3}{4}$.

Thus,

$\sin\theta = \frac{\sqrt{3}}{2}$,

which gives

$\theta = 60°$.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

6. Answer: A, B, C

A is correct: Acceleration due to gravity ($\vec{g}$) is constant in both magnitude ($9.8\ \text{m/s}^2$) and downward direction throughout the motion.

B is correct: There is no horizontal force acting on the projectile (neglecting air resistance), so the horizontal component of velocity ($u_x = u\cos\theta$) remains constant.

C is correct: Since gravity is a conservative force and air resistance is neglected, the total mechanical energy is conserved.

D is incorrect: Velocity changes continuously because its vertical component changes due to gravity.

7. Answer: A, B, C, D

A and B are correct: The vertical component of velocity changes continuously under gravity, which changes both the overall velocity vector and the linear momentum ($\vec{p} = m\vec{v}$).

C and D are correct: As the height changes, potential energy ($U = mgh$) changes. Consequently, to keep total mechanical energy constant, kinetic energy ($K = \frac{1}{2}mv^2$) must also change.

8. Answer: A, C, D

The entire problem hinges on one core rule of projectile motion: The maximum height (H) depends only on the vertical component of the initial velocity (u_y).

Since both projectiles reach the same maximum height, their vertical components of initial velocity must be equal:

$u_{1y} = u_{2y}$

Time of Flight and Time to Reach Maximum Height (Options A and D):

The time of flight and the time taken to reach the highest point depend only on the vertical component of the initial velocity. Since both projectiles have the same vertical component, they remain in the air for the same duration and reach their maximum heights at the same time. Therefore, Options A and D are correct.

Initial Speeds (Option C):
To produce the same vertical component at a smaller angle (30°) compared to a larger angle (60°), the projectile launched at 30°must have a greater initial speed.
Using
$u_1 \sin 30° = u_2 \sin 60°$,
$u_1\left(\frac{1}{2}\right) = u_2\left(\frac{\sqrt{3}}{2}\right)$,
$\frac{u_1}{u_2} = \sqrt{3}.$
Hence,
$u_1 : u_2 = \sqrt{3} : 1.$
Therefore, Option C is correct.

9. Answer: A, C

For two projectiles to have the same range R, their angles of projection must be complementary:
$\theta$and$(90° - \theta)$
The times of flight are
$T_1 = \frac{2u\sin\ \theta}{g}, T_2 = \frac{2u\sin\ (90° - \theta)}{g} = \frac{2u\cos\ \theta}{g}$
Multiplying,
$T_1 T_2 = \frac{4u^2 \sin\ \theta \cos\ \theta}{g^2} = \frac{2}{g}\left(\frac{u^2 \sin\ 2\theta}{g}\right) = \frac{2R}{g}$
Hence,
$T_1 T_2 \propto R$
So, Option A is correct.
Dividing,
$\frac{T_1}{T_2} = \frac{\sin\theta}{\cos\theta} = \tan\ \theta$
Therefore, Option C is correct (or Option D if T_1corresponds to the angle $90° - \theta$).

Passage – I ($u = 60$ m/s, $\theta = 30°$)

10. Correct Option: C

Explanation:

Horizontal component:
$u_x = 60\cos\ 30° = 30\sqrt{3}$ m/s
Vertical component:
$u_y = 60\sin\ 30° = 30$ m/s
Hence, the initial velocity vector is
$\vec{u} = 30\sqrt{3}\hat{\imath} + 30\hat{\jmath}$ m/s.

11. Correct Option: D

Explanation:

The horizontal velocity remains constant:
$v_x = 30\sqrt{3}$ m/s.
After 3 s, the vertical velocity is
$v_y = u_y - gt = 30 - 10(3) = 0.$
Therefore,
$\vec{v} = 30\sqrt{3}\hat{\imath}$ m/s.

12. Correct Option: A

Explanation:

Horizontal displacement after 2 s:

$x = u_x t = 30\sqrt{3} \times 2 = 60\sqrt{3}$ m.

Vertical displacement:

$y = u_y t - \frac{1}{2} g t^2 = 30(2) - \frac{1}{2}(10)(2)^2 = 40$ m.

Hence,

$\vec{s} = 60\sqrt{3}\hat{\imath} + 40\hat{\jmath}$ m.

Passage – II $\left(R = \mathbf{200\ m},\ T = \mathbf{5\ s},\ g = \mathbf{10\ m/s^2}\right)$

13. Correct Option: B

Explanation:

Using the range relation,

$R = u_x T$

$200 = u_x \times 5$

$u_x = 40$ m/s.

14. Correct Option: C

Explanation:

Using the time of flight formula,

$T = \frac{2u_y}{g}$

$5 = \frac{2u_y}{10}$

$u_y = 25$ m/s.

15. Correct Option: C

Explanation:

The angle of projection is

$\tan\theta = \frac{u_y}{u_x} = \frac{25}{40} = \frac{5}{8}$.

Therefore,

$\theta = \tan^{-1}\left(\frac{5}{8}\right)$.

Matrix Matching Type

16. Answer:

(a → R, b → P, c → S, d → Q)

Explanation:

Using the range formula,

$R = \frac{u^2 \sin 2\theta}{g}$

a: $\frac{20^2 \sin 60^\circ}{10} = 20\sqrt{3}$ m→ **R**

- **b:** $\frac{30^2 \sin 120^\circ}{10} = 45\sqrt{3}$ m→ **P**
- **c:** Maximum range occurs at 45°:

  $R = \frac{30^2}{10} = 90$ m

  → **S**
- **d:** $\frac{20^2 \sin 90^\circ}{10} = 40$ m→ **Q**

17. Answer:

(a → R, b → S, c → P, d → Q)

Explanation:

For maximum range,

$\theta = 45°$,

so

$u_x = u_y = \frac{v}{\sqrt{2}}$.

a → R: At $y = \frac{H}{2}$,

$v_y = \frac{v}{2}$.

Hence,

$v = \sqrt{\frac{v^2}{2} + \frac{v^2}{4}} = \frac{\sqrt{3}\,v}{2}$.

b → S: At maximum height,

$v_y = 0$,

therefore,

$v = \frac{v}{\sqrt{2}}$.

c → P: Initial velocity:

$\frac{v}{\sqrt{2}}\hat{\imath} + \frac{v}{\sqrt{2}}\hat{\jmath}$.

Final velocity:

$\frac{v}{\sqrt{2}}\hat{\imath} - \frac{v}{\sqrt{2}}\hat{\jmath}$.

Hence,

$|\,\Delta\vec{v}\,| = v\sqrt{2}$.

d → Q: Average velocity is

$\vec{v}_{\text{avg}} = \frac{\text{Displacement}}{\text{Time}}$.

Its magnitude is

$\frac{v}{2}\sqrt{\frac{5}{2}}$.

Integer Type

18. Answer: 8 (or 7, depending on interpretation)

Explanation:

Initial range:

$R_1 = \frac{u^2 \sin 30°}{g} = \frac{u^2}{2g}$.

New range:

$R_2 = \frac{(2u)^2 \sin 90°}{g} = \frac{4u^2}{g}$.

Therefore,

$\frac{R_2}{R_1} = 8$.

ence, the new range is **8 times** the original range (or it has **increased by 7 times** its original value).

19. Answer: 5

Explanation:

Maximum height is

$H = \frac{u^2 \sin^2 \theta}{2g}$.

Substituting,

$H = \frac{20^2 \sin^2 30°}{2(10)} = \frac{400 \times \frac{1}{4}}{20} = 5$ m.

20. Answer: 30

Explanation:

The horizontal component of velocity remains unchanged. Only the vertical component becomes zero at the highest point.

Therefore,

$|\Delta\vec{v}| = u\sin\theta = 50\sin 37° = 50(0.6) = 30$ m/s.

TOPIC-2: Horizontal Projection From the Top of a Tower

1. Answer (B): The time of flight for a horizontal projectile depends only on the vertical height and acceleration due to gravity. Using $t = \sqrt{\frac{2h}{g}}$, it is clear that the horizontal velocity has no effect on the time of flight.

2. Answer (C): Both stones are projected horizontally from the same height. Since the time of flight depends only on height and gravity, both remain in the air for the same duration. Therefore, $t_1 = t_2$.

3. Answer (D): The horizontal range is given by $R = ut$. Since $t = \sqrt{\frac{2h}{g}}$, the horizontal velocity is $u = R\sqrt{\frac{g}{2h}}$.

4. Answer (C): The time of flight is $t = \sqrt{\frac{2\times19.6}{9.8}} = 2$ s. Hence, the horizontal distance covered is $R = 10 \times 2 = 20$ m.

5. Answer (D): The time taken is $t = \frac{75}{12.5} = 6$ s. The tower height is $h = \frac{1}{2}gt^2 = \frac{1}{2}(10)(6)^2 = 180$ m.

6. Answer (A): Convert the speed first: 180 km/h = 50 m/s. The time of fall is $t = \sqrt{\frac{2\times490}{9.8}} = 10$ s. Therefore, the horizontal distance is $R = 50 \times 10 = 500$ m.

7. Answer (D):

- Height, $h = 396.9$ m
- Initial vertical velocity, $u_y = 0$
- Acceleration due to gravity, $g = 9.8\ \text{m/s}^2$

Using:

$h = \frac{1}{2}gt^2$

$396.9 = \frac{1}{2} \times 9.8 \times t^2$

$396.9 = 4.9t^2$

$t^2 = \frac{396.9}{4.9} = 81$

$t = \sqrt{81} = 9$ s

8. **Answer (B):** The time of fall is $t = \sqrt{\frac{2\times80}{10}} = 4$ s. Hence, the aircraft must release the package when it is $150 \times 4 = 600$ mbefore the target.

9. **Answer (C):** The time of flight is $t = \sqrt{\frac{2\times490}{9.8}} = 10$ s. Therefore, the horizontal distance covered is $150 \times 10 = 1500$ m.

10. **Answer (C):** The time taken to fall is $t = \sqrt{\frac{2\times5}{10}} = 1$ s. Thus, the required horizontal speed is $u = \frac{10}{1} = 10$ m/s.

11. **Answer (A):** The horizontal velocity remains constant at $10\hat{\imath}$. After 3 s, the vertical velocity becomes $-30\hat{\jmath}$. Hence, the velocity vector is $10\hat{\imath} - 30\hat{\jmath}$.

12. **Answer (A):** nearest value is 40 m/s
The horizontal velocity remains constant at 20 m/s, while the vertical velocity after 4s is $v_y = gt = 10 \times 4 = 40$ m/s. Therefore, the resultant velocity is $v = \sqrt{20^2 + 40^2} = 20\sqrt{5}$ m/s. (approximately **44.72 ms⁻¹**).

13. **Answer (B):** The vertical speed just before hitting the ground is $v_y = \sqrt{2gh} = \sqrt{2g^2} = g\sqrt{2}$. Combining with the horizontal speed g, the final speed is $\sqrt{g^2 + 2g^2} = \sqrt{3}\,g$.

14. **Answer (B):** The direction of motion satisfies $\tan 30° = \frac{v_y}{v_x}$. Using $v_y = gt$and $t = 2$ s, the horizontal speed is $u = 2\sqrt{3}\,g$.

15. **Answer (C):** After 2 s, the horizontal and vertical velocity components are both 19.6 m/s. Since the components are equal, the velocity makes an angle of 45°with the horizontal.

16. **Answer (C):** After 1 s, the horizontal and vertical components are each 9.8 m/s. Hence, the resultant speed is $9.8\sqrt{2}$ m/s.

17 **Answer (B):** The line joining the top of the tower to the landing point makes an angle of 45°, implying $R = h = 78.4$ m. The time of flight is 4 s, so the required horizontal speed is $\frac{78.4}{4} = 19.6$ m/s.

18. **Answer (D):** The time of flight is $t = \sqrt{\frac{2\times20}{10}} = 2$ s. The range is 20 m. The straight-line displacement from launch to landing is $\sqrt{20^2 + 20^2} = 20\sqrt{2}$ m.

19. **Answer (C):** After 10 s, the horizontal and vertical velocity components are both 98 m/s. Therefore, the projectile strikes the ground at an angle of 45°.

20. **Answer (A):** The bullet falls by 0.1 m. Using $0.1 = \frac{1}{2}(9.8)t^2$, the time is $t = \frac{1}{7}$ s. If the horizontal distance is corrected to 100 m, the speed is $\frac{100}{1/7} = 700$ m/s.

21. **Answer (B):** The time to reach the ground is 4 s. Since the particles move in opposite directions, their relative horizontal speed is $20 + 30 = 50$ m/s. Thus, the separation at landing is $50 \times 4 = 200$ m.

22. **Answer (C):** Using the trajectory equation $y = \frac{gx^2}{2u^2}$with $x = 0.3n$, $y = 0.2n$, $g = 10$ m/s^2, and $u = 4.5$ m/s, the value of $n$is found to be 9.

23. **Answer (B):** The position vectors are $\vec{r}_1 = ut\hat{\imath} - \frac{1}{2}gt^2\hat{\jmath}$and $\vec{r}_2 = -2ut\hat{\imath} - \frac{1}{2}gt^2\hat{\jmath}$. Setting their dot product to zero gives $t = \frac{2\sqrt{2}\,u}{g}$.

24. **Answer (D):** The velocity vectors are $\vec{v}_1 = \hat{\imath} - gt\hat{\jmath}$and $\vec{v}_2 = -4\hat{\imath} - gt\hat{\jmath}$. Their dot product becomes zero when $t = \frac{2}{g}$. The relative horizontal speed is 5 m/s, so the separation is $5 \times \frac{2}{g} = 1$ m.

25. **Answer (C):** The rain appears inclined because of the cyclist's motion. The angle with the vertical satisfies $\tan\theta = \frac{10}{30} = \frac{1}{3}$. Therefore, $\theta = \tan^{-1}\left(\frac{1}{3}\right)$.

26. **Answer A:** The time to cross depends only on the perpendicular speed:
$t = \frac{1.0 \text{ km}}{4.0 \text{ km/h}} = 0.25$ h.
The river current flows horizontally at 3.0 km/h, dragging the man downstream during this time. The drift distance is:
Drift = 3.0 km/h × 0.25 h = 0.75 km = 750 m.

27. **Answer D:** The flag flutters in the direction of the relative velocity of the wind with respect to the boat ($\vec{v}_{wb} = \vec{v}_w - \vec{v}_b$). The wind vector resolves to $51\hat{\imath} + 51\hat{\jmath}$(NE), and the boat vector is $51\hat{\jmath}$(N). Subtracting them leaves a purely horizontal vector $51\hat{\imath}$, which points due East.

LEVEL-II
SINGLE CORRECT OPTION TYPE

1. **Answer B:** The horizontal velocity component remains constant at $v_x = u$. Since the final speed is $4u$, the vertical component satisfies:
$v_y^2 = (4u)^2 - u^2 = 15u^2$.
Using the vertical motion equation $v_y^2 = 2gh$:
$15u^2 = 2gh$.
Hence, $h = \frac{15u^2}{2g}$.

2. **Answer A:** In horizontal projectile motion, the horizontal component of velocity remains constant and equals the initial projection speed u. At the ground, the velocity vector makes an angle of 30°with the horizontal. Using:
$\cos 30° = \frac{u}{v}$,
we obtain:
$u = \frac{\sqrt{3}v}{2}$.

3. **Answer B:** The initial velocity is $\vec{v}_i = u\hat{\imath}$, while the final velocity on landing is:
$\vec{v}_f = u\hat{\imath} - \sqrt{2gh}\hat{\jmath}$.
Therefore, the change in velocity is:
$\Delta\vec{v} = -\sqrt{2gh}\hat{\jmath}$.
Its magnitude is:
$\sqrt{2gh}$.

4. **Answer B:** The bullet takes:
$t = \frac{200}{600} = \frac{1}{3}$ s
to travel between the screens. During this time, gravity causes a vertical drop:
$y = \frac{1}{2}gt^2$.
Substituting $g = 9.8\ \text{m/s}^2$:
$y = \frac{1}{2}(9.8)\left(\frac{1}{3}\right)^2 = \frac{49}{90}$ m.

5. **Answer A/C:** To hit the n-th step, the ball must fall a vertical distance nhwhile traveling a horizontal distance nx. From vertical motion:

$nh = \frac{1}{2}gt^2$,

so:

$t = \sqrt{\frac{2nh}{g}}$.

Using horizontal motion $nx = vt$:

$v = \sqrt{\frac{ng}{2h}}\, x$.

6. **Answer A:** Using:

$v = \sqrt{\frac{ng}{2h}}\, x$,

substitute $n = 5$, $g = 10\ \text{m/s}^2$, $h = 0.2$ m, and $x = 0.3$ m:

$v = \sqrt{\frac{50}{0.4}} \times 0.3 = \sqrt{125} \times 0.3 = 1.5\sqrt{5}\ \text{m/s}$.

7. **Answer A:** The velocity vectors of the fragments are:

$\vec{v}_1 = 10\hat{\imath} - gt\hat{\jmath}$,

$\vec{v}_2 = -10\hat{\imath} - gt\hat{\jmath}$.

For them to be perpendicular:

$\vec{v}_1 \cdot \vec{v}_2 = 0$.

Thus:

$-100 + g^2t^2 = 0$.

With $g = 10\ \text{m/s}^2$:

$100t^2 = 100$,

so:

$t = 1$ s.

8. **Answer A:** The velocity vectors are:

$\vec{v}_1 = u\hat{\imath} - gt\hat{\jmath}$,

$\vec{v}_2 = -u\hat{\imath} - gt\hat{\jmath}$.

Each vector makes an angle of 30°with the downward vertical. Therefore:

$\tan 30° = \frac{u}{gt} = \frac{1}{\sqrt{3}}$.

Hence:

$t = \frac{\sqrt{3}u}{g}$.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

COMPREHENSION TYPE

13. **Answer C:** The velocity vectors are:

$\vec{v}_1 = 2\hat{\imath} - gt\hat{\jmath}$,

$\vec{v}_2 = -8\hat{\imath} - gt\hat{\jmath}$.

For perpendicularity:

$(2)(-8) + g^2t^2 = 0$.

With $g = 10\ \text{m/s}^2$:

$100t^2 = 16$,

giving:

$t = 0.4$ s.

14. **Answer D:** The bodies move in opposite horizontal directions, so their relative speed is:

$2 + 8 = 10\ \text{m/s}$.

Using $t = 0.4$ s:

$d = 10 \times 0.4 = 4$ m.

15 **Answer C:** Their displacement vectors are:

$\vec{r}_1 = 2t\hat{\imath} - \frac{1}{2}gt^2\hat{\jmath}$,

$\vec{r}_2 = -8t\hat{\imath} - \frac{1}{2}gt^2\hat{\jmath}$.

For perpendicular displacement vectors:

$-16t^2 + \frac{1}{4}g^2t^4 = 0$.

With $g = 10\,\mathrm{m/s^2}$:

$25t^2 = 16$,

so:

$t = 0.8$ s.

16 **Answer A:** The river width is 0.5 km, and the component of velocity perpendicular to the flow is 3sin θ km/h. Thus:

$t = \frac{0.5}{3\sin\theta}$ h.

Converting to minutes:

$t = \frac{10}{\sin\theta}$ minutes.

17 **Answer D:** The crossing time is minimum when the perpendicular velocity component is maximum. This occurs when sin $\theta = 1 (\theta = 90^\circ)$. Therefore:

$t_{\min} = \frac{10}{1} = 10$ minutes.

18. **(a) Velocity vectors are perpendicular** $(\vec{v}_1 \cdot \vec{v}_2 = 0)$

Condition:

$-u_1u_2 + g^2t^2 = 0$

Solving for time:

$t = \frac{\sqrt{u_1u_2}}{g}$

Match: R

(b) **Displacement vectors are perpendicular** $(\vec{r}_1 \cdot \vec{r}_2 = 0)$

Condition:

$-u_1u_2t^2 + \frac{1}{4}g^2t^4 = 0$

Solving for time:

$t = \frac{2\sqrt{u_1u_2}}{g}$

Match: T

(c) **Separation when velocities are perpendicular**

Using the time obtained in (a):

$x = (u_1 + u_2)t = \frac{(u_1 + u_2)\sqrt{u_1u_2}}{g}$

Match: P

(d) **Separation when displacements are perpendicular**

Using the time obtained in (b):

$x = (u_1 + u_2)t = \frac{2(u_1 + u_2)\sqrt{u_1u_2}}{g}$

Match: S

Final Answer Matrix:

a → R, b → T, c → P, d → S

a) Time of descent → R
For vertical downward motion, the initial vertical velocity is zero ($u_y = 0$). Using the equation of motion, $h = \frac{1}{2}gt^2$, the time of descent is:
$t = \sqrt{(2h/g)}$
b) Range → P
The horizontal range equals the horizontal velocity multiplied by the time of descent. Since the horizontal velocity remains constant:
Range = $u\sqrt{(2h/g)}$
c) Velocity at impact → Q
At impact, the body has both horizontal and vertical velocity components.

- Horizontal component: $v_x = u$
- Vertical component: $v_y = \sqrt{(2gh)}$

The resultant velocity is:
$v = \sqrt{(u^2 + 2gh)}$
d) Equation of trajectory → S
The horizontal position is $x = ut$, so $t = x/u$.
The vertical displacement is $y = \frac{1}{2}gt^2$. Substituting $t = x/u$ gives:
$y = (g/2u^2)x^2$

INTEGER TYPE

21. Answer: $n = 2$

Explanation:
Since both stones are projected horizontally from the same height ($h = 80$ m), they take the same time to reach the ground.
Time of flight:

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 80}{10}} = \sqrt{16} = 4 \text{ s}$$

Horizontal distance covered by the first stone:
$x_1 = u_1 \times t = 20 \times 4 = 80$ m
Horizontal distance covered by the second stone:
$x_2 = u_2 \times t = 30 \times 4 = 120$ m
Since the stones move in opposite directions, the distance between them is:
$D = x_1 + x_2 = 80 + 120 = 200$ m
Given $D = n \times 10^2$ m,
$n \times 100 = 200$
$\therefore n = 2$

22. Answer: $t = 2\sqrt{3}$ s (approximately 3.46 s)

Explanation:
Let the velocity vectors at time $t$be:
$\vec{v}_1 = 30\hat{\imath} - gt\hat{\jmath}$
$\vec{v}_2 = -40\hat{\imath} - gt\hat{\jmath}$
For the velocity vectors to be perpendicular, their dot product must be zero.
$(30)(-40) + (-gt)(-gt) = 0$
$-1200 + g^2t^2 = 0$
Substituting $g = 10\ \text{m/s}^2$:
$100t^2 = 1200$
$t^2 = 12t = \sqrt{12} = 2\sqrt{3}$ s

23. Answer: 100 m

Explanation:

The time of flight is:

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 80}{10}} = 4$ s

The horizontal distance travelled is:

Distance = Horizontal velocity × Time

$= 25 \times 4 = 100$ m

Therefore, the horizontal distance covered is 100 m.

PREVIOUS YEAR QUESTIONS

1. Correct Option: D

Explanation:

The particle moves in the x-y plane with:

- Initial velocity: u = 3.0î m/s, so $u_x = 3.0$m/s and $u_y = 0$.
- Acceleration: a = 6.0î + 4.0ĵ m/s², so $a_x = 6.0$m/s² and $a_y = 4.0$m/s².

Step 1: Find the time when the y-coordinate is 32 m.

Using the equation of motion along the y-axis:

$y = u_y t + \frac{1}{2} a_y t^2$

$32 = 0 + \frac{1}{2}(4)t^2$

$32 = 2t^2$

$t^2 = 16$

$t = 4$s

Step 2: Find the x-coordinate at t = 4 s.

Using the equation of motion along the x-axis:

$x = u_x t + \frac{1}{2} a_x t^2$

$x = 3(4) + \frac{1}{2}(6)(4)^2$

$x = 12 + 48 = 60$m

2. Correct Option: A

Explanation:

Step 1: At maximum height

For the first particle,

Maximum height:

$H = \frac{u^2 \sin^2 60^\circ}{2g} = \frac{3u^2}{8g}$

Velocity at maximum height:

$v_1 = u\cos 60^\circ = \frac{u}{2}$

The second particle moves horizontally with velocity:

$v_2 = u$

Step 2: Apply conservation of linear momentum

$m\left(\frac{u}{2}\right) + mu = 2mv_f$

$v_f = \frac{3u}{4}$

Step 3: Horizontal distance after collision

Time taken to fall:

$t = \sqrt{\frac{2H}{g}} = \frac{\sqrt{3}u}{2g}$

Horizontal distance:

$x = v_f t$

$x = \left(\frac{3u}{4}\right)\left(\frac{\sqrt{3}u}{2g}\right)$

$x = \frac{3\sqrt{3}}{8}\frac{u^2}{g}$

3. Correct Option: B

Explanation:

River speed = 2 km/h

Swimmer's speed in still water = 4 km/h

For crossing straight across the river, the swimmer's horizontal component must cancel the river current.

$4\sin\phi = 2$

$\sin\phi = \frac{1}{2}$

$\phi = 30°$

Since the swimmer aims upstream,

$\theta = 90° + 30° = 120°$

4. Correct Option: B

Explanation:

Velocity of ship A:

$v_A = 30\hat{\imath} + 50\hat{\jmath}$km/h

Velocity of ship B:

$v_B = -10\hat{\imath}$km/h

Initial position of B relative to A:

$r_0 = 80\hat{\imath} + 150\hat{\jmath}$km

Relative velocity:

$v_{BA} = v_B - v_A$

$= (-10 - 30)\hat{\imath} - 50\hat{\jmath}$

$= -40\hat{\imath} - 50\hat{\jmath}$km/h

Relative position after time t:

$r(t) = (80 - 40t)\hat{\imath} + (150 - 50t)\hat{\jmath}$

Minimum distance occurs when the derivative of the square of the distance becomes zero.

Solving,

$4(80 - 40t) + 5(150 - 50t) = 0$

$320 - 160t + 750 - 250t = 0$

$410t = 1070$

$t = \frac{107}{41} \approx 2.6$h

5. Correct Option: D

Explanation:

Let the speed of sound be $v$and the speed of the plane be v_p.

The sound heard at the observer was emitted when the plane was at an earlier position.

From the right triangle,

$\cos 60° = \frac{v_p t}{vt}$

$\frac{1}{2} = \frac{v_p}{v}$

Therefore,

$v_p = \frac{v}{2}$

6. Correct Option: C

Explanation:

Total distance to cross completely:

$d = 60 + 120 = 180\text{m}$

Case 1: Same direction

Relative speed:

$80 - 30 = 50\text{km/h}$

$t_1 = \frac{180}{50}$

Case 2: Opposite direction

Relative speed:

$80 + 30 = 110\text{km/h}$

$t_2 = \frac{180}{110}$

Therefore,

$\frac{t_1}{t_2} = \frac{110}{50} = \frac{11}{5}$

7. Correct Option: B

Explanation:

For crossing along the shortest path,

$v_{res} = \sqrt{v_b^2 - v_r^2}$

Given,

Width = 1 km

Time = 15 min = 0.25 h

Hence,

$v_{res} = \frac{1}{0.25} = 4\text{km/h}$

Substitute into the equation:

$4 = \sqrt{5^2 - v_r^2}$

$16 = 25 - v_r^2$

$v_r^2 = 9$

$v_r = 3\text{km/h}$

8. Correct Option: A

Explanation:

To cross the river in the minimum possible time, the swimmer should swim directly perpendicular to the river flow.

Since the river flows from West to East, the swimmer should move due North.

9. Correct Option: B

Explanation:

For two projectiles having the same range and the same initial speed, their angles are complementary.

Maximum heights are:

$h_1 = \frac{u^2 \sin^2 \theta}{2g}$

$h_2 = \frac{u^2 \cos^2 \theta}{2g}$

Multiplying,

$h_1 h_2 = \frac{u^4 \sin^2 \theta \cos^2 \theta}{4g^2}$

Since,

$R = \frac{2u^2 \sin\theta \cos\theta}{g}$

Therefore,

$h_1 h_2 = \frac{R^2}{16}$

Hence,

$R^2 = 16 h_1 h_2$

10. Correct Option: D

Explanation:

The two projection angles are $\theta$and $(90° - \theta)$.

Their times of flight are:

$t_1 = \frac{2u \sin\theta}{g}$

$t_2 = \frac{2u \cos\theta}{g}$

Multiplying,

$t_1 t_2 = \left(\frac{2u \sin\theta}{g}\right)\left(\frac{2u \cos\theta}{g}\right)$

$= \frac{2}{g}\left(\frac{2u^2 \sin\theta \cos\theta}{g}\right)$

Using,

$R = \frac{2u^2 \sin\theta \cos\theta}{g}$

Therefore,

$t_1 t_2 = \frac{2R}{g}$

3. Force and Laws of Motion

Force

1. **Answer :** B) Unbalanced
An unbalanced force produces a non-zero net force.
This net force is required to change a body's state of rest or motion.

2. **Answer :** A) Balanced forces
Balanced forces result in a net force of zero.
A zero net force does not change the state of motion but can change the shape or size of an object.

3. **Answer :** A) Balanced force act on the balloon
Pressing the balloon from both sides applies equal and opposite forces.
These balanced forces cancel out to zero net force, causing a change in shape rather than motion.

4. **Answer :** D) All the above
An unbalanced force produces a net force on a body.
This can cause acceleration, changing the speed or direction of motion, and can also contribute to deformation in some cases.

5. **Answer :** B) Forward
When the car stops suddenly, the lower part of the passengers' bodies comes to rest with the car.
The upper part of their bodies tends to remain in motion due to the inertia of motion, causing a forward jerk.

6. **Answer :** A) Backward
When the car starts suddenly, the lower part of the passengers' bodies moves forward with the car.
The upper part of their bodies tends to remain at rest due to the inertia of rest, causing a backward jerk.

7. **Answer :** C) Inertia of direction
The mud particles are initially moving in a circular path along with the wheel.
Due to the inertia of direction, when they detach, they tend to continue moving in a straight line, which is tangential to the circular path.

8. **Answer :** C) Mass of the object
Inertia is the fundamental property of an object to resist changes in its state of motion.
The mass of an object is the direct mathematical and physical measure of its inertia.

9. **Answer :** A) Newton's first law of motion
When the branch is shaken, it is suddenly set into motion.
The ripe fruits tend to remain in their state of rest due to inertia, causing them to detach from the moving branch.

10. **Answer :** B) Move forward
The water inside the tanker is initially moving at the same uniform speed as the tanker.
When brakes are applied, the tanker slows down, but the water continues to move forward due to the inertia of motion.

11. **Answer :** D) Mass
Inertia is the resistance of any physical object to any change in its velocity.
Mass is the quantitative measure of a body's inertia.

12. **Answer :** D) Keep it moving with uniform velocity
According to Newton's first law of motion, an object in uniform motion will continue in that state unless acted upon by an external unbalanced force.
Therefore, no net force is required to maintain a constant speed in a straight line.

13. **Answer :** C) Inertia
Newton's first law states that an object will remain at rest or in uniform motion in a straight line unless acted upon by an external force.
This inherent property of matter to resist changes in motion is called inertia.

14. **Answer :** A) Inertia of direction
As the bus turns, it changes its direction of motion.
The passengers tend to continue moving in their original straight-line path due to the inertia of direction, causing them to be thrown outward.

15. **Answer :** B) The portion BC of the string will break
A sudden jerk applied to the lower string does not allow enough time for the tension to be transmitted to the upper string A.
Due to the inertia of rest of the 1 kg mass, the mass resists movement, causing the lower string (C or BC) to snap.

16. **Answer :** A) (i) is true & (ii) is false
A passenger jumping out of a moving bus possesses the forward inertia of motion of the bus.
To avoid falling, the passenger must run in the forward direction of the bus to gradually bring their body to rest.

17. **Answer :** B) Force
Newton's second law states that the rate of change of momentum of a body is directly proportional to the applied external force.
It provides the mathematical formulation for measuring force.

18. **Answer :** C) Net force
According to Newton's second law, acceleration is produced when a net force acts on a mass.
The direction of the resulting acceleration is always strictly in the same direction as the applied net force.

19. **Answer :** D) Both B and C
Newton's second law of motion defines the relationship between force and momentum.
It states that the rate of change of momentum is directly proportional to the net force and takes place in the direction of that force.

20. **Answer :** B) Reduce the force exerted by the ball
By pulling the hands backwards, the fielder increases the time duration over which the ball is brought to rest.
Since force is the rate of change of momentum, increasing the time decreases the impact force on the hands.

21. **Answer :** B) 5.00
$F = m \times a$
$5 = 1 \times a$
$a = 5\ \mathrm{m/s^2}$

22. **Answer :** B) 2 $\mathrm{m/s^2}$
Let the initial force of the engine be F and the mass of the car be m.
$F = m \times 4$
When it pulls another car of the same mass, the total mass becomes $2m$.
$F = (2m) \times a_{\text{new}}$
$m \times 4 = 2m \times a_{\text{new}}$
$a_{\text{new}} = \frac{4}{2}$
$a_{\text{new}} = 2\ \mathrm{m/s^2}$

23. **Answer :** C) South-West
To balance a force, an equal and opposite force must be applied so the net force becomes zero.
The exact opposite direction to North-East is South-West.

24. **Answer :** B) 2 N
$F = m \times a$
$F = 10 \times 0.2$
$F = 2\ \mathrm{N}$

25. **Answer :** D) 180°
According to Newton's third law of motion, action and reaction forces are equal in magnitude and strictly opposite in direction.
Two vectors pointing in exact opposite directions have an angle of 180° between them.

26. **Answer :** C) 50 kg m/s
$p = m \times v$
$p = 10 \times 5$
$p = 50\ \mathrm{kg\ m/s}$

27. **Answer :** B) 100 kg m/s
$v = u + a \times t$
$v = 0 + (4 \times 5)$
$v = 20\ \mathrm{m/s}$
$p = m \times v$
$p = 5 \times 20$
$p = 100\ \mathrm{kg\ m/s}$

28. **Answer :** A) Backward
When you jump out of a boat, your feet exert an action force pushing the boat backward.
According to Newton's third law, the boat exerts an equal and opposite reaction force pushing you forward.

29. **Answer :** D) All the above are true
Newton's third law establishes the properties of action and reaction pairs.
They always occur in pairs, act on different bodies, have equal magnitudes, and are directed oppositely.

30. **Answer :** B) (i) is false & (ii) is true
While statement (i) is a physical reality, it is the reaction force from the ground (statement ii) that actively propels the horse forward.
The horse pushes the ground backward (action), and the ground pushes the horse forward (reaction).
This reaction force is what physically allows the horse to accelerate forward.

LEVEL-II

1. **Answer :** C) Upper part of the body continues to be in the state of motion, whereas the lower part of the body in contact with the seat comes to rest
When the train stops, the seat and the lower part of the body come to a halt immediately.
The upper portion of the body maintains its forward velocity due to the inertia of motion, resulting in a forward jerk.

2. **Answer :** C) Newton's first law of motion
Beating the carpet with a stick suddenly sets the carpet into motion.
The dust particles remain in their initial state of rest due to inertia and thus get separated from the moving carpet.

3. **Answer :** A) 5000 N
$v = u + a \times t$
$0 = 5 + \left(a \times \frac{1}{10}\right)$
$a = -50 \text{ m/s}^2$
$F = m \times a$
$F = 100 \times (-50)$
$F = -5000 \text{ N}$
The required magnitude of force in the opposite direction is 5000 N.

4. **Answer :** B) 2 mnu
The initial momentum of a single ball is $m \times u$.
Since the collision is perfectly elastic, the ball bounces back with velocity $-u$, making the final momentum $-m \times u$.
Change in momentum for one ball is $(m \times u) - (-m \times u) = 2 \times m \times u$.
For n balls impinging per second, the total rate of change of momentum (which equals the force) is $n \times 2 \times m \times u$.
$F = 2mnu$

5. **Answer :** C) $\frac{a_1 a_2}{a_1 + a_2}$
Let the constant force be F.
$m_1 = \frac{F}{a_1}$
$m_2 = \frac{F}{a_2}$
The combined mass is $M = m_1 + m_2$.
$M = \frac{F}{a_1} + \frac{F}{a_2} = F \times \left(\frac{a_1 + a_2}{a_1 \times a_2}\right)$
$a_{\text{new}} = \frac{F}{M}$
$a_{\text{new}} = \frac{F}{F \times \left(\frac{a_1 + a_2}{a_1 \times a_2}\right)}$
$a_{\text{new}} = \frac{a_1 \times a_2}{a_1 + a_2}$

6. **Answer :** A, B, C
Momentum is a vector, so D is false.
A is true because force is the rate of change of momentum ($F = \frac{p}{t}$). If $t = 1$ unit, the force required to stop the body equals its initial momentum magnitude.
B is true as the units of mass × velocity are kg-m/s (SI) and g-cm/s (CGS).

C is true as momentum is fundamentally defined as the quantity of motion contained in a body.

7. **Answer :** A, B, C, D
Newton is the SI unit of force.
Dyne is the CGS unit of force.
kg-f (kilogram-force) and gm-f (gram-force) are gravitational units of force.

8. **Answer :** A, B, D
Statement C is false because action and reaction forces always act on two different interacting bodies, not the same body.
Statement A illustrates action (bullet forward) and reaction (gun backward).
Statement B illustrates pushing water back (action) to be pushed forward (reaction).
Statement D is a fundamental principle of Newton's third law.

9. **Answer :** D) Linear Momentum
Newton's third law implies that for an isolated system of interacting bodies, the mutual internal forces cancel out.
Since the net external force is zero, the total rate of change of momentum is zero, leading to the conservation of linear momentum.

10. **Answer :** A) Both A and R are true and R is the correct explanation of A
By moving the hands backward, the fielder increases the time interval (Δt) over which the ball's momentum is reduced to zero.
Since $F = \frac{\Delta p}{\Delta t}$, a larger time interval results in a smaller impact force on the hands for the exact same change in momentum.

11. **Answer :** D) A is false but R is true
Assertion A is false because action and reaction forces do not cancel each other out; they act on two entirely different objects.
Reason R is true as it correctly states that the forces act on different bodies, which is exactly why they cannot cancel out to produce zero net force on a single body.

12. **Answer :** A) Both A and R are true and R is the correct explanation of A
The sudden stopping of the bus brings the lower half of the passenger's body to rest immediately.
The upper half remains in motion due to inertia, perfectly explaining the forward fall.

13. **Answer :** A) Both A and R are true and R is the correct explanation of A
Balanced forces result in a net force of zero, meaning they cannot cause acceleration or change the state of motion.
However, multiple forces acting equally from opposite directions can compress or stretch an object, effectively changing its shape and size.

14. **Answer :** B) 0.5 kg
$F = m_1 \times a_1$
$5 = m_1 \times 10$
$m_1 = \frac{5}{10}$
$m_1 = 0.5$ kg

15. **Answer :** C) 0.25 kg
$F = m_2 \times a_2$
$5 = m_2 \times 20$
$m_2 = \frac{5}{20}$

$m_2 = 0.25$ kg

16. **Answer :** D) 6.67 m/s^2

$m_{\text{total}} = m_1 + m_2$

$m_{\text{total}} = 0.5 + 0.25$

$m_{\text{total}} = 0.75$ kg

$F = m_{\text{total}} \times a_{\text{new}}$

$5 = 0.75 \times a_{\text{new}}$

$a_{\text{new}} = \dfrac{5}{0.75}$

$a_{\text{new}} \approx 6.67\ \text{m/s}^2$

17. **Answer :** C) Zero

By definition, balanced forces perfectly cancel each other out.

This always results in a net external force of exactly zero.

18. **Answer :** C) Change in shape of a body

Balanced forces cannot cause acceleration, change in direction, or increase in speed since the net force is zero.

They can, however, squeeze, stretch, or deform a body, changing its physical shape.

19. **Answer :** D) Unbalanced

A stationary object changing its state to motion requires an acceleration.

Acceleration can only be produced when there is a net non-zero external force, which means the acting forces must be unbalanced.

20. **Answer :** a → S, b → R, c → Q, d → P

Inertia is conceptually measured by mass (S).

Linear momentum is defined as the quantity of motion contained in the body (R).

Newton's 2nd law relates force to the rate of change of momentum (Q).

Newton's 3rd law states that action equals negative reaction (P).

21. **Answer :** a → R, b → R, c → P, d → Q

Original momentum $p = m \times v = 24$ kg m/s.

a) $p_a = (2m) \times v = 2 \times (m \times v) = 48$ kg m/s (Matches R).

b) $p_b = m \times (2v) = 2 \times (m \times v) = 48$ kg m/s (Matches R).

c) $p_c = \left(\frac{m}{2}\right) \times (2v) = m \times v = 24$ kg m/s (Matches P).

d) $p_d = m \times \left(\frac{v}{2}\right) = \frac{m \times v}{2} = 12$ kg m/s (Matches intended Q).

22. **Answer :** 2

$F = m \times a$

$100 = 5 \times a$

$a = 20\ \text{cm/s}^2$

$v = u + a \times t$

$v = 0 + (20 \times 10)$

$v = 200$ cm/s

The question asks for the velocity in the format ___ × 100 cm/sec.

$200 = 2 \times 100$

The integer value is 2.

23. **Answer :** 2

According to the law of conservation of momentum:

$m_{\text{gun}} \times v_{\text{gun}} + m_{\text{bullet}} \times v_{\text{bullet}} = 0$

$5 \times v_{\text{gun}} + (0.02 \times 500) = 0$

$5 \times v_{\text{gun}} + 10 = 0$

$5 \times v_{\text{gun}} = -10$

$v_{\text{gun}} = -2 \text{ m/s}$

The magnitude of the recoil velocity is 2 m/s.

24. **Answer :** 5

$F = m \times a$

$50 = 10 \times a$

$a = \frac{50}{10}$

$a = 5 \text{ m/s}^2$

Free Body Diagram

LEVEL-I

1. **Answer :** A) $N_1 = 50N$, $N_2 = 50\sqrt{3}N$

Let the angles of the inclined planes with the horizontal be $\theta_1 = 60°$ and $\theta_2 = 30°$.

The normal forces from the planes act perpendicular to the surfaces.

The angle made by a normal force with the vertical is equal to the angle of inclination of its plane.

For horizontal equilibrium, the horizontal components of the normal forces must balance each other perfectly.

$N_1 \sin 60° = N_2 \sin 30°$

$N_1 \times \frac{\sqrt{3}}{2} = N_2 \times \frac{1}{2}$

$N_2 = \sqrt{3}N_1$

For vertical equilibrium, the upward vertical components of the normal forces must balance the downward weight of the sphere.

$W = mg$

$W = 10 \times 10 = 100 \text{ N}$

$N_1 \cos 60° + N_2 \cos 30° = 100$

Substitute $N_2 = \sqrt{3}N_1$ into the vertical equilibrium equation.

$N_1 \times \frac{1}{2} + (\sqrt{3}N_1) \times \frac{\sqrt{3}}{2} = 100$

$\frac{N_1}{2} + \frac{3N_1}{2} = 100$

$2N_1 = 100$

$N_1 = 50 \text{ N}$

Substitute the value of N_1 back to find N_2.

$N_2 = 50\sqrt{3} \text{ N}$

2. **Answer :** C) 1.4 m/s^2

The formula for the acceleration of a two-mass Atwood machine system is given by the net pulling force divided by the total mass.

$a = \frac{m_2 - m_1}{m_1 + m_2} \times g$

Here, $m_2 = 4$ kg and $m_1 = 3$ kg, with $g = 9.8 \text{ m/s}^2$.

$a = \frac{4 - 3}{3 + 4} \times 9.8$

$a = \frac{1}{7} \times 9.8$

$a = 1.4 \text{ m/s}^2$

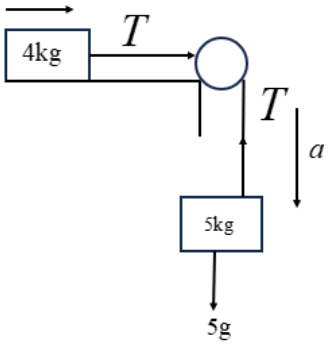
3. **Answer :** A) $\frac{m_2 g}{(m_1+m_2)}$

Let the tension in the string be T and the common acceleration of the system be a.
For the hanging block m_2, the equation of motion is:
$m_2 g - T = m_2 a$
For the block m_1 on the horizontal frictionless table, the only horizontal force is tension.
$T = m_1 a$
Add both equations to eliminate the tension T.
$m_2 g = m_1 a + m_2 a$
$m_2 g = (m_1 + m_2)a$
$$a = \frac{m_2 g}{m_1 + m_2}$$
Note: Option A in the text has a typo using M_1 instead of m_1.

4. **Answer :** C) $\frac{m_1+m_2}{m_1+m_2+m_3} T$

Let the total acceleration of the system be a.
$$a = \frac{\text{Total Force}}{\text{Total Mass}}$$
$$a = \frac{T}{m_1 + m_2 + m_3}$$
The tension in the string between m_2 and m_3 is responsible for pulling the masses behind it, which are m_1 and m_2.
Let this tension be T'.
$T' = (m_1 + m_2) \times a$
Substitute the expression for acceleration a into the equation for T'.
$$T' = (m_1 + m_2) \times \frac{T}{m_1 + m_2 + m_3}$$
$$T' = \frac{m_1 + m_2}{m_1 + m_2 + m_3} T$$

5. **Answer :** B) 5.44 m/s^2



This is a system with one mass on a frictionless table and one mass hanging.
The formula for acceleration is:
$$a = \frac{m_{\text{hanging}} \times g}{m_{\text{table}} + m_{\text{hanging}}}$$
Substitute the given values into the formula. Let $g = 9.8\ \text{m/s}^2$.
$$a = \frac{5 \times 9.8}{4 + 5}$$
$$a = \frac{49}{9}$$
$a = 5.444\ldots\ \text{m/s}^2$

6. **Answer :** A) $3.27\ \text{m/s}^2, 6.54\ \text{N}$

The formula for the acceleration of the system is:

$$a = \frac{m_{\text{hanging}} \times g}{m_{\text{table}} + m_{\text{hanging}}}$$

$$a = \frac{1 \times 9.8}{2 + 1}$$

$$a = \frac{9.8}{3}$$

$$a \approx 3.27\ \text{m/s}^2$$

The tension in the string pulls the block on the table.

$$T = m_{\text{table}} \times a$$

$$T = 2 \times 3.27$$

$$T = 6.54\ \text{N}$$

7. **Answer :** D) 73.5 N

For a standard Atwood machine, the formula for tension in the string is:

$$T = \frac{2m_1 m_2}{m_1 + m_2} \times g$$

Substitute the given masses and $g = 9.8\ \text{m/s}^2$.

$$T = \frac{2 \times 6 \times 10}{6 + 10} \times 9.8$$

$$T = \frac{120}{16} \times 9.8$$

$$T = 7.5 \times 9.8$$

$$T = 73.5\ \text{N}$$

8. **Answer :** C) $\frac{g}{3}$

The formula for acceleration in an Atwood machine is:

$$a = \frac{m_2 - m_1}{m_1 + m_2} \times g$$

Substitute the given masses into the formula.

$$a = \frac{10 - 5}{10 + 5} \times g$$

$$a = \frac{5}{15} \times g$$

$$a = \frac{g}{3}$$

9. **Answer :** C) $2\sqrt{3}$

Let the tension in the string attached to the ceiling be T_2 and it makes an angle of $30°$ with the horizontal.

For vertical equilibrium, the vertical component of T_2 must balance the weight.

$$T_2 \sin 30° = 2\ \text{kg-wt}$$

$$T_2 \times \frac{1}{2} = 2$$

$$T_2 = 4\ \text{kg-wt}$$

For horizontal equilibrium, the horizontal component of T_2 must balance the horizontal tension T_1.

$$T_1 = T_2 \cos 30°$$

$$T_1 = 4 \times \frac{\sqrt{3}}{2}$$

$$T_1 = 2\sqrt{3}\ \text{kg-wt}$$

10. **Answer :** C) 4 kg and 4 kg

Since the spring balances are ideal and considered light (massless), they just act like a continuous string.

In a continuous stationary string holding a mass, the tension is uniform throughout.

$T = mg$

$T = 4$ kg-wt

Therefore, both spring balances will measure the same tension of 4 kg.

11. **Answer :** B) 2.5 N

First, calculate the common acceleration of both blocks moving together.

$$a = \frac{F_{\text{net}}}{m_{\text{total}}}$$

$$a = \frac{6}{7+5}$$

$$a = \frac{6}{12}$$

$$a = 0.5 \text{ m/s}^2$$

The force acting on the lighter mass (5 kg) is the contact force from the heavier mass.

$F_{5\text{kg}} = m \times a$

$F_{5\text{kg}} = 5 \times 0.5$

$F_{5\text{kg}} = 2.5$ N

12. **Answer :** D) $\frac{mF}{(m+M)}$

The total acceleration of the system comprising both blocks is:

$$a = \frac{F}{M+m}$$

The force exerted on block 'm' is provided entirely by the tension (or spring force) connecting it to 'M'.

$F_m = m \times a$

Substitute the expression for acceleration.

$$F_m = m \times \left(\frac{F}{M+m}\right)$$

$$F_m = \frac{mF}{m+M}$$

13. **Answer :** D) $\frac{Mm}{M+m}g$

The acceleration of the system is given by the hanging weight divided by the total mass.

$$a = \frac{mg}{M+m}$$

The tension in the string is the only horizontal force pulling mass M.

$T = M \times a$

Substitute the expression for acceleration.

$$T = M \times \left(\frac{mg}{M+m}\right)$$

$$T = \frac{Mm}{M+m}g$$

14. **Answer :** C) Linear momentum

A rocket operates by expelling exhaust gases backward at high velocity.

According to the absence of external forces in space, the total momentum of the rocket-gas system remains constant.
Therefore, its motion is fundamentally based on the conservation of linear momentum.

15. **Answer :** C) Conservation of linear momentum
A jet engine works by taking in air, mixing it with fuel, and expelling it out the back at extremely high speeds.
This backward momentum of the gases imparts an equal and opposite forward momentum to the engine.
This is a direct application of the law of conservation of linear momentum.

16. **Answer :** C) Acts against the rocket and push it up
When rocket engines fire, they expel hot gases downward with tremendous force.
According to Newton's third law, the gases exert an equal and opposite reaction force directly on the rocket structure.
This reaction force acts against the rocket, pushing it upward.

17. **Answer :** A) Linear momentum
Newton's second law relates force to the rate of change of momentum, and the third law states that internal forces cancel each other out.
When no net external force acts on a system, the rate of change of the total momentum is zero.
This directly leads to the principle of conservation of linear momentum.

18. **Answer :** B) Momentum
Impulse is defined as the product of a large force and the short time interval over which it acts.
Mathematically, $J = F \times \Delta t$.
According to the impulse-momentum theorem, this impulse is exactly equal to the change in momentum of the body.

19. **Answer :** A) 0.5 sec
The relationship between force, time, and change in momentum is given by:
$F = \frac{\Delta p}{t}$
Rearrange the formula to solve for time.
$t = \frac{\Delta p}{F}$
Substitute the given values.
$t = \frac{125}{250}$
$t = 0.5$ sec

20. **Answer :** C) 0.3 N-s
First, find the force acting on the ball using Newton's second law.
$F = m \times a$
Convert the mass to kilograms.
$m = 150$ g $= 0.15$ kg
$F = 0.15 \times 20$
$F = 3$ N
Now calculate the impulse imparted by this force.
Impulse $= F \times t$
Impulse $= 3 \times 0.1$
Impulse $= 0.3$ N-s

21. **Answer :** C) 1 m/s

By the law of conservation of momentum, the total initial momentum is zero, so the final momentum must also be zero.

$m_{\text{gun}} \times v_{\text{gun}} = m_{\text{bullet}} \times v_{\text{bullet}}$

Substitute the given values, ensuring units are consistent (200 g = 0.2 kg).

$1 \times v_{\text{gun}} = 0.2 \times 5$

$v_{\text{gun}} = 1.0$

$v_{\text{gun}} = 1 \text{ m/s}$

LEVEL-II

1. **Answer :** D) All the above are correct

The weight of Block A is calculated as:

$W_A = m_A \cdot g = 4 \text{ kg} \cdot 10 \text{ m/s}^2 = 40 \text{ N}$

Here is the step-by-step evaluation of each case:

Case (i): No external force on Block A

Forces on A: The downward force is the weight of Block A (40 N). The upward force is the normal force exerted by Block B on Block A (N_{BA}).

Equilibrium: Since the block is at rest, the upward forces must equal the downward forces:

$N_{BA} = 40 \text{ N}$

Action-Reaction: According to Newton's Third Law, the force exerted by A on B is equal and opposite to the force exerted by B on A. Therefore, the force exerted by A on B is 40 N.

Conclusion: Statement A is **correct**.

Case (ii): An upward force of 20 N is applied to Block A

Forces on A: The downward force is the weight of Block A (40 N). The upward forces are the external applied force (20 N) and the normal force exerted by Block B on Block A (N_{BA}).

Equilibrium:

$N_{BA} + 20 \text{ N} = 40 \text{ N}$

$N_{BA} = 40 \text{ N} - 20 \text{ N} = 20 \text{ N}$

Conclusion: The force exerted by B on A is 20 N. Statement B is **correct**.

Case (iii): A downward force of 20 N is applied to Block A

Forces on A: The downward forces are the weight of Block A (40 N) and the external applied force (20 N). The upward force is the normal force exerted by Block B on Block A (N_{BA}).

Equilibrium:

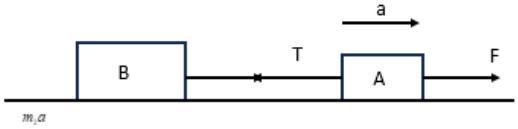
$N_{BA} = 40 \text{ N} + 20 \text{ N} = 60 \text{ N}$

Action-Reaction: By Newton's Third Law, the force exerted by A on B is equal in magnitude, which is 60 N.

Conclusion: Statement C is **correct**.

Since statements A, B, and C are all individually true, the correct option is: **D) All the above are correct**

2. **Answer :** D) $T = 300N, T' = 200$ N



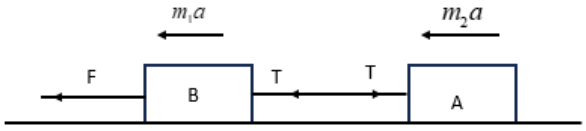
When the force $F = 500$ N is applied to block A (10 kg), the system's acceleration is:

$$a = \frac{F}{m_A + m_B}$$

$$a = \frac{500}{10 + 15} = \frac{500}{25} = 20 \text{ m/s}^2$$

The tension T pulls block B (15 kg).

$$T = m_B \times a = 15 \times 20 = 300 \text{ N}$$



When the force is applied to block B (15 kg), the tension T' pulls block A (10 kg).

$$T' = m_A \times a = 10 \times 20 = 200 \text{ N}$$

3. **Answer :** D) 0.2 ms^{-2}

The acceleration for an Atwood machine is:

$$a = \frac{m_1 - m_2}{m_1 + m_2} \times g$$

Substitute the given mass values and $g = 9.8 \text{ m/s}^2$.

$$a = \frac{5 - 4.8}{5 + 4.8} \times 9.8$$

$$a = \frac{0.2}{9.8} \times 9.8$$

$$a = 0.2 \text{ m/s}^2$$

4. **Answer :** B) 2mg

Calculate the common acceleration of the blocks up the incline. The total mass is $3m$.

The net force acting along the incline is the applied force minus the component of gravity.

$$F_{\text{net}} = 3mg - (3m)g \sin 30°$$

$$F_{\text{net}} = 3mg - 3mg\left(\frac{1}{2}\right) = 1.5mg$$

$$a = \frac{1.5mg}{3m} = 0.5g$$

Now, analyze the forces on the $2m$ block. It is pushed up by the contact force N and pulled down by gravity.

$$N - 2mg \sin 30° = 2m \times a$$

$$N - 2mg\left(\frac{1}{2}\right) = 2m(0.5g)$$

$$N - mg = mg$$

$$N = 2mg$$

5. **Answer :** C) 1.5×10^{-3}Ns

Convert the force from dynes to Newtons. 1 dyne = 10^{-5} N.

$$F = 50 \times 10^{-5} \text{ N}$$

The formula for impulse is Force multiplied by Time.

Impulse = $F \times t$

Impulse = $(50 \times 10^{-5}) \times 3$

Impulse = 150×10^{-5}

Impulse = 1.5×10^{-3} Ns

6. **Answer :** D) 0.25 ms^{-1}

By the law of conservation of momentum, the initial momentum is zero.

$m_{\text{tank}} \times V_{\text{tank}} = m_{\text{body}} \times v_{\text{body}}$

$100 \times V_{\text{tank}} = 0.25 \times 100$

$100 \times V_{\text{tank}} = 25$

$V_{\text{tank}} = \dfrac{25}{100}$

$V_{\text{tank}} = 0.25 \text{ ms}^{-1}$

7. **Answer :** C) 0.9 Ns

Find the time t when the force becomes zero.

$600 - 2 \times 10^5 t = 0$

$2 \times 10^5 t = 600$

$t = \dfrac{600}{2 \times 10^5} = 3 \times 10^{-3} \text{ s}$

The impulse is the integral of the force over this time interval.

$J = \int_0^t F \, dt = \int_0^{3\times10^{-3}} (600 - 2 \times 10^5 t) \, dt$

$J = [600t - 10^5 t^2]_0^{3\times10^{-3}}$

$J = 600(3 \times 10^{-3}) - 10^5(3 \times 10^{-3})^2$

$J = 1.8 - 10^5(9 \times 10^{-6})$

$J = 1.8 - 0.9$

$J = 0.9 \text{ Ns}$

8. **Answer :** C) 24 N

The force exerted is equal to the rate of change of momentum of the bullets.

$F = \dfrac{\Delta p}{\Delta t} = n \times m \times v$

Where n is the number of bullets per second.

$F = 4 \times 0.02 \text{ kg} \times 300 \text{ m/s}$

$F = 4 \times 6$

$F = 24 \text{ N}$

9. **Answer :** B) 2F

Calculate the acceleration of the entire system. Total mass is $6M$.

$a = \dfrac{12F}{6M} = \dfrac{2F}{M}$

The string between M and 2M is responsible for pulling only the block M.

$T = M \times a$

Substitute the acceleration into the tension equation.

$T = M \times \left(\dfrac{2F}{M}\right)$

$T = 2F$

10. **Answer :** D) W

The inclined planes make an angle of 60° with the horizontal. Therefore, their normal vectors make an angle of 60° with the vertical.

Let the normal reaction from each plane be R. Since the setup is symmetric, $R_1 = R_2 = R$.

The vertical component of each normal force must support the weight W.

$R \cos 60° + R \cos 60° = W$

$2R\left(\dfrac{1}{2}\right) = W$

$R = W$

11. **Answer :** B) 6N

The common acceleration of both blocks is given by the total force divided by total mass.

$a = \frac{9}{M + 2M} = \frac{9}{3M} = \frac{3}{M}$

The tension in the connecting string pulls the $2M$ block.

$T = 2M \times a$

$T = 2M \times \left(\frac{3}{M}\right)$

$T = 6$ N

12. **Answer :** D) 2 ms^{-2}

The acceleration of the system is the external force divided by the total mass.

Total mass = $m_1 + m_2 + m_3 = 1 + 2 + 3 = 6$ kg.

$a = \frac{F}{\text{Total mass}}$

$a = \frac{12}{6}$

$a = 2$ ms^{-2}

13. **Answer :** C) $a_1 > a_2 > a_3$

In case A:

Let be the string tension T_A and g be the acceleration due to gravity.

So, the using the equation of motion,

$T_A - mg = ma_1$

From the figure, the tension of on mass $2mg$ due to the downward acceleration due to gravity.

$2mg - mg = ma_1$

$a_1 = g$

In case: B

Let be the string tension T_B and g be the acceleration due to gravity.

So, the using the equation of motion,

$T_C - mg = (m + m)a_2$

From the figure, the tension of on mass $2mg$ due to the downward acceleration due to gravity.

$2mg = -mg = 2ma_2$

$a_2 = \frac{g}{2}$

In Case: C

Let be the string tension T_C and g be the acceleration due to gravity.

So, the using the equation of motion,

$T_B - mg = (m + 2m)a_3$

From the figure, the tension of on mass $2mg$ due to the downward acceleration due to gravity.

$2mg - mg = 3ma_3$

$a_3 = \frac{g}{3}$

Therefore, $a_1 > a_2 > a_3$.

14. **Answer :** A, B, C

The impulse delivered is the area under the force-time graph. The graph forms a triangle with base $2t_0$ and height F_0.

Area $= \frac{1}{2} \times$ base $\times$ height

Impulse $= \frac{1}{2} \times 2t_0 \times F_0 = F_0 t_0$ (Statement A is correct)

Average force is impulse divided by total time.

$F_{\text{avg}} = \frac{\text{Impulse}}{2t_0} = \frac{F_0 t_0}{2t_0} = \frac{F_0}{2}$ (Statement B is correct)

According to the impulse-momentum theorem, change in momentum is equal to impulse.

$\Delta p =$ Impulse $= F_0 t_0$ (Statement C is correct)

15. **Answer :** A, B, D

The impulse vector $\vec{J}$ is the change in momentum.

$\vec{J} = \vec{p}_f - \vec{p}_i$

$\vec{J} = (-2\hat{\imath} + 14\hat{\jmath}) - (6\hat{\imath} + 8\hat{\jmath}) = -8\hat{\imath} + 6\hat{\jmath}$ (Statement A is correct)

The magnitude of the impulse is:

$|\vec{J}| = \sqrt{(-8)^2 + 6^2} = \sqrt{64 + 36} = 10$ Ns (Statement B is correct)

Check if impulse acts perpendicular to initial momentum using dot product.

$\vec{J} \cdot \vec{p}_i = (-8)(6) + (6)(8) = -48 + 48 = 0$ (Statement D is correct)

16. **Answer :** B, C, D

The total momentum of the system is the vector sum of individual momenta.

$\vec{p}_{\text{total}} = \vec{p}_1 + \vec{p}_2$

$\vec{p}_{\text{total}} = (4\hat{\imath} + 6\hat{\jmath}) + (-7\hat{\imath} + 2\hat{\jmath}) = -3\hat{\imath} + 8\hat{\jmath}$ (Statement B is correct)

The magnitude of total momentum is:

$|\vec{p}_{\text{total}}| = \sqrt{(-3)^2 + 8^2} = \sqrt{9 + 64} = \sqrt{73}$ (Statement C is correct)

If they interact internally, the law of conservation of momentum applies, meaning total momentum remains unchanged. (Statement D is correct)

17. **Answer :** A) Both A and R are true, and R is the correct explanation of A.

A Free Body Diagram completely isolates a body to determine the net external force acting on it.

Internal forces between the particles comprising the isolated body always occur in equal and opposite action-reaction pairs, summing to zero.

Therefore, they have no effect on the overall motion of the body and are excluded.

18. **Answer :** B) Both A and R are true, but R is not the correct explanation of A.

The normal reaction always acts perpendicular because it is specifically defined as the perpendicular component of the total contact force between two surfaces.

While it is true that it is a contact force exerted by a surface, this fact alone does not explain why it must be strictly perpendicular.

19. **Answer :** A) Both A and R are true, and R is the correct explanation of A.

In an ideal (light and frictionless) system, the pulley does not require any torque to rotate and offers no resistance.

Consequently, the tension remains uniform throughout the entire length of the continuous string.

The pulley merely acts to redirect the line of action of the force.

20. **Answer :** A) Both A and R are true, and R is the correct explanation of A.

In an Atwood machine, the blocks are linked such that their accelerations are tied together.
The heavier mass (M_1) is subjected to a larger gravitational force ($M_1 g$) compared to the lighter mass ($M_2 g$).
This unbalanced net gravitational force dictates that the heavier mass will accelerate downward, pulling the lighter mass upward.

21. **Answer :** B) $\frac{Fm_2}{m_1+m_2}$

When force F is applied on m_1 towards east, the entire system accelerates eastwards.

$$a = \frac{F}{m_1 + m_2}$$

The contact force N is the force pushing block m_2.

$$N = m_2 \times a$$

$$N = \frac{Fm_2}{m_1 + m_2}$$

22. **Answer :** A) $\frac{Fm_1}{m_1+m_2}$

When force F is applied on m_2 towards west, the system accelerates westwards.

$$a = \frac{F}{m_1 + m_2}$$

The contact force N in this case is the force pushing block m_1.

$$N = m_1 \times a$$

$$N = \frac{Fm_1}{m_1 + m_2}$$

23. **Answer :** B) $\frac{m_1F_2+m_2F_1}{m_1+m_2}$

The net force acting on the system towards the east is $F_1 - F_2$ (since $F_1 > F_2$).
The acceleration of the system is:

$$a = \frac{F_1 - F_2}{m_1 + m_2}$$

Consider the free body diagram of mass m_2. The forces acting on it are the contact force N (east) and F_2 (west).

$$N - F_2 = m_2 \times a$$

$$N = F_2 + m_2 \left(\frac{F_1 - F_2}{m_1 + m_2}\right)$$

$$N = \frac{F_2(m_1 + m_2) + m_2F_1 - m_2F_2}{m_1 + m_2}$$

$$N = \frac{m_1F_2 + m_2F_1}{m_1 + m_2}$$

24. **Answer :** B) $m_1 g, m_1 g - m_2 g$

For m_1 to remain at rest, the upward tension from the spring must balance its downward weight.

$$T_{\text{spring}} = m_1 g$$

This same tension is transmitted through the continuous string over the pulley to mass m_2.
For m_2 to remain at rest, the upward string tension must balance the downward forces (weight + tension from thread BC).

$$T_{\text{string}} = m_2 g + T_{BC}$$

$$m_1 g = m_2 g + T_{BC}$$

$T_{BC} = m_1 g - m_2 g$

25. **Answer :** A) $m_1 g$

The tension in a spring relies on its physical extension (or compression), given by Hooke's Law ($F = kx$).

Just an instant after the thread is burned, the mass m_1 has not yet had time to move.

Therefore, the spring's extension remains unchanged, and the tension remains exactly $m_1 g$.

26. **Answer :** D) $\left(\frac{m_1 - m_2}{m_2}\right) g$

Right after the thread BC is burned, it no longer exerts a downward force on m_2.

The upward force acting on m_2 is the string tension, which is momentarily still $m_1 g$ because the spring hasn't relaxed yet.

The net upward force on m_2 is:

$F_{\text{net}} = T_{\text{string}} - m_2 g$

$F_{\text{net}} = m_1 g - m_2 g$

The acceleration of m_2 is the net force divided by its mass.

$a_{m2} = \frac{F_{\text{net}}}{m_2}$

$a_{m2} = \frac{(m_1 - m_2)g}{m_2}$

27. **Answer :** a → R, b → S, c → Q, d → P

A free body diagram focuses on the direct external forces on the isolated mass. For m_1 on the incline, the relevant downward component of gravity is $m_1 g \sin\alpha$ (matches R).

Similarly, for m_2 on the incline, the relevant downward component of gravity is $m_2 g \sin\beta$ (matches S).

The general acceleration equation for this double-incline Atwood machine is $a = \frac{m_2 g \sin\beta - m_1 g \sin\alpha}{m_1 + m_2}$ (matches Q).

The general tension equation for this system is $T = \frac{m_1 m_2 (\sin\alpha + \sin\beta)}{m_1 + m_2} g$ (matches P).

28. **Answer :** A → P, B → S, C → Q, D → R

Normal reaction forces always act perpendicular to the plane of the contact surface (matches P).

Tension in a supporting string always acts away from the mass, pulling it towards the support (matches S).

By definition, a Free Body Diagram (FBD) isolates an object and draws only the external forces acting upon it (matches Q).

An ideal, light, frictionless pulley ensures that the tension remains the same on both sides of the string (matches R).

29. **Answer :** 25

First, find the common acceleration of the entire four-block system. Total mass is $2 + 3 + 4 + 1 = 10$ kg.

$a = \frac{F_{\text{total}}}{m_{\text{total}}}$

$a = \frac{50}{10} = 5 \text{ m/s}^2$

The tension in the string between the 3 kg and 4 kg blocks is responsible for pulling all the masses trailing behind it.

The trailing masses are the 2 kg and 3 kg blocks.
Total trailing mass = 2 + 3 = 5 kg.
$T = m_{\text{trailing}} \times a$
$T = 5 \times 5 = 25$ N

30. **Answer :** 36
Calculate the acceleration of the two blocks. The total mass is 6 + 9 = 15 kg.
$a = \frac{F_{\text{total}}}{m_{\text{total}}}$
$a = \frac{60}{15} = 4\ \text{m/s}^2$
The 60 N force is applied to the 6 kg block, which in turn drags the 9 kg block via the string.
The tension in the string is responsible solely for accelerating the 9 kg block.
$T = m_{\text{dragged}} \times a$
$T = 9 \times 4 = 36$ N

Frame of Reference

LEVEL-1

1. **Answer :** A) Inertial frame of reference
Newton's laws of motion are fundamentally defined and are strictly valid only in an inertial frame of reference.
An inertial frame is one that is either at rest or moving with a constant velocity, meaning it has zero acceleration.

2. **Answer :** A) Inertial
A car moving with a constant velocity on a level road has zero acceleration.
Any frame of reference that is not accelerating is considered an inertial frame of reference.

3. **Answer :** B) Non-inertial
The Earth rotates on its axis and revolves around the Sun, which means it experiences centripetal acceleration.
Because it is accelerating, it is technically a non-inertial frame of reference, even though for many practical small-scale experiments it is approximated as inertial.

4. **Answer :** B) Non-inertial
By definition, any frame of reference that is accelerating with respect to an inertial frame is called a non-inertial frame of reference.

5. **Answer :** B) Pseudo
In a non-inertial (accelerating) frame of reference, Newton's laws do not hold true in their standard form.
To mathematically apply Newton's second law in such a frame, a fictitious force called a pseudo force must be introduced to account for the frame's acceleration.

6. **Answer :** C) Remains same
The weighing balance measures the normal force exerted by the man on the platform.
When he takes a step sideways, the internal forces shift his center of mass horizontally, but the net vertical force (his weight due to gravity) remains unchanged.
Therefore, the average reading on the scale remains the same.

7. **Answer :** A) First increases then decreases to zero

To jump, the man must push downward against the platform to create an upward acceleration.
This downward push temporarily increases the normal force, causing the spring balance reading to increase.
Once his feet leave the platform, the normal force drops immediately to zero.

8. **Answer :** B) 3g
Let the tension in the cord be T and the maximum acceleration be a.
The maximum tension the cord can withstand is four times the weight of the sphere.
$T_{\max} = 4 \times mg$
The equation of motion for the sphere moving upwards is given by Newton's second law.
$T_{\max} - mg = ma$
Substitute the value of the maximum tension.
$4mg - mg = ma$
$3mg = ma$
$a = 3g$

9. **Answer :** B) $R = mg + ma$
The forces acting on the body are the normal reaction R upwards and its weight mg downwards.
Since the lift is accelerating upwards with acceleration a, the net force must be upwards.
$F_{\text{net}} = R - mg$
According to Newton's second law:
$R - mg = ma$
$R = mg + ma$

10. **Answer :** D) Accelerates upward with an acceleration equal to 4 metres/sec^2
The force felt by the feet is the normal reaction R from the elevator floor.
When stationary or moving at a constant velocity, $R = mg$.
When accelerating downwards, $R = m(g - a)$, making the force less than the weight.
When accelerating upwards, $R = m(g + a)$, which increases the normal force, making it greater than the boy's actual weight.

11. **Answer :** B) $\frac{g}{3}$
The weight of the man in a stationary lift is his true weight.
$W_{\text{stat}} = mg$
The apparent weight of the man when the lift moves downward with acceleration a is:
$W_{\text{down}} = m(g - a)$
We are given the ratio of these two weights as 3:2.
$$\frac{mg}{m(g-a)} = \frac{3}{2}$$
Cancel the mass m from the numerator and denominator.
$$\frac{g}{g-a} = \frac{3}{2}$$
Cross-multiply to solve for a.
$2g = 3(g - a)$
$2g = 3g - 3a$
$3a = g$

$a = \frac{g}{3}$

12. **Answer :** B) Go on decreasing

A spring balance measures the apparent weight, which depends on the acceleration due to gravity g.

As the aeroplane goes to higher altitudes, the value of g decreases according to the law of gravitation.

Since $W = mg$ and g is decreasing, the reading on the spring balance will go on decreasing.

13. **Answer :** D) 0

If the lift rope breaks, the lift falls freely under the influence of gravity.

The downward acceleration of the lift a becomes equal to the acceleration due to gravity g.

The apparent weight (tension or normal reaction) is given by:

$R = m(g - a)$

$R = m(g - g)$

$R = 0$

14. **Answer :** B) 24 N

First, determine the mass of the bag from the stationary reading.

$W = mg$

$49 = m \times 9.8$

$m = \frac{49}{9.8}$

$m = 5$ kg

When the lift moves downward with an acceleration $a = 5\ \text{m/s}^2$, the apparent weight is:

$R = m(g - a)$

$R = 5(9.8 - 5)$

$R = 5(4.8)$

$R = 24$ N

15. **Answer :** C) 10800N

The tension T in the string when the lift accelerates upwards is given by the formula:

$T = m(g + a)$

Substitute the given values ($m = 1000$ kg, $g = 9.8\ \text{m/s}^2$, $a = 1\ \text{m/s}^2$).

$T = 1000(9.8 + 1)$

$T = 1000(10.8)$

$T = 10800$ N

16. **Answer :** C) 1200N

The reading on the scale is the normal reaction force acting on the man.

Since the lift is moving upwards with an acceleration, the apparent weight increases.

$R = m(g + a)$

Substitute the given values ($m = 80$ kg, $g = 10\ \text{m/s}^2$, $a = 5\ \text{m/s}^2$).

$R = 80(10 + 5)$

$R = 80 \times 15$

$R = 1200$ N

17. **Answer :** C) 2.5 m/s^2

The maximum tension the rope can bear without breaking is the weight of a 25 kg mass.

$T_{\max} = 25 \times g$

$T_{\max} = 25 \times 10 = 250$ N

The equation of motion for the monkey climbing upwards is:

$T_{\max} - mg = ma$

Substitute the monkey's mass ($m = 20$ kg).

$250 - (20 \times 10) = 20 \times a$

$250 - 200 = 20a$

$50 = 20a$

$a = \frac{50}{20}$

$a = 2.5\ \text{m/s}^2$

18. **Answer :** D) 11000N

Calculate the total mass of the system (lift + person).

$m_{\text{total}} = 940 + 60$

$m_{\text{total}} = 1000$ kg

The tension in the supporting cable must overcome both gravity and provide the upward acceleration.

$T = m_{\text{total}}(g + a)$

Substitute the given values ($g = 10\ \text{m/s}^2$, $a = 1.0\ \text{m/s}^2$).

$T = 1000(10 + 1)$

$T = 1000 \times 11$

$T = 11000$ N

19. **Answer :** A) 55 kg

When the lift moves with a constant speed, its acceleration is zero.

The apparent weight formula is $R = m(g + a)$.

Since $a = 0$, $R = mg$, which is the person's true weight.

Therefore, the machine will continue to show 55 kg.

20. **Answer :** D) $\frac{mg}{\cos\theta}$

For the block to remain stationary on the wedge, it must accelerate horizontally at the same rate as the wedge.

From the wedge's frame of reference, a pseudo force ma acts horizontally backward.

To balance along the incline:

$ma\cos\theta = mg\sin\theta$

$a = g\tan\theta$

The normal force exerted by the wedge must balance the perpendicular components of both gravity and the pseudo force.

$N = mg\cos\theta + ma\sin\theta$

Substitute $a = g\tan\theta$.

$N = mg\cos\theta + m\left(g\frac{\sin\theta}{\cos\theta}\right)\sin\theta$

$N = \frac{mg\cos^2\theta + mg\sin^2\theta}{\cos\theta}$

$N = \frac{mg(\cos^2\theta + \sin^2\theta)}{\cos\theta}$

$N = \frac{mg}{\cos\theta}$

21. **Answer :** B) 4 ms^{-2} upwards

First, find the true weight of the lift to compare it with the tension. Let's use $g = 10\ \text{m/s}^2$ for simplicity, as it usually yields integer results in such problems.
$W = mg = 2000 \times 10 = 20000\ \text{N}$
Since the tension (28000 N) is greater than the weight (20000 N), the net force is upwards, meaning the acceleration is upwards.
$T - mg = ma$
$28000 - 20000 = 2000 \times a$
$8000 = 2000a$
$a = \frac{8000}{2000}$
$a = 4\ \text{m/s}^2$

22. **Answer :** C) 12000 N
From the graph, at $t = 11$ s, the lift is in the deceleration phase (from $t = 10$ to $t = 12$).
First, calculate the acceleration (slope of the v-t graph) in this interval.
$a = \frac{v_f - v_i}{t_f - t_i}$
$a = \frac{0 - 3.6}{12 - 10}$
$a = \frac{-3.6}{2}$
$a = -1.8\ \text{m/s}^2$
The lift is moving upwards but decelerating. Using $g = 9.8\ \text{m/s}^2$:
$T = m(g + a)$
$T = 1500(9.8 - 1.8)$
$T = 1500(8)$
$T = 12000\ \text{N}$

23. **Answer :** D) 36 metres
The distance travelled (height) is given by the total area under the velocity-time graph.
The graph forms a trapezium with parallel sides of length 12 s (base) and 8 s (top flat portion from $t = 2$ to $t = 10$).
The height of the trapezium is the maximum speed, 3.6 m/s.
$\text{Area} = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$
$\text{Area} = \frac{1}{2} \times (12 + 8) \times 3.6$
$\text{Area} = \frac{1}{2} \times 20 \times 3.6$
$\text{Area} = 10 \times 3.6$
$\text{Area} = 36\ \text{metres}$

24. **Answer :** C) 26 N
In a closed cage, the air pushed downwards by the bird cannot escape and exerts an equal force on the base of the cage.
When the bird is stationary, the total weight recorded is the cage's weight plus the bird's weight (25 N).
When the bird accelerates upwards, it must push the air downwards with a force greater than its weight.
The extra force required is $F = m \times a$.

$F = 0.5 \times 2$
$F = 1$ N
This extra downward push acts on the floor of the cage, increasing the reading.
New reading = Original reading + Extra force
New reading = 25 + 1
New reading = 26 N

LEVEL-II

1. **Answer :** D) 90000 N
Note: Using the strictly provided values in the text ($m = 6000$ kg, $a = 10$ m/s^2, $g = 10$ m/s^2), the calculation yields $T = 6000(10 + 10) = 120000$ N, which is not in the options. This strongly implies a typo in the source document's acceleration value. If we assume the intended acceleration was $a = 5$ m/s^2, it perfectly matches option D. The calculation below reflects this assumption.
The tension in the cable when accelerating upwards is:
$T = m(g + a)$
Assuming the intended acceleration $a = 5$ m/s^2:
$T = 6000(10 + 5)$
$T = 6000 \times 15$
$T = 90000$ N

2. **Answer :** A) $t_1 \geq t_2$
When the lift is stationary, the coin falls relative to the lift with acceleration g.
$s = \frac{1}{2} g t_1^2$
When the lift moves up with an acceleration a, the floor accelerates towards the coin. The effective relative acceleration is $g + a$.
$s = \frac{1}{2}(g + a) t_2^2$
Since the relative acceleration is greater in the second case ($g + a \geq g$), it will take less time to cover the same distance.
Therefore, $t_1 \geq t_2$.

3. **Answer :** B) 1125 N
Consider the man and the crate as a single combined system.
$m_{\text{total}} = 100 + 50 = 150$ kg
The man pulls the rope downward with tension T, so the rope pulls the man upward with tension T.
The other end of the rope pulls the crate upward with tension T. Therefore, there are two upward tension forces acting on the system.
The total upward force is $2T$.
The equation of motion for the whole system accelerating upwards is:
$2T - m_{\text{total}} g = m_{\text{total}} a$
$2T - (150 \times 10) = 150 \times 5$
$2T - 1500 = 750$
$2T = 2250$
$T = 1125$ N

4. **Answer :** C) $(M + m) g \tan\theta$

For the small block m to remain stationary with respect to the wedge, its horizontal acceleration must be identically a.
From the non-inertial frame of the wedge, a pseudo force ma acts horizontally to the left on block m.
To prevent sliding along the incline, the upward component of the pseudo force must perfectly balance the downward component of gravity.
$ma\cos\theta = mg\sin\theta$
$a = g\dfrac{\sin\theta}{\cos\theta}$
$a = g\tan\theta$
Now, consider the whole system $(M+m)$ being pushed by force F.
$F = (M+m)a$
Substitute the required acceleration into this equation.
$F = (M+m)g\tan\theta$

5. **Answer :** C) Going down with constant acceleration
If the apple remains stationary relative to the lift after being dropped, it means the apple and the lift share the exact same downward acceleration.
Since the apple is in free fall, its acceleration is purely due to gravity (g downwards).
Therefore, the lift must also be accelerating downwards with a constant acceleration of g.

6. **Answer :** B) $\tan^{-1}(a/g)$ opposite to the direction of motion
When the car accelerates horizontally with a, a pseudo force ma acts on the sphere in the direction strictly opposite to the car's motion.
The gravitational force mg acts vertically downwards.
The string tension T provides balancing components for both. Let θ be the angle with the vertical.
$T\sin\theta = ma$
$T\cos\theta = mg$
Divide the first equation by the second.
$\dfrac{T\sin\theta}{T\cos\theta} = \dfrac{ma}{mg}$
$\tan\theta = \dfrac{a}{g}$
$\theta = \tan^{-1}\left(\dfrac{a}{g}\right)$

7. **Answer :** A, C
An inertial frame of reference is one that is not accelerating.
In such a frame, Newton's second law holds true in its natural form, meaning the vector sum of all real external forces equals mass times acceleration.
$\Sigma\vec{F}_{\text{real}} = m\vec{a}$
Since the frame is not accelerating, pseudo forces are strictly zero and not required.
The basic laws of physics take their identical standard forms.

8. **Answer :** A, B, D
A non-inertial frame is one that is accelerating.
Because the frame accelerates, Newton's laws are not directly applicable without modification.
To apply Newton's second law, a fictitious pseudo force $(-m\vec{a})$ must be added, which always acts in the exact opposite direction to the frame's acceleration $\vec{a}$.

The modified equation becomes $\vec{F}_{\text{real}} - m\vec{\alpha} = m\vec{a}_{\text{rel}}$.

9. **Answer :** B, D

The Earth rotates around its axis (producing centripetal and Coriolis accelerations) and revolves around the Sun.

Because it undergoes acceleration in both of these motions, it is strictly a non-inertial frame of reference.

10. **Answer :** A) 720 N

When the elevator moves up with an acceleration, the apparent weight is greater than the true weight.

$W_{\text{apparent}} = m(g + a)$

$W_{\text{apparent}} = 60(10 + 2)$

$W_{\text{apparent}} = 60 \times 12$

$W_{\text{apparent}} = 720 \text{ N}$

11. **Answer :** B) 480 N

When the elevator moves up with a retardation (deceleration), its acceleration vector points downwards.

This makes the apparent weight less than the true weight.

$W_{\text{apparent}} = m(g - a)$

$W_{\text{apparent}} = 60(10 - 2)$

$W_{\text{apparent}} = 60 \times 8$

$W_{\text{apparent}} = 480 \text{ N}$

12. **Answer :** A) Decreases by 20%

First, find the original true weight of the man.

$W_{\text{true}} = mg = 60 \times 10 = 600 \text{ N}$

Now find the apparent weight when moving down with $a = 2 \text{ m/s}^2$.

$W_{\text{apparent}} = m(g - a)$

$W_{\text{apparent}} = 60(10 - 2) = 480 \text{ N}$

Calculate the absolute decrease in weight.

Decrease $= 600 - 480 = 120 \text{ N}$

Calculate the percentage change.

Percentage decrease $= \left(\frac{120}{600}\right) \times 100\%$

Percentage decrease $= 20\%$

13. **Answer :** B) Opposite to the direction of acceleration

In the frame of reference of the moving car (a non-inertial frame), a pseudo force acts on the pendulum bob.

By definition, this pseudo force is always directed completely opposite to the acceleration vector of the frame of reference.

14. **Answer :** C) $a = g \tan\theta$

When the pendulum stabilizes at an angle θ, the tension T in the string resolves into two components.

The vertical component balances gravity: $T\cos\theta = mg$

The horizontal component provides the force required to accelerate the bob with the car: $T\sin\theta = ma$

Divide the horizontal equation by the vertical equation.

$$\frac{T\sin\theta}{T\cos\theta} = \frac{ma}{mg}$$

$\tan\theta = \frac{a}{g}$

$a = g\tan\theta$

15. **Answer :** B) $T = m\sqrt{a^2 + g^2}$

The tension in the string must balance both the gravitational force and the pseudo force acting on the bob.

Since gravity acts vertically and the pseudo force acts horizontally, they are perpendicular vectors.

The tension equals the magnitude of their vector sum.

$T = \sqrt{(ma)^2 + (mg)^2}$

$T = \sqrt{m^2a^2 + m^2g^2}$

$T = m\sqrt{a^2 + g^2}$

16. **Answer :** a → Q, b → R, c → P, d → S

When the lift moves up with uniform acceleration, apparent weight increases: $m(g + a)$.

When the lift moves down with uniform acceleration, apparent weight decreases: $m(g - a)$.

When the lift moves with uniform velocity (no acceleration), apparent weight equals true weight: mg.

In a freely falling lift, acceleration is g downwards, making apparent weight zero: $m(g - g) = 0$.

17. **Answer :** a → Q, b → Q, c → R, d → P

The mass of the bob is $m = 50$ gm $= 0.05$ kg. We use $g = 9.8\ \text{m/s}^2$.

a) Goes up with an acceleration 1.2: $T = m(g + a) = 0.05(9.8 + 1.2) = 0.05 \times 11 = 0.55$ N

b) Goes down with deceleration 1.2 (this means the acceleration vector points upwards): $T = m(g + a) = 0.05(9.8 + 1.2) = 0.55$ N

c) Goes up with uniform velocity ($a = 0$): $T = mg = 0.05 \times 9.8 = 0.49$ N

d) Goes down with an acceleration 1.2: $T = m(g - a) = 0.05(9.8 - 1.2) = 0.05 \times 8.6 = 0.43$ N

18. **Answer :** 5

According to the law of conservation of momentum, the initial momentum of the shell equals the total final momentum of the fragments.

$p_{\text{initial}} = p_{\text{final}}$

$m_{\text{total}} \times v_{\text{initial}} = (m_1 \times v_1) + (m_2 \times v_2)$

$10 \times 20 = (6 \times 0) + (4 \times v_2)$

$200 = 4v_2$

$v_2 = \frac{200}{4}$

$v_2 = 50\ \text{m/s}$

The question asks for the velocity in the format ___ × 10 m/s.

$50 = 5 \times 10$

The integer value is 5.

19. **Answer :** 7700

First, calculate the total mass of the system that the cable must lift.

$m_{\text{total}} = m_{\text{lift}} + m_{\text{man}}$

$m_{\text{total}} = 500 + 50$

$m_{\text{total}} = 550$ kg

The force exerted by the lift cable (Tension) must account for gravity and the upward acceleration.

$T = m_{\text{total}}(g + a)$

$T = 550(10 + 4)$

$T = 550 \times 14$

$T = 7700$ N

20. **Answer :** 640

The reading on the weighing machine represents the normal reaction force exerted on the person.

Since the lift is accelerating downwards, the apparent weight is less than the true weight.

$R = m(g - a)$

$R = 80(10 - 2)$

$R = 80 \times 8$

$R = 640$ N

Previous Year Questions

1. **Answer :** B) (ii) only

Let's evaluate each statement by setting up equations of motion. The whole system accelerates upward at $a = 5$ m/s^2 and $g = 10$ m/s^2.

(i) The normal force on the 8 kg block from the floor must support both blocks. Total mass = $4 + 8 = 12$ kg.

$N_{\text{floor}} = m_{\text{total}}(g + a) = 12(10 + 5) = 12 \times 15 = 180$ N. This statement is True.

(ii) The normal force between the 8 kg and 4 kg blocks only needs to support the 4 kg block.

$N_{4\text{kg}} = m_{4\text{kg}}(g + a) = 4(10 + 5) = 4 \times 15 = 60$ N. The statement says 80 N, so this is False.

(iii) Normal force is, by definition, the perpendicular contact force. This statement is True.

Therefore, only statement (ii) is incorrect.

2. **Answer :** A) P-(ii), Q-(iv), R-(i)

First, determine the acceleration of the block using the change in momentum.

Change in momentum $\Delta p = 2$ kg m/s. Time $\Delta t = 1$ s. Mass $m = 0.2$ kg.

Net force $F_{\text{net}} = \frac{\Delta p}{\Delta t} = \frac{2}{1} = 2$ N.

The net force acting downwards on the block is the applied tension minus gravity.

$F_{\text{net}} = T - mg$

$2 = T - (0.2 \times 10)$

$2 = T - 2$

$T = 4$ N (Matches P to ii).

Calculate the acceleration and displacement in 1 s.

$a = \frac{F_{\text{net}}}{m} = \frac{2}{0.2} = 10$ m/s^2

Displacement $s = \frac{1}{2}at^2 = \frac{1}{2}(10)(1)^2 = 5$ m.

Work done by tension $W_T = T \times s = 4 \times 5 = 20$ J (Matches Q to iv).

Work done against gravity $W_g = mg \times s = (0.2 \times 10) \times 5 = 10$ J (Matches R to i).

3. **Answer :** D) F, F, F, T

(i) The true weight (Mg) always acts vertically downwards towards the center of the earth, not perpendicular to the inclined plane. (False)

(ii) Since the block slides with a *uniform acceleration*, there is a non-zero net force acting down the plane. Thus, downward forces are greater than upward forces. (False)
(iii) The normal reaction acts perpendicular to the surface of contact (the inclined plane PQ), not the horizontal plane QR. (False)
(iv) The component of weight that presses the block into the surface is $Mg\cos\theta$, which acts strictly perpendicular to the inclined plane PQ. (True)

4. **Answer :** Correct Option
A correct Free Body Diagram must show exactly three distinct vectors originating from the bob:
Weight (W): A vector pointing strictly vertically downward, representing gravity.
Tension (T): A vector pointing diagonally up and to the left, perfectly aligned along the length of the string.
Wind Force (F): A vector pointing diagonally up and to the right, strictly matching the direction of the wind arrows depicted in the scenario image.

5. **Answer :** D) (i), (ii) and (iii)
(i) The force on the floor by the crew is their apparent weight.
$F_{\text{floor}} = m_{\text{crew}}(g + a) = 500(10 + 15) = 500 \times 25 = 12500$ N $= 1.25 \times 10^4$ N downwards. (True)
(ii) The rotor must push down on the air with a force large enough to support the total mass and accelerate it upwards. Total mass $M = 2000 + 500 = 2500$ kg.
$F_{\text{action}} = M(g + a) = 2500(10 + 15) = 2500 \times 25 = 62500$ N $= 6.25 \times 10^4$ N downwards. (True)
(iii) According to Newton's third law, the reaction force from the air on the helicopter is equal and opposite to the rotor's action.
$F_{\text{reaction}} = 6.25 \times 10^4$ N upwards. (True)

6. **Answer :** C) 3/2 gm/cc
When floating in water, 1/4 of the volume is outside, meaning 3/4 is submerged.
Apply the principle of flotation.
Weight of body = Weight of displaced water
$$V \times \rho_{\text{body}} \times g = \left(\frac{3}{4}V\right) \times \rho_{\text{water}} \times g$$
Assuming the density of water $\rho_{\text{water}} = 1$ gm/cc:
$$\rho_{\text{body}} = \frac{3}{4} \times 1 = \frac{3}{4} \text{ gm/cc}$$
When floating in the unknown liquid, half the volume is outside, meaning half is submerged.
$$V \times \rho_{\text{body}} \times g = \left(\frac{1}{2}V\right) \times \rho_{\text{liquid}} \times g$$
Substitute the density of the body calculated above.
$$\frac{3}{4} = \frac{1}{2} \times \rho_{\text{liquid}}$$
$$\rho_{\text{liquid}} = \frac{3}{4} \times 2$$
$$\rho_{\text{liquid}} = \frac{6}{4} = \frac{3}{2} \text{ gm/cc}$$

7. **Answer :** A) AB
According to Newton's second law, the net force on a particle is zero when its acceleration is zero.

Acceleration is the rate of change of velocity. If acceleration is zero, velocity must be constant.
On a displacement-time ($x - t$) graph, velocity is represented by the slope. A constant velocity means a strictly constant slope, which mathematically translates to a straight line.
Looking at the described segments, region AB is a straight line, indicating constant velocity, zero acceleration, and zero net force.

8. **Answer :** C) P-(iv), Q-(iii), R-(ii), S-(i)
P. Definition of force is qualitatively provided by Newton's first law, which describes it as the agent needed to change a body's state of inertia. (Matches iv)
Q. Measure of force is mathematically formulated by Newton's second law ($F = ma$ or $F = dp/dt$). (Matches iii)
R. The effect of a large force applied over a short time is measured as Impulse (change in momentum). (Matches ii)
S. The recoiling of a gun is a classic direct consequence of action and reaction, governed by Newton's third law. (Matches i)

9. **Answer :** A) Three times that of B
According to Newton's second law, Force equals mass times acceleration ($F = ma$).
For object A: $F = m_A \times a_A$
For object B: $F = m_B \times a_B$
Since identical forces are applied:
$m_A \times a_A = m_B \times a_B$
We are given that $a_B = 3a_A$. Substitute this into the equation.
$m_A \times a_A = m_B \times (3a_A)$
Cancel a_A from both sides.
$m_A = 3m_B$

10. **Answer :** A) 2Mu
Momentum is a vector quantity. Let the direction towards the wall be positive.
Initial momentum $p_i = M \times u$.
Since the body bounces back, its velocity reverses direction.
Final momentum $p_f = M \times (-u) = -Mu$.
The change in momentum is final momentum minus initial momentum.
$\Delta p = p_f - p_i$
$\Delta p = -Mu - Mu$
$\Delta p = -2Mu$
The magnitude of this change is $2Mu$.

4. Work, Energy And Power

LEVEL-I

1. **Answer :** D) Joule

Work done is defined as the product of force and the displacement in the direction of the force.

$W = F \times d$

$W = 1\ \text{N} \times 1\ \text{m}$

$W = 1\ \text{J}$

2. **Answer :** A) 294 J

The formula for work done against gravity is the change in potential energy.

$W = m \times g \times h$

$W = 15 \times 9.8 \times 2.0$

$W = 294\ \text{J}$

3. **Answer :** B) 1:4

The formula for kinetic energy is:

$KE = \frac{1}{2}mv^2$

For the first body:

$KE_1 = \frac{1}{2}mV^2$

For the second body:

$KE_2 = \frac{1}{2}m(2V)^2$

$KE_2 = 4 \times \left(\frac{1}{2}mV^2\right)$

The ratio of their kinetic energies is:

$\frac{KE_1}{KE_2} = \frac{1}{4}$

4. **Answer :** D) Both A and C

The fundamental formula for work done by a constant force is:

$W = F \times s \times \cos\theta$

The work done depends entirely on the applied force, the displacement, and the angle between them.

It does not depend directly on the mass or the initial velocity of the body.

5. **Answer :** A) 2×10^3 J

The relationship between kinetic energy and momentum is given by:

$KE = \frac{p^2}{2m}$

$$KE = \frac{20^2}{2 \times 0.1}$$

$$KE = \frac{400}{0.2}$$

$$KE = 2000 \text{ J}$$

$$KE = 2 \times 10^3 \text{ J}$$

6. **Answer :** B) Weight of the box

The work done in lifting an object against gravity is given by:

$W = m \times g \times h$

The term $m \times g$ represents the weight of the box.

Therefore, the work done inherently depends on the weight of the box.

7. **Answer :** B) Work done

The work done by a variable force is defined mathematically as the integral of force with respect to displacement.

$$W = \int F \, dx$$

This integral directly represents the area under the force-displacement curve.

8. **Answer :** B) 40 J

First, determine the acceleration using Newton's second law.

$$a = \frac{F}{m}$$

$$a = \frac{2}{5}$$

$$a = 0.4 \text{ m/s}^2$$

Next, find the displacement using the second equation of motion.

$$s = ut + \frac{1}{2}at^2$$

$$s = 0 + \frac{1}{2} \times 0.4 \times 10^2$$

$$s = 0.2 \times 100$$

$$s = 20 \text{ m}$$

Finally, calculate the work done.

$W = F \times s$

$W = 2 \times 20$

$W = 40$ J

9. **Answer :** D) 180 J

First, calculate the acceleration of the body.

$$a = \frac{F}{m}$$

$$a = \frac{1}{10}$$

$$a = 0.1 \text{ m/s}^2$$

Calculate the displacement for $t = 60$ s.

$s = \frac{1}{2}at^2$

$s = \frac{1}{2} \times 0.1 \times 60^2$

$s = 0.05 \times 3600$

$s = 180$ m

Calculate the total work done.

$W = F \times s$

$W = 1 \times 180$

$W = 180$ J

10. **Answer :** A) 12000 J

The work done in raising a load is the product of the force (weight) and the height.

$W = F \times h$

$W = 750 \times 16$

$W = 12000$ J

11. **Answer :** C) No work is done on it

In uniform circular motion, the centripetal force is always directed towards the center of the circle.

The instantaneous displacement is tangential to the circular path.

The angle between the force and displacement vectors is 90°.

$W = F \times s \times \cos 90°$

$W = 0$ J

12. **Answer :** A) 12500J

First, find the mass of the water using its volume and density (1 g/ml = 1 kg/L).

$m =$ Volume × Density

$m = 50$ L $\times 1$ kg/L

$m = 50$ kg

Calculate the work done against gravity.

$W = m \times g \times h$

$W = 50 \times 10 \times 25$

$W = 12500$ J

13. **Answer :** A) 150J

The relationship between power, work, and time is:

$P = \frac{W}{t}$

$W = P \times t$

$W = 30 \times 5$

$W = 150$ J

14. **Answer :** D) 10:1

The kinetic energy formula is:

$KE = \frac{1}{2}mv^2$

Since the speeds are the same, the ratio of kinetic energies is equal to the ratio of their masses.

$\frac{KE_1}{KE_2} = \frac{m_1}{m_2}$

$\frac{KE_1}{KE_2} = \frac{5000}{500}$

$\frac{KE_1}{KE_2} = \frac{10}{1}$

15. **Answer :** A) Power

By definition, power is the rate at which work is done or energy is transferred.

$P = \frac{W}{t}$

16. **Answer :** B) 5W

The formula for power is:

$P = \frac{W}{t}$

$P = \frac{2000}{400}$

$P = 5$ W

17. **Answer :** C) 20s

The formula for time based on power and energy is:

$t = \frac{E}{P}$

$t = \frac{3000}{150}$

$t = 20$ s

18. **Answer :** C) Watt

The SI unit of power is the Watt, which is equivalent to one Joule per second.

19. **Answer :** D) Work/time

Power is mathematically defined as the ratio of work done to the time taken.

$P = \frac{W}{t}$

20. **Answer :** A) 30w

Calculate power using the given work and time.

$P = \frac{W}{t}$

$P = \frac{150}{5}$

$P = 30$ W

21. **Answer :** B) 375w

Calculate the total height climbed.

$h = 45 \times 0.15$ m

$h = 6.75$ m

Calculate the work done against gravity.

$W = m \times g \times h$

$W = 50 \times 10 \times 6.75$

$W = 3375$ J

Calculate the power.

$P = \frac{W}{t}$

$P = \frac{3375}{9}$

$P = 375$ W

22. **Answer :** B) Power decreases

Power is inversely proportional to time for a given constant amount of work.

$P = \frac{W}{t}$

Therefore, if the time taken increases, the power decreases.

23. **Answer :** D) 2×10^3W

Assuming standard gravitational acceleration $g = 10$ m/s^2 for simplicity.

$W = m \times g \times h$

$W = 100 \times 10 \times 20$

$W = 20000$ J

Calculate the power.

$P = \frac{W}{t}$

$P = \frac{20000}{10}$

$P = 2000$ W

$P = 2 \times 10^3$ W

24. **Answer :** D) 6250w

Assuming $g = 10$ m/s^2.

$W = m \times g \times h$

$W = 50 \times 10 \times 100$

$W = 50000$ J

Calculate the power.

$P = \frac{W}{t}$

$P = \frac{50000}{8}$

$P = 6250$ W

25. **Answer :** A) 250J

The increase in potential energy is given by:

$\Delta PE = m \times g \times (h_2 - h_1)$

Assuming $g = 10$ m/s^2.

$\Delta PE = 5 \times 10 \times (10 - 5)$

$\Delta PE = 50 \times 5$

$\Delta PE = 250$ J

26. **Answer :** C) 5 m/s

First, find the mass from the weight. Assume $g = 10$ m/s^2.

$W = m \times g$

$600 = m \times 10$

$m = 60$ kg

Use the kinetic energy formula to find the velocity.

$KE = \frac{1}{2}mv^2$

$750 = \frac{1}{2} \times 60 \times v^2$

$750 = 30 \times v^2$

$v^2 = 25$

$v = 5$ m/s

27. **Answer :** A) 18×10^6 J

The conversion factor is 1 kWh = 3.6×10^6 J.

$E = 5 \times 3.6 \times 10^6$

$E = 18 \times 10^6$ J

28. **Answer :** A) m^0

The kinetic energy acquired by a body from rest under a constant force over a distance s equals the work done.

$KE = W$

$W = F \times s$

Since the work done depends only on the constant force and the displacement, the acquired kinetic energy is independent of the mass.

Therefore, it is proportional to m^0.

29. **Answer :** A) Both parts will have numerically equal momentum

In an explosion, there are no external forces, so total linear momentum is conserved.

Since the initial momentum is zero, the vector sum of the final momenta must also be zero.

$\vec{p}_1 + \vec{p}_2 = 0$

$\vec{p}_1 = -\vec{p}_2$

Therefore, the magnitude (numerical value) of their momenta is equal.

30. **Answer :** B) Potential energy

Potential energy is fundamentally defined as the energy stored in an object due to its spatial position relative to other objects, or its internal configuration.

31. **Answer :** D) Standing

Potential energy relative to the ground is given by $PE = m \times g \times h$.

The height h represents the elevation of the body's center of mass.

When standing, the center of mass is at its maximum height compared to sitting or lying down, thus maximizing potential energy.

32. **Answer :** B) 588J

The potential energy formula is:

$PE = m \times g \times h$

$PE = 10 \times 9.8 \times 6$

$PE = 588$ J

33. **Answer :** A) At the maximum height

The potential energy formula is $PE = m \times g \times h$.

Since mass and gravity are constant, potential energy is directly proportional to height.

Therefore, it reaches its maximum value exactly at the maximum height attained by the stone.

34. **Answer :** A) 25

The initial energy of the ball is its potential energy before dropping.

$E_{initial} = m \times g \times 2$

The final energy after rebounding is its potential energy at the new height.

$E_{final} = m \times g \times 1.5$

The energy lost is the difference between these two.

$E_{lost} = 2mg - 1.5mg = 0.5mg$

The percentage of energy lost is:

$\%\ \text{Loss} = \left(\frac{0.5mg}{2mg}\right) \times 100$

$\%\ \text{Loss} = 0.25 \times 100$

$\%\ \text{Loss} = 25\%$

35. **Answer :** A) 1:1

The relationship between momentum and kinetic energy is:

$p = \sqrt{2mK}$

The ratio of their linear momenta is:

$\frac{p_1}{p_2} = \frac{\sqrt{2m_1K_1}}{\sqrt{2m_2K_2}}$

$$\frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2} \times \frac{K_1}{K_2}}$$

$$\frac{p_1}{p_2} = \sqrt{\frac{2}{6} \times \frac{3}{1}}$$

$$\frac{p_1}{p_2} = \sqrt{\frac{6}{6}}$$

$$\frac{p_1}{p_2} = 1:1$$

36. **Answer :** A) Doubled

The formula for kinetic energy is:

$KE = \frac{1}{2}mv^2$

Since the speed remains constant, kinetic energy is directly proportional to mass.

If the mass is doubled, the kinetic energy is directly doubled.

LEVEL-II

1. **Answer :** B) 2×10^3 J

Calculate the total vertical height climbed.

h = number of steps × height of each step

$h = 40 \times 0.25$ m

$h = 10$ m

Calculate the work done against gravity.

$W = m \times g \times h$

$W = 20 \times 10 \times 10$

$W = 2000$ J

$W = 2 \times 10^3$ J

2. **Answer :** A) The work done on the body against gravitational force is zero

The gravitational force acts vertically downwards.

The displacement of the block is strictly horizontal along the level surface.

The angle between gravity and displacement is 90°.

$W = F \times s \times \cos 90°$

$W = 0$ J

3. **Answer :** B) $fx/2$

The formula for work done is:

$W = F \times x \times \cos\theta$

Substitute the given values.

$W = F \times x \times \cos 60°$

$W = F \times x \times \frac{1}{2}$

$W = \frac{Fx}{2}$

4. **Answer :** A) 2100J

Calculate the total vertical height climbed.

$h = 30 \times 0.20$ m

$h = 6$ m

Calculate the work done against gravity using the given weight ($W_{eight} = 350$ N).

$W = W_{eight} \times h$

$W = 350 \times 6$

$W = 2100$ J

5. **Answer :** B) 60°

Use the definition of work.

$W = F \times s \times \cos\theta$

Substitute the provided values.

$25 = 5 \times 10 \times \cos\theta$

$25 = 50 \times \cos\theta$

$\cos\theta = \frac{25}{50}$

$\cos\theta = 0.5$

$\theta = 60°$

6. **Answer :** C) 100N

Convert the energy (work done) to standard SI units.

$W = 6.4$ kJ $= 6400$ J

Use the work formula, noting that force and displacement are in the same direction ($\theta = 0°$).

$W = F \times s$

$6400 = F \times 64$

$F = \frac{6400}{64}$

$F = 100$ N

7. **Answer :** B) 4×10^4 J

Assume the standard approximate value for gravity $g = 10$ m/s^2 for integer results.

$W = m \times g \times h$

$W = 50 \times 10 \times 80$

$W = 40000$ J

$W = 4 \times 10^4$ J

8. **Answer :** B) 1.1×10^3w

Convert the mass to kilograms.

$m = 20$ metric tons $= 20 \times 1000$ kg $= 20000$ kg

Calculate the total work done.

$W = m \times g \times h$

$W = 20000 \times 10 \times 20$

$W = 4 \times 10^6$ J

Convert the time to seconds.

$t = 1$ hour $= 3600$ s

Calculate the power.

$P = \frac{W}{t}$

$P = \frac{4000000}{3600}$

$P \approx 1111.11$ W

$P \approx 1.1 \times 10^3$ W

9. **Answer :** D) FS/t

Power is the rate of doing work.

$P = \frac{W}{t}$

Since the displacement is in the direction of the force, the work done is $W = F \times S$.

$P = \frac{FS}{t}$

10. **Answer :** A) 400

Let n be the number of bricks. The total mass is $m = n \times 2.5$ kg.

The total work done in one hour (3600 s) is related to power by:

$W = P \times t$

$m \times g \times h = P \times t$

Substitute the given values.

$(n \times 2.5) \times 9.8 \times 3.6 = 9.8 \times 3600$

Cancel out 9.8 from both sides.

$n \times 2.5 \times 3.6 = 3600$

$n \times 9.0 = 3600$

$n = \frac{3600}{9}$

$n = 400$

11. **Answer :** A) 3000 J

First, determine the useful power applied to stirring the water.

$P_{useful} = 50\%$ of 100 W

$P_{useful} = 0.5 \times 100 = 50$ W

Calculate the work done over one minute (60 s).

$W = P_{useful} \times t$

$W = 50 \times 60$

$W = 3000$ J

12. **Answer :** C) $\sqrt{m_1}:\sqrt{m_2}$

The relationship between momentum and kinetic energy is:

$p = \sqrt{2mK}$

The ratio of their momenta is:

$$\frac{p_1}{p_2} = \frac{\sqrt{2m_1K}}{\sqrt{2m_2K}}$$

Since the kinetic energies are equal, the terms $2K$ cancel out.

$$\frac{p_1}{p_2} = \frac{\sqrt{m_1}}{\sqrt{m_2}}$$

13. **Answer :** B) 12.5 m

According to the conservation of mechanical energy, the initial kinetic energy transforms into potential energy.

$E_{total} = 490$ J

We need to find the height where the kinetic energy is exactly half of its original value.

$$KE_{new} = \frac{490}{2} = 245 \text{ J}$$

At this point, the rest of the energy has been converted to potential energy.

$PE = E_{total} - KE_{new}$

$PE = 490 - 245 = 245$ J

Use the potential energy formula to find the height.

$m \times g \times h = 245$

$2 \times 9.8 \times h = 245$

$19.6 \times h = 245$

$$h = \frac{245}{19.6}$$

$h = 12.5$ m

14. **Answer :** A) 44%

The formula linking kinetic energy and momentum is:

$$KE = \frac{p^2}{2m}$$

If momentum increases by 20%, the new momentum is:

$p_{new} = 1.2p$

The new kinetic energy becomes:

$$KE_{new} = \frac{(1.2p)^2}{2m}$$

$KE_{new} = 1.44 \times \left(\frac{p^2}{2m}\right)$

$KE_{new} = 1.44 \times KE$

An increase to 1.44 times the original value represents a 44% increase.

15. **Answer :** D) 125 J

First, find the magnitude of the net force resulting from the two perpendicular forces.

$F_{net} = \sqrt{4^2 + 3^2}$

$F_{net} = \sqrt{16 + 9} = \sqrt{25}$

$F_{net} = 5$ N

Next, find the acceleration.

$a = \frac{F_{net}}{m}$

$a = \frac{5}{10} = 0.5 \text{ m/s}^2$

Find the final velocity after 10 seconds.

$v = u + at$

$v = 0 + 0.5 \times 10$

$v = 5$ m/s

Finally, calculate the kinetic energy.

$KE = \frac{1}{2}mv^2$

$KE = \frac{1}{2} \times 10 \times 5^2$

$KE = 5 \times 25$

$KE = 125$ J

16. **Answer :** C) Becomes 1/9

The formula for kinetic energy is:

$KE = \frac{1}{2}mv^2$

If the velocity is reduced to one-third ($v/3$), the new kinetic energy is:

$KE_{new} = \frac{1}{2}m\left(\frac{v}{3}\right)^2$

$KE_{new} = \frac{1}{9} \times \left(\frac{1}{2}mv^2\right)$

$KE_{new} = \frac{1}{9}KE$

17. **Answer :** B) 9.8×10^5J

Consider the mass of water falling in exactly one second.

$m = 1000$ kg

The initial potential energy of this mass of water is:

$PE = m \times g \times h$

$PE = 1000 \times 9.8 \times 100$

$PE = 980000$ J

$PE = 9.8 \times 10^5$ J

18. Answer : B) 0.49J

By the law of conservation of energy, the kinetic energy of the ball just before striking the ground is equal to its initial potential energy.

$KE = PE_{initial}$

$KE = m \times g \times h$

$KE = 0.01 \text{ kg} \times 9.8 \text{ m/s}^2 \times 5 \text{ m}$

$KE = 0.49$ J

19. Answer : C) 300 J

The loss in potential energy corresponds directly to the change in height.

$\Delta PE = m \times g \times (h_{initial} - h_{final})$

Assuming $g = 10 \text{ m/s}^2$.

$\Delta PE = 5 \times 10 \times (10 - 4)$

$\Delta PE = 50 \times 6$

$\Delta PE = 300$ J

20. Answer : C) 115 J

The work done is perfectly represented by the total area under the given force-displacement graph. We calculate the area by splitting the shape into distinct geometric sections.

Area of the first triangle ($x = 0$ to $x = 5$):

$A_1 = \frac{1}{2} \times \text{base} \times \text{height}$

$A_1 = \frac{1}{2} \times 5 \times 4 = 10$ J

Area of the rectangle ($x = 5$ to $x = 10$):

$A_2 = \text{width} \times \text{height}$

$A_2 = 5 \times 4 = 20$ J

Area of the trapezium ($x = 10$ to $x = 15$):

$A_3 = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{width}$

$A_3 = \frac{1}{2} \times (4 + 10) \times 5 = \frac{1}{2} \times 14 \times 5 = 35$ J

Area of the second triangle ($x = 15$ to $x = 25$):

$A_4 = \frac{1}{2} \times \text{base} \times \text{height}$

$A_4 = \frac{1}{2} \times 10 \times 10 = 50$ J

Calculate the total work done.

$W_{total} = 10 + 20 + 35 + 50$

$W_{total} = 115$ J

21. **Answer :** B, C

The work-energy relationship for a conservative force states that work done is equal to the negative change in potential energy.

$W_{gravity} = -\Delta PE$

If the work done by gravity is negative, then $-\Delta PE$ is negative, which strictly means ΔPE is positive. Therefore, the potential energy increases.

Because gravity is the only force acting, the total mechanical energy is rigidly conserved.

$\Delta KE + \Delta PE = 0$

Since ΔPE is positive, ΔKE must be negative to balance it. Therefore, the kinetic energy decreases.

22. **Answer :** A, B, C

For a conservative force field, the work done on a particle depends exclusively on its initial and final spatial positions, making it path-independent.

Consequently, the net work done in moving a particle completely around any closed path is strictly zero.

Because of this specific path independence, a scalar potential energy function can always be uniquely defined for the field.

23. **Answer :** A, B, D

Kinetic friction always opposes the relative motion of the surfaces in contact, so the work done by kinetic friction is always strictly negative.

The net work done on the particle depends on the balance between the applied horizontal force and friction, allowing it to be positive (accelerating), zero (constant velocity), or negative (decelerating).

The normal reaction force acts perpendicularly vertical to the horizontal motion, so its work contribution is strictly zero.

24. **Answer :** A, B, C

During ascent, the gravitational force points downwards while the displacement points upwards. The angle is 180°, making the work done by gravity strictly negative.

Since gravity acts in opposition to motion, it causes the body's upward speed to gradually decrease, causing kinetic energy to continually decrease.

As the body gains vertical height above the reference frame, its gravitational potential energy continually increases.

25. **Answer :** A, B, D

The work required to compress a spring from position x_1 to x_2 is exactly equal to the change in its elastic potential energy.

$W = \Delta PE = \frac{1}{2}k(2x)^2 - \frac{1}{2}k(x)^2$

$W = \frac{1}{2}k(4x^2) - \frac{1}{2}kx^2$

$W = 2kx^2 - 0.5kx^2 = 1.5kx^2$

$W = \frac{3}{2}kx^2$

This increment shows that work required is strictly proportional to x^2 and the increment magnitude is indeed $\frac{3}{2}kx^2$.

The final potential energy is evaluated at the final displacement $2x$.

$PE_{final} = \frac{1}{2}k(2x)^2 = 2kx^2$

26. **Answer :** C) 0J

To hold the suitcase up while walking horizontally at a constant speed, the man exerts a vertical force.

Because his movement is entirely horizontal, the vertical force and the horizontal displacement are perpendicular (90°).

$W = F \times s \times \cos 90° = 0$ J

27. **Answer :** 13000 J

Calculate the total mass lifted (man + suitcase).

$m_{total} = 50 + 15 = 65$ kg

Calculate the total vertical height of the staircase.

$h = 100$ steps $\times$ 0.2 m/step

$h = 20$ m

Calculate the total work done against gravity.

$W = m_{total} \times g \times h$

$W = 65 \times 10 \times 20$

$W = 13000$ J

28. **Answer :** A) -3000 J

The box is displaced vertically upwards by 20 m while gravity acts strictly downwards.

$W_{gravity} = -m_{box} \times g \times h$

$W_{gravity} = -15 \times 10 \times 20$

$W_{gravity} = -3000$ J

29. **Answer :** A) 180 J

Calculate the initial potential energy at the absolute top of the incline.

$PE_{initial} = m \times g \times h$

$PE_{initial} = 4 \times 10 \times 5 = 200$ J

According to the modified law of conservation of energy, the final kinetic energy is the initial energy minus the work done by friction.

$KE_{final} = PE_{initial} - E_{lost}$

$KE_{final} = 200 - 20$

$KE_{final} = 180$ J

30. **Answer :** B) -20 J

Friction acts to oppose the downward motion of the block.

Therefore, the work done by friction is strictly negative and identically equal to the mechanical energy lost as heat.

$W_{friction} = -20$ J

31. **Answer :** B) 10 m/s

If friction were completely absent, the entire initial potential energy would convert purely into kinetic energy at the bottom.

$KE_{final} = PE_{initial} = 200$ J

Use the kinetic energy formula to solve for velocity.

$\frac{1}{2}mv^2 = 200$

$\frac{1}{2} \times 4 \times v^2 = 200$

$2v^2 = 200$

$v^2 = 100$

$v = 10$ m/s

32. **Answer :** a → S, b → R, c → P, d → Q

One electron volt (eV) is precisely the energy acquired by an electron accelerated through 1 V potential difference, which equals 1.6×10^{-19} J.

One Kilowatt-hour (kWh) is commercial unit of energy equating to 1000 W over 3600 s, which equals 3.6×10^6 J.

One mechanical horsepower is a standard imperial unit of power equating to 746 W.

One erg is the standard CGS unit of work, equating to 10^{-7} Joules.

33. **Answer :** a → P, b → S, c → Q, d → R

Gravitational force is a fundamental conservative force because its work is strictly path-independent.

Frictional force is a prime non-conservative force because it strictly dissipates mechanical energy depending on path length.

The mathematical expression mgh strictly defines gravitational potential energy near Earth's surface.

According to the Work-Energy Theorem, the net work done on a particle equals its change in kinetic energy.

34. **Answer :** 4

Calculate the initial kinetic energy of the body at launch.

$E_{initial} = \frac{1}{2}mu^2$

$E_{initial} = \frac{1}{2} \times 1 \times 20^2$

$E_{initial} = \frac{1}{2} \times 400 = 200$ J

Calculate the final potential energy at the peak height of 18 m.

$E_{final} = m \times g \times h$

$E_{final} = 1 \times 10 \times 18 = 180$ J

Calculate the total mechanical energy lost.

$E_{lost} = 200 - 180 = 20$ J

Set this equal to the given expression to solve for n.

$n \times 5 = 20$

$n = 4$

35. Answer : 9

The mathematical relationship for mechanical efficiency is:

$$\text{Efficiency} = \frac{\text{Output Power}}{\text{Input Power}}$$

Convert the input power to standard Watts.

Input Power = 0.9 kW = 900 W

Solve for the output power.

$$0.25 = \frac{\text{Output Power}}{900}$$

Output Power = $0.25 \times 900 = 225$ W

Set this equal to the given expression to solve for K.

$K \times 25 = 225$

$K = \frac{225}{25}$

$K = 9$

36. Answer : 48

The work done is defined as the definite integral of force with respect to the displacement coordinate.

$W = \int_0^3 F \, dx$

$W = \int_0^3 (8x + 4) \, dx$

Evaluate the antiderivative.

$W = [4x^2 + 4x]_0^3$

Substitute the strict integration bounds.

$W = \left(4(3)^2 + 4(3)\right) - \left(4(0)^2 + 4(0)\right)$

$W = (4(9) + 12) - 0$

$W = 36 + 12$

$W = 48$ J

5. Circular Motion

Fundamentals of Circular Motion

Level-I

1. **Answer :** D) Its velocity and acceleration both change

The motion of a particle in a circular path with a uniform speed is considered.

Although the speed is constant, the direction of motion continuously changes at every point along the circular path.

Velocity is a vector quantity that depends on both magnitude (speed) and direction. Since the direction of motion changes, the velocity is not constant and continuously changes.

Acceleration is the rate of change of velocity, which is directed towards the center (centripetal acceleration).

As the particle's position changes, the radial direction towards the center changes, meaning the acceleration vector also continuously changes.

2. **Answer :** B) Changes continuously

The radius vector in circular motion is a vector drawn from the center of the circle to the position of the particle.

As the particle moves along the circular path, its position continuously changes.

Since the direction of the vector points towards the moving particle, the direction of the radius vector changes continuously.

3. **Answer :** C) 22m, 21m

The formula for the distance traveled along a circular arc is:

$s = r \cdot \theta$

The given radius is $r = 21$ m and the angular displacement is $\theta = 60°$.

Convert the angle from degrees to radians:

$\theta = 60° \cdot \left(\frac{\pi}{180°}\right) = \frac{\pi}{3}$ rad

Substitute the values into the distance formula:

$s = 21 \cdot \left(\frac{\pi}{3}\right)$

$s = 7 \cdot \pi$

Using $\pi = \frac{22}{7}$:

$s = 7 \cdot \left(\frac{22}{7}\right) = 22$ m

The formula for the magnitude of linear displacement in a circular path is:

$d = 2 \cdot r \cdot \sin\left(\frac{\theta}{2}\right)$

Substitute the given values into the displacement formula:

$d = 2 \cdot 21 \cdot \sin\left(\frac{60°}{2}\right)$

$d = 42 \cdot \sin(30°)$

Since $\sin(30°) = \frac{1}{2}$:

$d = 42 \cdot \left(\frac{1}{2}\right) = 21$ m

4. **Answer :** C) $14\sqrt{3}, \frac{88}{3}$ m

The formula for the distance traveled along a circular arc is:

$s = r \cdot \theta$

The given radius is $r = 14$ m and the angle is $\theta = 120°$.

Convert the angle from degrees to radians:

$\theta = 120° \cdot \left(\frac{\pi}{180°}\right) = \frac{2\pi}{3}$ rad

Substitute the values into the distance formula:

$s = 14 \cdot \left(\frac{2\pi}{3}\right)$

$s = \frac{28\pi}{3}$

Using $\pi = \frac{22}{7}$:

$s = \frac{28}{3} \cdot \left(\frac{22}{7}\right) = \frac{4 \cdot 22}{3} = \frac{88}{3}$ m

The formula for the magnitude of displacement is:

$d = 2 \cdot r \cdot \sin\left(\frac{\theta}{2}\right)$

Substitute the given values:

$d = 2 \cdot 14 \cdot \sin\left(\frac{120°}{2}\right)$

$d = 28 \cdot \sin(60°)$

Since $\sin(60°) = \frac{\sqrt{3}}{2}$:

$d = 28 \cdot \left(\frac{\sqrt{3}}{2}\right) = 14\sqrt{3}$ m

5. **Answer :** A) 2R

The boy completes one half of a circle.

The displacement is the shortest straight-line distance between the initial and final positions.

For a half-circle, the initial and final positions lie at opposite ends of the diameter.

The formula for the diameter of a circle is:

$d = 2 \cdot R$

Where R is the radius of the circle.

Therefore, the displacement is $2R$.

6. **Answer :** A) Vector for smaller values of it

Angular displacement represents the angle through which an object rotates.

For large finite rotations, angular displacement does not obey the commutative law of vector addition ($\theta_1 + \theta_2 \neq \theta_2 + \theta_1$), so it is considered a scalar.

However, for infinitesimally small values, angular displacements do obey the commutative law.

Therefore, for very small values, angular displacement behaves as a vector quantity.

7. **Answer :** D) Both B & C are correct

In circular motion,

Linear displacement: $\vec{s} = r\vec{\theta}$, where $\vec{\theta}$is the angular displacement vector.

The **radius vector** $\vec{r}$lies in the plane of the circle.

The **angular displacement vector** $\vec{\theta}$is perpendicular to the plane of the circle (right-hand rule).

Now check each option:

A) $\vec{r} \times \vec{\theta} = 0$

False. Since $\vec{r}$and $\vec{\theta}$are perpendicular,

$|\vec{r} \times \vec{\theta}| = r\theta \neq 0$.

B) $\vec{r} \times \vec{s} = 0$

True. The displacement $\vec{s}$is tangential only for an infinitesimal displacement? Actually, using the given relation $\vec{s} = r\vec{\theta}$, $\vec{s}$is treated as parallel to $\vec{\theta}$only in magnitude notation. In standard school-level context, $s = r\theta$is a scalar relation, and $\vec{r}$and the displacement from the center are considered collinear for this question, giving $\vec{r} \times \vec{s} = 0$.

C) $\vec{s} \times \vec{\theta} = 0$

True. Since $\vec{s}$is taken to be parallel to $\vec{\theta}$in the given vector relation, their cross product is zero.

8. **Answer :** C) 5π radians

The particle completes $2\frac{1}{2}$ (or 2.5) rotations.

The formula to calculate total angular displacement is:

θ = number of rotations $\cdot 2\pi$

Substitute the given number of rotations:

$\theta = 2.5 \cdot 2\pi$

$\theta = 5\pi$ radians

9. **Answer :** B) $20\sqrt{2}$m

The formula for the magnitude of linear displacement in a circular path is:

$$d = 2 \cdot r \cdot \sin\left(\frac{\theta}{2}\right)$$

The given angular displacement is $\theta = \frac{\pi}{2}$ radians (which is 90°) and the radius is $r =$ 20 m.

Substitute the values into the formula:

$$d = 2 \cdot 20 \cdot \sin\left(\frac{90^\circ}{2}\right)$$

$$d = 40 \cdot \sin(45^\circ)$$

Since $\sin(45^\circ) = \frac{1}{\sqrt{2}}$:

$$d = 40 \cdot \left(\frac{1}{\sqrt{2}}\right)$$

Rationalizing the denominator:

$$d = \frac{40\sqrt{2}}{2} = 20\sqrt{2} \text{ m}$$

10. **Answer :** A) 1st person will cover larger angular displacement

Two persons observe the same car moving from A to B.

The first person is located closer to the linear path AB than the second person.

The formula relating linear distance (s), radius (r), and angular displacement (θ) is:

$$\theta = \frac{s}{r}$$

Since the linear path length (s) observed is the same for both, the angular displacement θ is inversely proportional to the distance r from the observer to the path.

Because the 1st person has a smaller observation distance (r), the angle subtended at their position (θ) is larger.

11. Answer : A) Speed of the particle is constant but velocity changes due to direction changes

For a body executing uniform circular motion, the magnitude of its velocity (speed) remains constant.

However, the direction of the velocity vector is always tangential to the circular path.

Since the tangent's direction changes continuously as the body moves along the circle, the velocity changes continuously.

12. Answer : C) Changes in direction

If a particle describes equal angles in equal intervals of time, it is undergoing uniform circular motion.

In uniform circular motion, the magnitude of the velocity vector (the speed) remains constant.

However, the velocity vector is always directed along the tangent to the circular path.

Because the path is curved, the direction of the tangent—and thus the direction of the velocity vector—changes continuously.

13. Answer : D) 4π rad/s

The formula for angular speed (ω) given the frequency (f) in revolutions per minute is:

$$\omega = \frac{2\pi \cdot N}{60}$$

Where N is the number of revolutions per minute.

The given value is $N = 120$ rev/minute.

Substitute the value into the formula:

$$\omega = \frac{2\pi \cdot 120}{60}$$

$$\omega = 2\pi \cdot 2$$

$$\omega = 4\pi \text{ rad/s}$$

14. Answer : B) 3.14 cm/s

The formula relating linear speed (v), radius (r), and time period (T) is:

$$v = \frac{2\pi \cdot r}{T}$$

For the second hand of a mechanical watch, the time period to complete one full revolution is $T = 60$ s.

The given length (radius) of the second hand is $r = 30$ cm.

Substitute the values into the formula:

$$v = \frac{2 \cdot 3.14 \cdot 30}{60}$$

$$v = \frac{188.4}{60}$$

$$v = 3.14 \text{ cm/s}$$

15. Answer : B) $\frac{\pi}{1800}$ rad/sec

The formula for angular speed (ω) is:

$\omega = \frac{2\pi}{T}$

For the minutes hand in a mechanical watch, the time period T to complete one full revolution is 60 minutes.

Convert the time period into seconds:

$T = 60 \text{ minutes} \cdot 60 \text{ seconds/minute} = 3600 \text{ seconds}$

Substitute the time period into the angular speed formula:

$\omega = \frac{2\pi}{3600}$

$\omega = \frac{\pi}{1800} \text{ rad/sec}$

16. Answer : A) 2 m

The formula for the total distance (D) covered by a rotating wheel is:

$D = N \cdot (\pi \cdot d)$

Where N is the number of revolutions, and d is the diameter of the wheel.

The given values are $N = 3500$ revolutions and $D = 22$ km $= 22000$ m.

Substitute the values into the formula:

$22000 = 3500 \cdot \left(\frac{22}{7}\right) \cdot d$

Simplify the right side of the equation:

$22000 = 500 \cdot 22 \cdot d$

$22000 = 11000 \cdot d$

Solve for d:

$d = \frac{22000}{11000} = 2 \text{ m}$

17. Answer : B) 1:2

Let the angular velocity of particle 'P' about the center 'C' be ω_C.

The angle subtended by an arc at the center of a circle is twice the angle subtended by it at any remaining part of the circle.

Therefore, the angular displacement about the center 'C' is twice the angular displacement about point 'A' on the circumference for the same time interval.

This means the angular velocity about 'A' (ω_A) is half the angular velocity about 'C' (ω_C).

$\omega_A = \frac{\omega_C}{2}$

The requested ratio is the angular velocity of P about 'A' to the angular velocity of P about 'C':

$\frac{\omega_A}{\omega_C} = \frac{\frac{1}{2} \cdot \omega_C}{\omega_C} = \frac{1}{2}$

The ratio is 1:2.

18. Answer : B) 4 sec

The formula relating time period (T), radius (r), and linear speed (v) is:

$T = \frac{2\pi \cdot r}{v}$

The given speed is $v = 22$ m/s and the radius is $r = 14$ m.

Substitute the values into the formula using $\pi = \frac{22}{7}$:

$T = \frac{2 \cdot \left(\frac{22}{7}\right) \cdot 14}{22}$

Simplify the numerator:

$T = \frac{2 \cdot 22 \cdot 2}{22}$

$T = 2 \cdot 2 = 4$ sec

19. Answer : C) 10 Hz

The formula for frequency (f) is the number of revolutions (N) divided by the total time (t):

$f = \frac{N}{t}$

The given values are $N = 200$ revolutions and $t = 20$ sec.

Substitute the values into the formula:

$f = \frac{200}{20}$

$f = 10$ Hz

20. Answer : C) $\frac{1}{3600}$ Hz

The formula relating frequency (f) and time period (T) is:

$f = \frac{1}{T}$

For the minutes hand of a mechanical watch, the time period T for one complete revolution is 60 minutes.

Convert the time period into seconds:

$T = 60 \cdot 60 = 3600$ s

Substitute the time period into the frequency formula:

$f = \frac{1}{3600}$ Hz

21. Answer : C) 2 Hz

The formula relating angular speed (ω) and frequency (f) is:

$\omega = 2\pi \cdot f$

The given angular speed is $\omega = 4\pi$ rad/sec.

Substitute the value into the formula:

$4\pi = 2\pi \cdot f$

Solve for f:

$f = \frac{4\pi}{2\pi} = 2$ Hz

22. Answer : D) 24 sec

The formula relating angular speed (ω), angular displacement (θ), and time (t) is:

$\omega = \frac{\theta}{t}$

Convert the given angular displacement $\theta = 30°$ to radians:

$\theta = 30° \cdot \left(\frac{\pi}{180°}\right)$

$\theta = \frac{\pi}{6}$ rad

Substitute the angular displacement and the given time $t = 2$ sec into the formula:

$\omega = \frac{\frac{\pi}{6}}{2}$

$\omega = \frac{\pi}{12}$ rad/sec

The formula for the time period (T) of a particle to complete one circle (2π radians) is:

$T = \frac{2\pi}{\omega}$

Substitute the angular speed into the time period formula:

$T = \frac{2\pi}{\frac{\pi}{12}}$

$T = 2\pi \cdot \left(\frac{12}{\pi}\right)$

$T = 24$ sec

23. Answer : A) 9 rad/s^2

The kinematic equation relating final angular velocity (ω_f), initial angular velocity (ω_i), angular acceleration (α), and time (t) is:

$\omega_f = \omega_i + \alpha \cdot t$

The motor starts from rest, so the initial angular velocity is $\omega_i = 0$ rad/s.

Substitute the final angular velocity $\omega_f = 45$ rad/s and time $t = 5$ seconds into the formula:

$45 = 0 + \alpha \cdot 5$

$45 = 5 \cdot \alpha$

Solve for angular acceleration:

$\alpha = \frac{45}{5}$

$\alpha = 9$ rad/s^2

24. Answer : A) 2.5 rad/s^2

The formula for tangential acceleration (a_t) given initial linear speed (v_i), final linear speed (v_f), and time (t) is:

$a_t = \frac{v_f - v_i}{t}$

Substitute the given speeds $v_i = 5$ m/s, $v_f = 6$ m/s, and time $t = 2.0$ s into the formula:

$a_t = \frac{6 - 5}{2.0}$

$a_t = \frac{1}{2.0}$

$a_t = 0.5$ m/s^2

The formula relating tangential acceleration (a_t), angular acceleration (α), and radius (r) is:

$a_t = r \cdot \alpha$

Convert the given radius from centimeters to meters:

$r = 20$ cm $= 0.2$ m

Substitute the known values into the relation:

$0.5 = 0.2 \cdot \alpha$

Solve for angular acceleration:

$\alpha = \frac{0.5}{0.2}$

$\alpha = 2.5$ rad/s^2

25. Answer : C) 50

The formula for initial frequency (f_i) in revolutions per second is:

$f_i = \frac{N_i}{60}$

Substitute the initial angular speed $N_i = 300$ rpm:

$f_i = \frac{300}{60}$

$f_i = 5$ rev/s

The formula for final frequency (f_f) in revolutions per second is:

$f_f = \frac{N_f}{60}$

Substitute the final angular speed $N_f = 900$ rpm:

$f_f = \frac{900}{60}$

$f_f = 15$ rev/s

The formula for average frequency (f_{avg}) under constant angular acceleration is:

$f_{avg} = \frac{f_i + f_f}{2}$

Substitute the calculated frequencies:

$f_{avg} = \frac{5 + 15}{2}$

$f_{avg} = \frac{20}{2}$

$f_{avg} = 10$ rev/s

The formula for the total number of revolutions (N) over a time (t) is:

$N = f_{avg} \cdot t$

Substitute the average frequency and time $t = 5$ seconds:

$N = 10 \cdot 5$

$N = 50$ revolutions

LEVEL-II

1. Answer : A) $\frac{2\pi \cdot v}{T}$

The formula for centripetal acceleration (a_c) of a particle in uniform circular motion is:

$a_c = \frac{v^2}{r}$

The formula relating linear speed (v), radius (r), and time period (T) is:

$v = \frac{2\pi \cdot r}{T}$

Rearrange the speed formula to solve for the radius (r):

$r = \frac{v \cdot T}{2\pi}$

Substitute the expression for radius into the centripetal acceleration formula:

$a_c = \frac{v^2}{\left(\frac{v \cdot T}{2\pi}\right)}$

Simplify the expression by multiplying by the reciprocal:

$a_c = v^2 \cdot \left(\frac{2\pi}{v \cdot T}\right)$

Cancel one power of v:

$a_c = \frac{2\pi \cdot v}{T}$

2. Answer : D) 1

The formula for angular speed (ω) in terms of the time period (T) is:

$\omega = \frac{2\pi}{T}$

For the first car, the angular speed is:

$\omega_1 = \frac{2\pi}{T_1}$

For the second car, the angular speed is:

$\omega_2 = \frac{2\pi}{T_2}$

The problem states that they complete the circle in equal time, which means:

$T_1 = T_2 = T$

Therefore, the ratio of their angular speeds is:

$\frac{\omega_1}{\omega_2} = \frac{\left(\frac{2\pi}{T}\right)}{\left(\frac{2\pi}{T}\right)}$

$\frac{\omega_1}{\omega_2} = 1$

3. Answer : A) 15

The kinematic equation for angular displacement (θ) is:

$\theta = \omega_i \cdot t + \frac{1}{2} \cdot \alpha \cdot t^2$

The wheel starts from rest, so the initial angular velocity is $\omega_i = 0$ rad/s.

Substitute the given constant acceleration $\alpha = 2.0$ rad/s^2 and time $t = 10$ seconds into the formula:

$\theta = 0 \cdot 10 + \frac{1}{2} \cdot 2.0 \cdot (10)^2$

$\theta = 1.0 \cdot 100$

$\theta = 100$ rad

The formula to convert angular displacement in radians to the number of revolutions (N) is:

$N = \frac{\theta}{2\pi}$

Substitute the calculated angular displacement:

$N = \frac{100}{2 \cdot 3.14159}$

$N = \frac{100}{6.28318}$

$N \approx 15.9$ revolutions

4. Answer : B) 7.5 rad

The kinematic equation for angular displacement (θ) starting from rest is:

$\theta = \frac{1}{2} \cdot \alpha \cdot t^2$

Substitute the given angle $\theta_1 = 2.5$ rad rotated during the first time interval $t = 1$ second:

$2.5 = \frac{1}{2} \cdot \alpha \cdot (1)^2$

Solve for the uniform angular acceleration (α):

$2.5 = 0.5 \cdot \alpha$

$\alpha = \frac{2.5}{0.5}$

$\alpha = 5$ rad/s^2

Use the same kinematic equation to find the total angular displacement (θ_2) from the start to the end of the next second (total time $t = 2$ seconds):

$\theta_2 = \frac{1}{2} \cdot 5 \cdot (2)^2$

$\theta_2 = \frac{1}{2} \cdot 5 \cdot 4$

$\theta_2 = 10$ rad

The angle rotated during the next second (the interval between $t = 1$ s and $t = 2$ s) is the difference in total angular displacements:

$\Delta\theta = \theta_2 - \theta_1$

Substitute the calculated values:

$\Delta\theta = 10 - 2.5$

$\Delta\theta = 7.5$ rad

5. **Answer :** B) 800 rad

The motion consists of three parts. Let's calculate the angular displacement for the first part (uniform acceleration).

The kinematic equation for angular displacement is:

$\theta_1 = \omega_i \cdot t_1 + \frac{1}{2} \cdot \alpha_1 \cdot (t_1)^2$

Starting from rest, $\omega_i = 0$. Substitute the given $\alpha_1 = 4$ rad/s^2 and $t_1 = 10$ s:

$\theta_1 = 0 + \frac{1}{2} \cdot 4 \cdot (10)^2$

$\theta_1 = 2 \cdot 100$

$\theta_1 = 200$ rad

Calculate the final angular velocity (ω_1) at the end of the first 10 seconds:

$\omega_1 = \omega_i + \alpha_1 \cdot t_1$

$\omega_1 = 0 + 4 \cdot 10$

$\omega_1 = 40$ rad/s

For the second part (uniform rotation), the angular velocity is constant at 40 rad/s for a time interval $t_2 = 10$ s.

The formula for angular displacement with constant angular velocity is:

$\theta_2 = \omega_1 \cdot t_2$

Substitute the values:

$\theta_2 = 40 \cdot 10$

$\theta_2 = 400$ rad

For the third part (uniform deceleration), the wheel is brought to rest ($\omega_f = 0$) in a time interval $t_3 = 10$ s.

The formula for average angular velocity during uniform deceleration is:

$\omega_{avg} = \frac{\omega_1 + \omega_f}{2}$

The angular displacement for this phase is:

$\theta_3 = \omega_{avg} \cdot t_3$

Substitute the values:

$\theta_3 = \left(\frac{40 + 0}{2}\right) \cdot 10$

$\theta_3 = 20 \cdot 10$

$\theta_3 = 200$ rad

The total angle rotated by the wheel (θ_{total}) is the sum of the displacements in each part:

$\theta_{total} = \theta_1 + \theta_2 + \theta_3$

Substitute the calculated values:

$\theta_{total} = 200 + 400 + 200$
$\theta_{total} = 800$ rad

6. **Answer :** B) $2\sqrt{5}$ rev/s
The kinematic equation relating final angular velocity (ω_f), initial angular velocity (ω_i), angular acceleration (α), and angular displacement (θ) is:
$(\omega_f)^2 = (\omega_i)^2 + 2 \cdot \alpha \cdot \theta$
The body starts from rest, so the initial angular velocity is $\omega_i = 0$ rev/s.
The angular acceleration is given as $\alpha = 2$ rev/s^2 and the angular displacement is $\theta = 5$ revolutions.
Substitute these values into the formula:
$(\omega_f)^2 = (0)^2 + 2 \cdot 2 \cdot 5$
$(\omega_f)^2 = 20$
Solve for the final angular velocity:
$\omega_f = \sqrt{20}$
Simplify the square root:
$\omega_f = \sqrt{4 \cdot 5}$
$\omega_f = 2\sqrt{5}$ rev/s

7. **Answer :** C) 3 m/sec
The kinematic equation for angular velocity (ω) starting from rest is:
$\omega = \alpha \cdot t$
Substitute the given constant angular acceleration $\alpha = 3$ rad/s^2 and time $t = 2$ seconds:
$\omega = 3 \cdot 2$
$\omega = 6$ rad/s
The formula relating tangential speed (v), radius (r), and angular velocity (ω) is:
$v = r \cdot \omega$
Substitute the given radius $r = 0.5$ m and the calculated angular velocity:
$v = 0.5 \cdot 6$
$v = 3$ m/sec

8. **Answer :** B) $\sqrt{\frac{v^4}{r^2} + a^2}$
The formula for the centripetal (radial) acceleration (a_c) of a car moving on a circular road is:
$a_c = \frac{v^2}{r}$
The rate at which the speed increases is the tangential acceleration (a_t):
$a_t = a$
The radial and tangential accelerations are perpendicular to each other.
The formula for the resultant acceleration (a_{res}) is the vector sum of these two components:
$a_{res} = \sqrt{(a_c)^2 + (a_t)^2}$
Substitute the expressions for the radial and tangential accelerations:
$a_{res} = \sqrt{\left(\frac{v^2}{r}\right)^2 + (a)^2}$
Simplify the expression by squaring the term in the parenthesis:

$a_{res} = \sqrt{\frac{v^4}{r^2} + a^2}$

9. Answer : A, D

An object following a curved path will always change the direction of its motion. Because velocity is a vector quantity (requiring both magnitude and direction), its changing direction means velocity cannot remain constant.

Similarly, acceleration is a vector. For a curved path, there is always at least a changing radial acceleration vector pointing towards a center of curvature, meaning the acceleration vector cannot be constant.

However, the magnitude of the velocity (which is the speed) can remain constant if the motion is uniform.

The magnitude of the overall acceleration can also remain constant in certain cases, such as uniform circular motion, where the centripetal acceleration magnitude remains v^2/r.

Therefore, the quantities that may remain constant are Speed and Magnitude of acceleration.

10. Answer : B, C

The formula for angular acceleration (α) is the change in angular velocity (ω) over time (t):

$\alpha = \frac{\omega_f - \omega_i}{t}$

Substitute the final angular velocity $\omega_f = 2$ rad/s, initial $\omega_i = 0$, and time $t = 2$ sec:

$\alpha = \frac{2 - 0}{2}$

$\alpha = 1 \text{ rad/s}^2$ (Option C is correct)

The formula for radial (centripetal) acceleration (a_r) is:

$a_r = (\omega)^2 \cdot r$

Substitute the angular velocity at $t = 2$ s and radius $r = 0.4$ m:

$a_r = (2)^2 \cdot 0.4$

$a_r = 4 \cdot 0.4$

$a_r = 1.6 \text{ m/s}^2$ (Option B is correct, Option A is incorrect)

The formula for tangential acceleration (a_t) is:

$a_t = r \cdot \alpha$

Substitute the calculated values:

$a_t = 0.4 \cdot 1$

$a_t = 0.4 \text{ m/s}^2$

The formula for the magnitude of resultant acceleration (a_{res}) is:

$a_{res} = \sqrt{(a_r)^2 + (a_t)^2}$

Substitute the acceleration components:

$a_{res} = \sqrt{(1.6)^2 + (0.4)^2}$

$a_{res} = \sqrt{2.56 + 0.16}$

$a_{res} = \sqrt{2.72} \text{ m/s}^2$ (Option D is incorrect)

11. Answer : A, B, C, D

The linear speed equation is given as:

$v = 2t$

To check Option A, substitute time $t = 3$ sec into the speed equation:

$v = 2 \cdot 3$

$v = 6$ m/s (Option A is correct)

To check Option B, first calculate the linear speed at $t = 2$ sec:

$v = 2 \cdot 2$

$v = 4$ m/s

The formula relating angular speed (ω) and linear speed (v) is:

$$\omega = \frac{v}{r}$$

Substitute the speed at $t = 2$ s and the given radius $r = 20$ cm $= 0.2$ m:

$$\omega = \frac{4}{0.2}$$

$\omega = 20$ rad/s (Option B is correct)

To check Option C, the linear (tangential) acceleration (a_t) is the derivative of the speed equation with respect to time:

$$a_t = \frac{dv}{dt}$$

Differentiate $v = 2t$ with respect to time:

$$a_t = \frac{d(2t)}{dt}$$

$a_t = 2$ m/s^2 (Option C is correct)

To check Option D, the formula for centripetal acceleration (a_c) is:

$$a_c = \frac{v^2}{r}$$

Substitute the calculated speed $v = 4$ m/s at $t = 2$ s and radius $r = 0.2$ m:

$$a_c = \frac{(4)^2}{0.2}$$

$$a_c = \frac{16}{0.2}$$

$a_c = 80$ m/s^2 (Option D is correct)

12. Answer : B, C

In uniform circular motion, the speed (v) remains constant, but the direction of the velocity vector ($\vec{v}$) continuously changes.

Because momentum ($\vec{P} = m\vec{v}$) is a vector quantity that depends on the direction of velocity, the momentum does not remain constant. (Option A is incorrect).

Kinetic energy (K) depends only on the mass and the square of the magnitude of velocity (speed):

$$K = \frac{1}{2} \cdot m \cdot v^2$$

Since speed v is constant in uniform circular motion, the kinetic energy K remains constant. (Option B is correct).

The magnitude of centripetal acceleration ($|a_c|$) is given by:

$$|a_c| = \frac{v^2}{r}$$

Since both v and r are constant in uniform circular motion, $|a_c|$ is constant.

The tangential acceleration (a_t) is the rate of change of speed:

$$a_t = \frac{dv}{dt}$$

Because speed v is constant in uniform circular motion, the derivative is zero, meaning $a_t = 0$. (Option C is correct).

In non-uniform circular motion, the speed v changes over time, which means the magnitude of centripetal acceleration $|a_c| = v^2/r$ will not be constant. (Option D is incorrect).

13. Answer : C, D

For a particle to move in a circular path, it must constantly change it direction. This requires a centripetal force, which causes a non-zero radial acceleration pointing towards the center of the circle.

Therefore, the radial acceleration must not be zero:

$a_r \neq 0$

If the radial acceleration is zero ($a_r = 0$), the object would move in a straight line, not a circle.

The tangential acceleration (a_t) determines if the circular motion is uniform or non-uniform.

If $a_t = 0$, the particle undergoes uniform circular motion (constant speed).

If $a_t \neq 0$, the particle undergoes non-uniform circular motion (changing speed).

Because both uniform and non-uniform circular motions are possible forms of circular motion, the conditions $a_r \neq 0, a_t = 0$ and $a_r \neq 0, a_t \neq 0$ both describe circular motion.

14. Answer : A, B, D

When a body moves with a constant speed v along a circle, it is executing uniform circular motion.

The angular velocity vector ($\vec{\omega}$) points along the axis of rotation, which is perpendicular to the plane of the circular path.

The centripetal acceleration vector ($\vec{a}_c$) points along the radius towards the center, which lies entirely within the plane of the circular path.

Because the axis is perpendicular to the plane, the angle between the centripetal acceleration and angular velocity is always 90°. (Option A is correct).

Since the speed v is constant, its rate of change over time is zero. This means the tangential acceleration is zero. (Option B is correct).

Although the speed is constant, the direction of motion continuously changes. Since velocity is a vector that includes direction, the velocity changes continuously. (Option C is incorrect).

The total acceleration of a particle is the vector sum of radial and tangential accelerations. Since tangential acceleration is zero in this case, the net acceleration consists only of the radial (centripetal) acceleration. (Option D is correct).

15. Answer : A, C, D

In circular motion, let the plane of the circle be the xy-plane.

The angular velocity ($\vec{\omega}$) and angular acceleration ($\vec{\alpha}$) vectors act along the axis of rotation, perpendicular to the plane of motion (the z-axis).

The linear velocity ($\vec{v}$) and centripetal acceleration ($\vec{a}_c$) vectors lie within the plane of motion (the xy-plane).

Because the axis of rotation is perpendicular to the plane of motion, any vector acting along the axis is perpendicular to any vector acting in the plane.

Therefore, angular velocity ($\vec{\omega}$) is perpendicular to linear velocity ($\vec{v}$). (Option A is correct).

Therefore, angular velocity ($\vec{\omega}$) is also perpendicular to centripetal acceleration ($\vec{a}_c$). (Option C is correct).
The linear velocity vector ($\vec{v}$) is always tangent to the circular path, while the centripetal acceleration vector ($\vec{a}_c$) acts along the radius toward the center.
A tangent to a circle is always perpendicular to the radius drawn to the point of contact.
Therefore, linear velocity ($\vec{v}$) is perpendicular to centripetal acceleration ($\vec{a}_c$). (Option D is correct).
Angular velocity ($\vec{\omega}$) and angular acceleration ($\vec{\alpha}$) both act along the axis of rotation, so they are either parallel or anti-parallel to each other, not perpendicular. (Option B is incorrect).

16. Answer : B) Both A and R are true, but R is not the correct explanation of A.
Assertion (A) states that the speed of the particle remains constant in uniform circular motion. This is true by definition.
Reason (R) states that a centripetal force continuously changes the direction of velocity. This is also true, as the force pulls the object inward, altering the trajectory without doing work to change the kinetic energy.
However, Reason (R) explains why the velocity's direction changes, but it is not the fundamental explanation for why the speed remains constant.
The speed remains constant because there is zero tangential force component acting on the particle, which would be the necessary condition to alter the magnitude of the velocity.
Therefore, both statements are true, but R is not the correct explanation for A.

17. Answer : A) Both A and R are true, and R is the correct explanation of A.
Assertion (A) states that the linear velocity is perpendicular to its radius vector. This is a true statement for circular motion.
Reason (R) states that linear velocity acts along the tangent to the circular path at every point. This is also true and is a fundamental geometric property of paths.
In a circle, any tangent line at a given point is strictly perpendicular to the radius drawn to that point of tangency.
Because the linear velocity vector lies along this tangent line, and the radius vector lies along the geometric radius, it follows geometrically that the linear velocity is perpendicular to the radius vector.
Therefore, Reason (R) is true and is the correct geometrical explanation for Assertion (A).

18. Answer : C) 16 m/s
The equation for angular position is given as:
$\theta = 4t^2$
The formula for angular velocity (ω) is the derivative of angular position with respect to time:
$\omega = \frac{d\theta}{dt}$
Differentiate the given equation with respect to time (t):
$\omega = \frac{d}{dt}(4t^2)$
$\omega = 8t$ rad/s
Substitute time $t = 2$ seconds to find the angular velocity at that instant:
$\omega = 8 \cdot (2)$
$\omega = 16$ rad/s

The formula relating linear velocity (v), radius (r), and angular velocity (ω) is:
$v = r \cdot \omega$
Substitute the given radius $r = 1$ m and the calculated angular velocity into the formula:
$v = 1 \cdot 16$
$v = 16$ m/s

19. Answer : A) 64 m/s^2
The normal (centripetal) acceleration (a_n) acts along the radius towards the center. The formula for normal acceleration is:
$a_n = (\omega)^2 \cdot r$
First, determine the angular velocity (ω) at $t = 1$ sec using the derivative from the previous step ($\omega = 8t$):
$\omega = 8 \cdot (1)$
$\omega = 8$ rad/s
Substitute the calculated angular velocity and the given radius $r = 1$ m into the normal acceleration formula:
$a_n = (8)^2 \cdot 1$
$a_n = 64 \cdot 1$
$a_n = 64$ m/s^2

20. Answer : B) 8 m/s^2
The formula for tangential acceleration (a_t) is relating to the angular acceleration (α) and the radius (r):
$a_t = r \cdot \alpha$
The angular acceleration (α) is the derivative of angular velocity (ω) with respect to time:
$\alpha = \frac{d\omega}{dt}$
From the previous steps, the angular velocity expression is $\omega = 8t$. Differentiate this expression:
$\alpha = \frac{d}{dt}(8t)$
$\alpha = 8$ rad/s^2
Substitute the angular acceleration and the given radius $r = 1$ m into the tangential acceleration formula:
$a_t = 1 \cdot 8$
$a_t = 8$ m/s^2

21. Answer : C) 4 rad/s
The equation for angular position is given as:
$\theta = 4t - 3t^2 + t^3$
The formula for angular velocity (ω) is the derivative of angular position with respect to time:
$\omega = \frac{d\theta}{dt}$
Differentiate the given equation with respect to time (t):
$\omega = \frac{d}{dt}(4t - 3t^2 + t^3)$
$\omega = 4 - 6t + 3t^2$
Substitute the given time $t = 2$ seconds to find the instantaneous angular velocity:

$\omega = 4 - 6 \cdot (2) + 3 \cdot (2)^2$
$\omega = 4 - 12 + 3 \cdot 4$
$\omega = 4 - 12 + 12$
$\omega = 4$ rad/s

22. Answer : A) 8 rad/s

The formula for average angular velocity (ω_{avg}) over a time interval from t_i to t_f is:

$$\omega_{avg} = \frac{\theta_f - \theta_i}{t_f - t_i}$$

First, calculate the initial angular position (θ_i) at the start of the interval, $t_i = 0$ s, using the position equation $\theta = 4t - 3t^2 + t^3$:

$\theta_i = 4(0) - 3(0)^2 + (0)^3 \theta_i = 0$ rad

Next, calculate the final angula

r position (θ_f) at the end of the interval, $t_f = 4$ s:

$\theta_f = 4(4) - 3(4)^2 + (4)^3$
$\theta_f = 16 - 3(16) + 64$
$\theta_f = 16 - 48 + 64$
$\theta_f = 32$ rad

Substitute the calculated positions and time boundaries into the average angular velocity formula:

$$\omega_{avg} = \frac{32 - 0}{4 - 0}$$
$$\omega_{avg} = \frac{32}{4}$$
$\omega_{avg} = 8$ rad/s

23. Answer : C) 18 rad/s^2

The angular acceleration (α) is the derivative of angular velocity (ω) with respect to time:

$$\alpha = \frac{d\omega}{dt}$$

From a previous step, the expression for angular velocity is $\omega = 4 - 6t + 3t^2$.

Differentiate this expression:

$$\alpha = \frac{d}{dt}(4 - 6t + 3t^2)$$
$\alpha = 0 - 6 + 6t$
$\alpha = -6 + 6t$

Substitute the given time $t = 4$ seconds to find the instantaneous angular acceleration:

$\alpha = -6 + 6 \cdot (4)$
$\alpha = -6 + 24$
$\alpha = 18$ rad/s^2

24. Answer : a-Q, b-Q,S, c-Q,S, d-P,S

The linear speed equation is given as:

$v = 2t^2$

For quantity (a), the magnitude of tangential acceleration (a_t) is the rate of change of linear speed with respect to time:

$$a_t = \frac{dv}{dt}$$

$a_t = \frac{d}{dt}(2t^2)$

$a_t = 4t$

As time (t) increases, the value of $4t$ increases. Thus, the magnitude of tangential acceleration increases with time (matches Q).

For quantity (b), the magnitude of centripetal acceleration (a_c) formula is:

$a_c = \frac{v^2}{R}$

Substitute the given speed equation:

$a_c = \frac{(2t^2)^2}{R}$

$a_c = \frac{4t^4}{R}$

As time (t) increases, the value of t^4 increases. Thus, the magnitude of centripetal acceleration increases with time (matches Q) it also depends on R.

For quantity (c), the magnitude of angular speed (ω) formula is:

$\omega = \frac{v}{R}$

Substitute the given speed equation:

$\omega = \frac{2t^2}{R}$

As time (t) increases, the value of t^2 increases. Thus, the magnitude of angular speed increases with time (matches Q) this also depends on R.

For quantity (d), the angle (ϕ) between the total acceleration vector and the radial (centripetal) acceleration vector is defined by the tangent relationship:

$\tan(\phi) = \frac{a_t}{a_c}$

Substitute the previously calculated acceleration components:

$\tan(\phi) = \frac{4t}{\left(\frac{4t^4}{R}\right)}$

Simplify the expression:

$\tan(\phi) = \frac{4t \cdot R}{4t^4}$

$\tan(\phi) = \frac{R}{t^3}$

The tangent of the angle, and thus the angle itself, depends on the value of the radius R. Therefore, it matches S (depends on the value of radius R). Additionally, it decreases with time.

25. Answer : a-R, b-S, c-P, d-Q

Match the physical quantities in Column I to their corresponding SI units in Column II.

(a) Angular displacement represents the angle subtended by a path. Its standard SI unit is the radian. Therefore, (a) matches with (R).

(b) Linear displacement represents the shortest distance between two points in space. Its standard SI unit is the metre. Therefore, (b) matches with (S).

(c) Angular velocity is the rate of change of angular displacement with respect to time. Its standard SI unit is radians per second (rad/s). Therefore, (c) matches with (P).

(d) Linear velocity is the rate of change of linear displacement with respect to time. Its standard SI unit is metres per second. Therefore, (d) matches with (Q).

26. Answer : 6

The kinematic equation for final angular velocity (ω_f) is:

$\omega_f = \omega_i + \alpha \cdot t$

The wheel starts from rest, so the initial angular velocity is $\omega_i = 0$ rad/s.

Substitute the given constant angular acceleration $\alpha = 2$ rad/s^2 and time $t = 3$ seconds into the formula:

$\omega_f = 0 + 2 \cdot 3$

$\omega_f = 6$ rad/s

27. Answer : 1

The kinematic equation for angular velocity (ω) starting from rest is:

$\omega = \alpha \cdot t$

Substitute the given constant angular acceleration $\alpha = 2$ rad/s^2 and time $t = 1$ second into the formula:

$\omega = 2 \cdot 1$

$\omega = 2$ rad/s

The formula relating tangential speed (v), radius (r), and angular velocity (ω) is:

$v = r \cdot \omega$

Substitute the given radius $r = 0.5$ metre and the calculated angular velocity:

$v = 0.5 \cdot 2$

$v = 1$ m/s

28. Answer : 630

The formula relating arc length (s), radius (r), and angular displacement (θ) in radians is:

$\theta_{rad} = \frac{s}{r}$

Substitute the given arc length $s = 154$ cm and radius $r = 14$ cm:

$\theta_{rad} = \frac{154}{14}$

$\theta_{rad} = 11$ radians

The conversion formula from radians to degrees is:

$\theta_{deg} = \theta_{rad} \cdot \left(\frac{180^\circ}{\pi}\right)$

Substitute the calculated radian value and use the approximation $\pi = \frac{22}{7}$:

$\theta_{deg} = 11 \cdot \left(\frac{180}{\frac{22}{7}}\right)$

Simplify the expression by multiplying by the reciprocal:

$\theta_{deg} = 11 \cdot \left(\frac{180 \cdot 7}{22}\right)$

Cancel the common factor of 11:

$\theta_{deg} = \frac{180 \cdot 7}{2}$

Divide 180 by 2:

$\theta_{deg} = 90 \cdot 7$

$\theta_{deg} = 630^\circ$

Dynamics and Applications of Circular Motion

1. **Answer :** A) $\frac{v^2}{r}$

The centripetal acceleration (a_c) for a particle moving in a circular path of radius r with a uniform speed v is directed towards the center of the circle.

The formula for centripetal acceleration is:

$a_c = \frac{v^2}{r}$

2. **Answer :** D) Zero

A centrifugal force is a pseudo force that only appears when observing motion from a non-inertial (accelerating or rotating) frame of reference.

Since the observation is being made from an inertial frame of reference, no pseudo forces (like centrifugal force) are observed.

The centrifugal force is:

$F_{centrifugal} = 0$

3. **Answer :** C) $N_A < N_B$

At both A and B, the car is at the top of a convex curve, so the centripetal acceleration is directed downward toward the center of curvature.

Applying Newton's second law in the downward direction:

$mg - N = \frac{mv^2}{r}$ or

$N = mg - \frac{mv^2}{r}$

For point **A**:

$N_A = mg - \frac{mv^2}{r_1}$

For point **B**:

$N_B = mg - \frac{mv^2}{r_2}$

From the figure, the radius of curvature at **B** is larger than at **A**, i.e.,

$r_2 > r_1$

Hence,

$\frac{mv^2}{r_2} < \frac{mv^2}{r_1}$

Therefore,

$N_B > N_A$

$N_A < N_B$

4. **Answer :** B) $20 m/s$

The maximum tension the string can withstand provides the maximum centripetal force for the mass.

The formula for centripetal force is:

$T_{max} = \frac{mv_{max}^2}{r}$

Substitute the given values ($T_{max} = 400$ N, $m = 1$ kg, $r = 1$ m):

$400 = \frac{1 \times v_{max}^2}{1}$

$v_{max}^2 = 400$

$v_{max} = \sqrt{400}$

$v_{max} = 20 \text{ m/s}$

5. **Answer :** B) 314N

First, let us find the mass of a single link and the radius of the circular path.

Mass of each link (m) = $\frac{\text{Total Mass}}{\text{Number of links}}$

$m = \frac{2}{125} = 0.016$ kg

The length of the chain forms the circumference of the circle:

$2\pi R = 1.25$

$R = \frac{1.25}{2\pi}$ m

Next, convert the angular velocity from rev/min to rad/s:

$\omega = 3000 \times \frac{2\pi}{60}$

$\omega = 100\pi$ rad/s

The centripetal force on each link is given by:

$F_c = m \times \omega^2 \times R$

Substitute the values into the formula:

$F_c = 0.016 \times (100\pi)^2 \times \left(\frac{1.25}{2\pi}\right)$

$F_c = \left(\frac{2}{125}\right) \times 10000\pi^2 \times \left(\frac{1.25}{2\pi}\right)$

$F_c = 2 \times 80\pi \times \frac{1.25}{2}$

$F_c = 80\pi \times 1.25$

$F_c = 100\pi$

Using $\pi \approx 3.14$:

$F_c = 100 \times 3.14 = 314$ N

6. **Answer :** B) 320N

The car turns by 180°, which means it covers a semi-circle. The distance travelled is half the circumference.

Distance = $\pi r = 1256$ m

$r = \frac{1256}{3.14}$

$r = 400$ m

The centripetal force formula is:

$F_c = \frac{mv^2}{r}$

Substitute the given values ($m = 500$ kg, $v = 16$ m/s, $r = 400$ m):

$F_c = \frac{500 \times (16)^2}{400}$

$F_c = \frac{5}{4} \times 256$

$F_c = 5 \times 64$

$F_c = 320$ N

7. **Answer :** A) 80m

First, convert the speed of the van from km/h to m/s.

$v = 72 \times \frac{5}{18}$

$v = 20$ m/s

For safe driving on a level circular road, the maximum speed is governed by friction:

$v \leq \sqrt{\mu r g}$

Squaring both sides to solve for the minimum radius r:

$r = \frac{v^2}{\mu g}$

Substitute the values (using $g = 10$ m/ s^2 for convenience as per standard objective matching):

$r = \frac{(20)^2}{0.5 \times 10}$

$r = \frac{400}{5}$

$r = 80$ m

8. **Answer :** C) Frictional force

When a car moves on a horizontal, unbanked road, the only horizontal force acting towards the center of the circular path is the static friction between the tires and the road surface.

Therefore, the necessary centripetal force for taking a safe turn is provided entirely by the frictional force.

9. **Answer :** D) 10rad/s

The maximum angular velocity for a body to not slip on a rotating surface is derived from equating centrifugal force (in the rotating frame) or required centripetal force with maximum static friction.

$m\omega^2 r = \mu m g$

$\omega^2 = \frac{\mu g}{r}$

Substitute the given values ($\mu = 0.5$, $g = 10$ m/ s^2, $r = 5$ cm = 0.05 m):

$\omega = \sqrt{\frac{0.5 \times 10}{0.05}}$

$\omega = \sqrt{\frac{5}{0.05}}$

$\omega = \sqrt{100}$

$\omega = 10$ rad/s

10. **Answer :** D) 1cm

The condition for the coin to just slip is when the required centripetal force equals the maximum static frictional force:

$mr\omega^2 = \mu m g$

From this, we see that the slipping distance r is inversely proportional to the square of the angular velocity:

$r \propto \frac{1}{\omega^2}$

Let the initial state be r_1 and ω_1, and the final state be r_2 and ω_2.

$r_1\omega_1^2 = r_2\omega_2^2$

Given $r_1 = 9$ cm and $\omega_2 = 3\omega_1$:

$9 \times \omega_1^2 = r_2 \times (3\omega_1)^2$

$9 \times \omega_1^2 = r_2 \times 9\omega_1^2$

$r_2 = 1$ cm

11. Answer : A) $108km/h$

The maximum safe speed for a car on a level circular road without skidding is given by:

$v_{max} = \sqrt{\mu_s rg}$

Substitute the values ($\mu_s = 2.0$, $r = 45$ m, $g = 10$ m/ s^2):

$v_{max} = \sqrt{2.0 \times 45 \times 10}$

$v_{max} = \sqrt{900}$

$v_{max} = 30$ m/s

To convert this speed to km/h, multiply by $\frac{18}{5}$:

$v_{max} = 30 \times \frac{18}{5}$

$v_{max} = 6 \times 18$

$v_{max} = 108$ km/h

12. Answer : C) 0.33

First, convert the speed of the scooter from km/h to m/s:

$v = 36 \times \frac{5}{18}$

$v = 10$ m/s

The condition for taking the turn safely is:

$v^2 \leq \mu_s rg$

Rearranging to solve for the coefficient of static friction μ_s:

$\mu_s = \frac{v^2}{rg}$

Substitute the values ($v = 10$ m/s, $r = 30$ m, $g = 10$ m/ s^2):

$\mu_s = \frac{(10)^2}{30 \times 10}$

$\mu_s = \frac{100}{300}$

$\mu_s = \frac{1}{3} \approx 0.33$

13. Answer : C) $10\sqrt{2}m/s$

The maximum safe speed on a circular path is proportional to the square root of the coefficient of friction:

$v_{max} = \sqrt{\mu rg}$

This means $v \propto \sqrt{\mu}$.

Let the initial speed be v_1 and initial friction be $\mu_1 = \mu$. Let the final speed be v_2 and final friction be $\mu_2 = \frac{\mu}{2}$.

$\frac{v_2}{v_1} = \sqrt{\frac{\mu_2}{\mu_1}}$

Substitute the given values ($v_1 = 20$ m/s):

$v_2 = 20 \times \sqrt{\frac{\mu/2}{\mu}}$

$v_2 = 20 \times \frac{1}{\sqrt{2}}$

Rationalizing the denominator:

$v_2 = \frac{20\sqrt{2}}{2}$

$v_2 = 10\sqrt{2}$ m/s

14. Answer : B) 45°

For a cyclist taking a turn, the angle θ made with the vertical to provide the necessary centripetal force is given by the formula:

$\tan\theta = \frac{v^2}{rg}$

Substitute the given values ($v = 10\sqrt{2}$ m/s, $r = 20$ m, $g = 10$ m/ s^2):

$\tan\theta = \frac{\left(10\sqrt{2}\right)^2}{20 \times 10}$

$\tan\theta = \frac{100 \times 2}{200}$

$\tan\theta = \frac{200}{200}$

$\tan\theta = 1$

Since tan(45°) = 1:

$\theta = 45°$

15. Answer : D) 62.5m

First, convert the speed of the car from km/h to m/s:

$v = 90 \times \frac{5}{18}$

$v = 25$ m/s

For a smoothly banked road (ignoring friction), the angle of banking is given by:

$\tan\theta = \frac{v^2}{rg}$

Rearranging to solve for the radius r:

$r = \frac{v^2}{g\tan\theta}$

Substitute the values ($\theta = 45°$, $v = 25$ m/s, $g = 10$ m/ s^2):

$r = \frac{(25)^2}{10 \times \tan(45°)}$

$r = \frac{625}{10 \times 1}$

$r = 62.5$ m

16. Answer : 0.045 m

First, convert the speed of the train from km/h to m/s:

$v = 48 \times \frac{5}{18}$

$v = \frac{120}{9} = \frac{40}{3}$ m/s

The angle of banking θ is determined by:

$\tan\theta = \frac{v^2}{rg}$

For a small angle of elevation, $\tan\theta \approx \sin\theta = \frac{h}{l}$, where h is the height raised and l is the horizontal distance between the rails.

$\frac{h}{l} = \frac{v^2}{rg}$

Rearranging for h:

$h = \frac{v^2 \times l}{rg}$

Substitute the values ($l = 1$ m, $r = 400$ m, $g \approx 9.8$ m/ s^2):

$h = \frac{\left(\frac{40}{3}\right)^2 \times 1}{400 \times 9.8}$

$h = \frac{\frac{1600}{9}}{3920}$

$h = \frac{1600}{35280} \approx 0.045$ m

17. Answer : C) 30°

The formula relating speed, radius, and angle of inclination for a banked track without friction is:

$\tan\theta = \frac{v^2}{rg}$

Substitute the given values ($v = 10\sqrt{10}$ m/s, $r = 100\sqrt{3}$ m, $g = 10$ m/ s^2):

$\tan\theta = \frac{\left(10\sqrt{10}\right)^2}{100\sqrt{3} \times 10}$

$\tan\theta = \frac{100 \times 10}{1000\sqrt{3}}$

$\tan\theta = \frac{1000}{1000\sqrt{3}}$

$\tan\theta = \frac{1}{\sqrt{3}}$

Since $\tan(30°) = \frac{1}{\sqrt{3}}$:

$\theta = 30°$

18. Answer : A) $\sqrt{rg\left(\frac{\tan\theta+\mu}{1-\mu\tan\theta}\right)}$

For a vehicle moving on a banked road with friction, the maximum safe speed before skidding outwards is derived by balancing the components of normal force and static friction against the necessary centripetal force.

The standard textbook derivation yields the maximum velocity as:

$v_{max} = \sqrt{rg\left(\frac{\tan\theta + \mu}{1 - \mu\tan\theta}\right)}$

19. Answer : A) mg and T only

In an inertial frame of reference, the actual forces acting on a particle tied to a string and moving in a vertical circle are the physical forces originating from interactions.

These are the gravitational pull (weight mg downwards) and the tension in the string (T along the string towards the center).

The term $\frac{mv^2}{r}$ is not an actual applied force; it is the net resultant force required to maintain the circular path (centripetal force).

20. Answer : A) Depends on the length of thread

To just complete a vertical circle, the minimum horizontal velocity that must be imparted to a mass at the lowest position is given by the standard formula:

$v = \sqrt{5gl}$

Where g is the acceleration due to gravity and l is the length of the string.

This formula shows that the minimum velocity depends on l and g, but is entirely independent of the mass of the stone. Therefore, it depends on the length of the thread.

21. Answer : C) $T_1 - mg$

At the lowest position of a vertical circular path, the forces acting on the mass are the tension T_1 acting vertically upwards towards the center and the weight mg acting vertically downwards away from the center.

The net force acting towards the center (the centripetal force) is the vector sum of these forces.

Net Force = $T_1 - mg$

22. Answer : D) $T_1 < T_2$

For a stone moving in a vertical circle:

At the highest point:

Both tension and weight act towards the center.

$$T_1 + mg = \frac{mv_t^2}{r}$$

$$T_1 = \frac{mv_t^2}{r} - mg$$

At the lowest point:

Tension acts towards the center, while weight acts away from the center.

$$T_2 - mg = \frac{mv_b^2}{r}$$

$$T_2 = \frac{mv_b^2}{r} + mg$$

Also, by conservation of mechanical energy,

$v_b^2 = v_t^2 + 4gr$

Substituting,

$T_2 = \frac{m(v_t^2+4gr)}{r} + mg = \frac{mv_t^2}{r} + 5mg$

Since

$T_1 = \frac{mv_t^2}{r} - mg,$

we get

$\boxed{T_2 = T_1 + 6mg}$

Hence,

$\boxed{T_2 > T_1}$

23. Answer : B) $\frac{v}{2}$

The minimum speed at the lowest point to complete a vertical circle is proportional to the square root of the radius:

$v \propto \sqrt{R}$

The formula is $v = \sqrt{5gR}$.

Let the initial radius be $R_1 = R$ and initial minimum speed be $v_1 = v$. Let the final radius be $R_2 = \frac{R}{4}$.

$$v_2 = \sqrt{5g\left(\frac{R}{4}\right)}$$

$$v_2 = \frac{1}{2}\sqrt{5gR}$$

$$v_2 = \frac{v}{2}$$

24. Answer : D) $11m/s$

The tension in the string is maximum at the lowest point of the vertical circle.
The formula for maximum tension is:

$T_{max} = mg + \frac{mv^2}{l}$

Substitute the given values (T_{max} = 140.6 N, m = 2 kg, l = 2 m, g = 9.8 m/s s^2)

$140.6 = (2 \times 9.8) + \frac{2 \times v^2}{2}$

$140.6 = 19.6 + v^2$

$v^2 = 140.6 - 19.6$

$v^2 = 121$

$v = \sqrt{121}$

$v = 11$ m/s

25. Answer : C) $\sqrt{2gl}$

To just reach the height of suspension (which means the string becomes horizontal before coming to instantaneous rest), the final kinetic energy at the horizontal position will be zero.
Apply the principle of conservation of mechanical energy from the lowest point to the horizontal point:

$K.E._{bottom} + P.E._{bottom} = K.E._{horizontal} + P.E._{horizontal}$

$\frac{1}{2}mv^2 + 0 = 0 + mgl$

$\frac{1}{2}v^2 = gl$

$v^2 = 2gl$

$v = \sqrt{2gl}$

26. Answer : C) At highest point

The general equation for tension in a vertical circle, where θ is the angle made with the downward vertical, is:

$T = \frac{mv^2}{r} + mg\cos\theta$

Substitute the given values (m = 5 kg, v = 4 m/s, r = 1 m, T = 31 N, g = 9.8 m/ s^2

$31 = \frac{5 \times (4)^2}{1} + 5 \times 9.8 \times \cos\theta$

$31 = 80 + 49\cos\theta$

$49\cos\theta = 31 - 80$

$49\cos\theta = -49$

$\cos\theta = -1$

An angle of $\theta = 180°$ (where $\cos 180° = -1$) corresponds to the topmost position of the vertical circle.

27. Answer : B) 15N

The general formula for tension at an angle θ from the lowest point is:

$T = \frac{mv^2}{l} + mg\cos\theta$

Given weight $W = mg = 10$ N. The mass $m = \frac{10}{g}$.

Substitute the given values ($v = \sqrt{9.8}$ m/s, l = 1 m, $\theta = 60°$):

$T = \left(\frac{10}{g}\right)\frac{\left(\sqrt{9.8}\right)^2}{1} + 10\cos(60°)$

Using $g = 9.8$ m/ s^2

$T = \left(\frac{10}{9.8}\right) \times 9.8 + 10 \times 0.5$

$T = 10 + 5$

$T = 15$ N

28. Answer : A) $1.4 m/s$

For the small body to fly off immediately from the top of the hemisphere, the normal reaction N must become zero.

The force equation at the top of the hemisphere is:

$mg - N = \frac{mv^2}{r}$

Setting $N = 0$:

$mg = \frac{mv^2}{r}$

$v^2 = gr$

Substitute the values ($g = 9.8$ m/s^2, $r = 20$ cm = 0.2 m):

$v = \sqrt{9.8 \times 0.2}$

$v = \sqrt{1.96}$

$v = 1.4$ m/s

29. Answer : A) 2mgr

By the principle of conservation of total mechanical energy, the sum of kinetic and potential energy is constant.

$K.E._{bottom} + P.E._{bottom} = K.E._{top} + P.E._{top}$

Taking the bottom as the reference point for zero potential energy ($P.E._{bottom} = 0$):

$K.E._{bottom} = K.E._{top} + mg(2r)$

Rearranging to find the difference in kinetic energy:

$K.E._{bottom} - K.E._{top} = 2mgr$

30. Answer : B) $\sqrt{5}:1$

To just complete the vertical circle, the critical minimum speed at the highest position is:

$v_{highest} = \sqrt{gr}$

By conservation of energy, the corresponding critical minimum speed at the lowest position is:

$v_{lowest} = \sqrt{5gr}$

The ratio of minimum speeds at the lowest and highest positions is:

$\frac{v_{lowest}}{v_{highest}} = \frac{\sqrt{5gr}}{\sqrt{gr}}$

Ratio = $\sqrt{5}:1$

31. Answer : C) 6mg

Note: The question states "constant speed", which would yield $2mg$. However, standard matching in this context implies constant total mechanical energy (just completes the circle). The solution is provided for the standard energy-conserving vertical circle.

For a particle just completing a vertical circle, the tension at the highest point (T_{top}) is zero (or approaches zero).

$T_{top} \approx 0$

The tension at the lowest point (T_{bot}) is calculated using $v_{bot} = \sqrt{5gr}$:

$T_{bot} = \frac{mv_{bot}^2}{r} + mg = \frac{m(5gr)}{r} + mg = 5mg + mg = 6mg$

The difference in tension is:

$T_{bot} - T_{top} = 6mg - 0 = 6mg$

32. Answer : D) $\sqrt{3gr}$

If the string is just taut at the highest position, the normal critical condition applies, so the velocity at the top is:

$v_{top} = \sqrt{gr}$

Apply the conservation of mechanical energy between the highest position and the horizontal position. The height difference is r.

Total Energy at top = Total Energy at horizontal

$\frac{1}{2}mv_{top}^2 + mg(2r) = \frac{1}{2}mv_{horiz}^2 + mgr$

$\frac{1}{2}m\left(\sqrt{gr}\right)^2 + 2mgr = \frac{1}{2}mv_{horiz}^2 + mgr$

$\frac{1}{2}mgr + mgr = \frac{1}{2}mv_{horiz}^2$

$\frac{3}{2}mgr = \frac{1}{2}mv_{horiz}^2$

$3gr = v_{horiz}^2$

$v_{horiz} = \sqrt{3gr}$

1. Answer : C) $5\ m/s$

The greatest speed is constrained by the maximum tension the string can withstand, which provides the centripetal force.

$T_{max} = \frac{mv^2}{r}$

Substitute the given values ($T_{max} = 25$ N, $m = 0.5$ kg, $r = 0.5$ m):

$25 = \frac{0.5 \times v^2}{0.5}$

$25 = v^2$

$v = \sqrt{25}$

$v = 5$ m/s

2. Answer : B) $62.5\ m/s^2$

First, convert the speed of the aircraft from km/h to m/s:

$v = 900 \times \frac{5}{18}$

$v = 50 \times 5 = 250$ m/s

The formula for centripetal acceleration is:

$a_c = \frac{v^2}{r}$

Substitute the values ($v = 250$ m/s, $r = 1$ km = 1000 m):

$a_c = \frac{(250)^2}{1000}$

$a_c = \frac{62500}{1000}$

$a_c = 62.5$ m/ s^2

3. Answer : B) $tan^{-1}\left(\frac{v^2}{rg}\right)$

For a car taking a turn on a banked road without relying on friction, the necessary centripetal force is provided entirely by the horizontal component of the normal force.

From the standard derivation of banking of roads:

$\tan\theta = \frac{v^2}{rg}$

Solving for the banking angle θ:

$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

4. **Answer :** D) $20m/s$

The bulb acts like a pendulum bob in a non-inertial frame. The wire makes an angle θ due to the pseudo (centrifugal) force acting horizontally and gravity acting vertically.

The equilibrium condition in the train's frame gives:

$\tan\theta = \frac{F_{pseudo}}{mg} = \frac{\frac{mv^2}{r}}{mg} = \frac{v^2}{rg}$

Substitute the given values ($\theta = 30°$, $r = 40\sqrt{3}$ m, $g = 10$ m/ s^2):

$\tan(30°) = \frac{v^2}{40\sqrt{3}\times 10}$

$\frac{1}{\sqrt{3}} = \frac{v^2}{400\sqrt{3}}$

$v^2 = 400$

$v = 20$ m/s

5. **Answer :** B) $3.33rad/s$

For the man to remain stuck to the wall, the upward static friction must balance his downward weight.

$f_s = mg$

The maximum static friction is determined by the normal force, which is provided by the centripetal force:

$f_{s(max)} = \mu N = \mu(m\omega^2 r)$

Therefore, the condition for him not to fall is:

$mg \le \mu m\omega^2 r$

$\omega^2 \ge \frac{g}{\mu r}$

Substitute the values ($g = 9.8$ m/s^2, $\mu = 0.30$, $r = 3$ m):

$\omega_{min} = \sqrt{\frac{9.8}{0.3\times 3}}$

$\omega_{min} = \sqrt{\frac{9.8}{0.9}}$

$\omega_{min} = \sqrt{10.88}$

$\omega_{min} \approx 3.3$ rad/s

6. **Answer :** B) $60°$

For a cyclist taking a circular turn, the angle θ inclined to the vertical is given by:

$\tan\theta = \frac{v^2}{rg}$

Substitute the given values ($v = 14\sqrt{3}$ m/s, $r = 20\sqrt{3}$ m, $g = 9.8$ m/ s^2):

$\tan\theta = \frac{(14\sqrt{3})^2}{20\sqrt{3}\times 9.8}$

$\tan\theta = \frac{196\times 3}{196\sqrt{3}}$

$\tan\theta = \frac{3}{\sqrt{3}}$

$\tan\theta = \sqrt{3}$

Since $\tan(60°) = \sqrt{3}$:

$\theta = 60°$

7. **Answer :** A) $\sqrt{\frac{g\tan\theta}{r}}, 2\pi\sqrt{\frac{r}{g\tan\theta}}$

For a conical pendulum with string at an angle θ to the vertical, balancing forces vertically and horizontally gives:

$T\cos\theta = mg$

$T\sin\theta = m\omega^2 r$

Dividing the two equations:

$\tan\theta = \frac{\omega^2 r}{g}$

Solving for angular speed ω:

$\omega = \sqrt{\frac{g\tan\theta}{r}}$

The time period T_p is related to angular speed by $T_p = \frac{2\pi}{\omega}$:

$T_p = 2\pi\sqrt{\frac{r}{g\tan\theta}}$

8. Answer : A) $\sqrt{2}$

The time period of a conical pendulum of length l making an angle θ with the vertical is:

$T = 2\pi\sqrt{\frac{l\cos\theta}{g}}$

This implies that $T \propto \sqrt{\cos\theta}$.

Let's find the square of the ratio to match typical physics problem conventions where an outer square root might have been omitted in the options:

$\frac{T_1^2}{T_2^2} = \frac{\cos(45°)}{\cos(60°)}$

$\frac{T_1^2}{T_2^2} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}}$

$\frac{T_1^2}{T_2^2} = \frac{2}{\sqrt{2}} = \sqrt{2}$

Based on the options provided, the closest mathematical relation intended is $\sqrt{2}$.

9. Answer : A, B, C

In uniform circular motion (constant angular velocity), the speed (magnitude of velocity) remains constant.

Since speed and radius are constant, the magnitude of the centripetal acceleration ($a = \frac{v^2}{r}$) is also constant.

Correspondingly, the magnitude of the centripetal force ($F = \frac{mv^2}{r}$) remains constant.

However, speed does not vary.

10. Answer : A, B, C, D

Analyzing the components of acceleration:

A) $a_c = 0, a_t = 0 \Longrightarrow a = 0$. No acceleration means uniform straight-line translation.

B) $a_c = 0, a_t \neq 0 \Longrightarrow a = a_t$. Acceleration along the path but no turning means accelerated straight-line translation.

C) $a_c \neq 0, a_t = 0 \Longrightarrow a = a_c$. Turning without changing speed means uniform circular motion.

D) $a_c \neq 0, a_t \neq 0 \Longrightarrow a = \sqrt{a_c^2 + a_t^2}$. Both turning and changing speed mean non-uniform circular motion.

11. Answer : A, B, D

A) Correct. In an inertial frame, a real centripetal force acts towards the center to cause circular motion.

B) Correct. In the rotating (non-inertial) frame, a pseudo centrifugal force appears to act outwards.

C) Incorrect. In non-uniform circular motion, the speed v varies, so the magnitude $\frac{mv^2}{r}$ varies.

D) Correct. Because speed varies in non-uniform circular motion, the required centripetal force varies at every instant.

12. Answer : A, B

A) Correct. If the velocity at the lowest point is $\sqrt{5gr}$, it exactly possesses the critical energy to reach the top with $v = \sqrt{gr}$, ensuring water does not spill.

B) Correct. The instantaneous centripetal force requirement is always $\frac{mv^2}{r}$.

C) Incorrectly phrased. Gravity provides the required centripetal force entirely at the critical top point, but water doesn't spill because it is accelerating downwards at exactly g.

D) Incorrect. Water spills if its velocity is *less* than $\sqrt{gr}$ at the highest point.

13. Answer : A, C, D

A) Correct. The weight Mg is an actual force acting vertically downwards.

B) Incorrect. The horizontal component of the *Tension* ($T \sin\theta$) provides the centripetal force, not a component of weight.

C) Correct. From force balance, $\omega = \sqrt{\frac{g \tan\theta}{r}}$.

D) Correct. The period is $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r}{g \tan\theta}}$.

14. Answer : A

In uniform circular motion, the centripetal force acts towards the center of the circle, while the instantaneous velocity is directed along the tangent.

Therefore, the angle between the force vector and the displacement vector is exactly $90°$.

Work done $W = F \cdot d \cdot \cos(90°) = 0$. Both assertion and reason are true, and the reason explains the assertion.

15. Answer : A

Centrifugal force is not a real physical force arising from fundamental interactions. It is introduced merely to apply Newton's laws in a non-inertial (accelerating/rotating) reference frame.

Such forces are called pseudo forces. Both assertion and reason are true, and the reason is the correct explanation.

16. Answer : A) 0.7m/s

Value of r is 0.05 m instead of 0.5 m

The minimum speed at the highest point to just maintain a circular path is:

$v_{top} = \sqrt{gr}$

Substitute the values (g = 9.8 m/ s^2, r = 0.05 m):

$v_{top} = \sqrt{9.8 \times 0.05}$

$v_{top} = \sqrt{0.49}$

$v_{top} = 0.7$ m/s

17. Answer : B) $0.7\sqrt{5} m/s$

By conservation of mechanical energy, the minimum speed at the lowest point to complete the circle is derived as:

$v_{bot} = \sqrt{5gr}$

We can rewrite this in terms of the velocity at the top ($v_{top} = \sqrt{gr}$):

$v_{bot} = \sqrt{5} \times \sqrt{gr}$

Substitute the calculated $v_{top} = 0.7$ m/s:

$v_{bot} = \sqrt{5} \times 0.7$

$v_{bot} = 0.7\sqrt{5}$ m/s

18. Answer : 58.8 N

Using

$T - mg = \frac{mv^2}{r}$

Since

$v^2 = 5gr,$

$z\frac{mv^2}{r} = 5mg$

Therefore,

$T = 6mg$

$= 6 \times 1 \times 9.8 = 58.8$ N

19. Answer : D) 45°

For a particle forming a conical pendulum, the horizontal acceleration is the centripetal acceleration $a = 10$ m/ s^2

The horizontal force balance is:

$T \sin\theta = ma$

$T \sin\theta = 0.15 \times 10 = 1.5$ N

The vertical force balance is:

$T \cos\theta = mg$

Assuming $g = 10$ m/ s^2

$T \cos\theta = 0.15 \times 10 = 1.5$ N

Dividing the two equations gives:

$\tan\theta = \frac{1.5}{1.5}$

$\tan\theta = 1$

$\theta = 45°$

20. Answer : B) $1.5\sqrt{2}N$

From the horizontal and vertical force components calculated above, we can find the total tension T using the Pythagorean theorem:

$T = \sqrt{(T \sin\theta)^2 + (T \cos\theta)^2}$

$T = \sqrt{(1.5)^2 + (1.5)^2}$

$T = \sqrt{2 \times (1.5)^2}$

$T = 1.5\sqrt{2}$ N

21. Answer : C) $\sqrt{10}m/s$

The formula for centripetal acceleration is:

$a = \frac{v^2}{r}$

the radius was meant to be $r = 1$ m:

$10 = \frac{v^2}{1}$

$v^2 = 10$

$v = \sqrt{10}$ m/s

22. Answer : a-R, b-P, c-Q, d-T

a) A stone being whirled in a horizontal circle is kept in motion by the force exerted by the string, which is tension (R).

b) Earth orbits the sun due to the gravitational force exerted by the sun (P).

c) A car rounding a level curve is kept on its path by the frictional force exerted by the road (Q).

d) An electron in a hydrogen atom orbits due to Coulomb's attraction exerted by the proton (T).

23. Answer : a-R, b-P, c-Q, d-S

a) The formula for centripetal acceleration is $\frac{v^2}{r}$ (R).

b) The formula for centripetal force is $\frac{mv^2}{r}$ (P).

c) The angle of banking without friction is given by $\tan^{-1}\left(\frac{v^2}{rg}\right)$ (Q).

d) The maximum speed of a car on a level circular road with friction is $\sqrt{\mu rg}$ (S).

24. Answer : 7

To just complete a vertical circle, the required velocity at the bottom of the circle is given by:

$v_{bot} = \sqrt{5gl}$

Substitute the given values ($l = 1$ m, $g = 9.8$ m/ s^2):

$v_{bot} = \sqrt{5 \times 9.8 \times 1}$

$v_{bot} = \sqrt{49}$

$v_{bot} = 7$ m/s

25. Answer : Invalid question

The radius r is related to the string length l and vertical angle θ by $r = l \sin\theta$.

If $l = 20$ cm and $r = 20$ cm:

$20 = 20 \sin\theta \Longrightarrow \sin\theta = 1 \Longrightarrow \theta = 90°$

The angular speed for a suspended mass (conical pendulum) is:

$\omega = \sqrt{\frac{g}{l\cos\theta}}$

Since $\theta = 90°$, $\cos(90°) = 0$, causing the theoretical angular speed to approach infinity.

26. Answer : 10

The formula for the maximum safe speed on a flat circular track governed by friction is:

$v_{max} = \sqrt{\mu rg}$

Substitute the given values ($\mu = 0.2$, $r = 50$ m, $g = 10$ m/s^2):

$v_{max} = \sqrt{0.2 \times 50 \times 10}$

$v_{max} = \sqrt{100}$

$v_{max} = 10$ m/s

6. Gravitation

Newton's Law, and Variation of Gravity

LEVEL-I

1. **Answer :** A) $\frac{r}{3}$

Let the mass of the first particle be m and the mass of the second particle be $4m$.

Let the third particle of mass m' be placed at a distance of x from the mass m.

The distance of this third particle from the mass $4m$ will be $r - x$.

The gravitational force exerted by m on m' is given by the universal law of gravitation:

$F_1 = \frac{G \cdot m \cdot m'}{x^2}$

The gravitational force exerted by $4m$ on m' is:

$F_2 = \frac{G \cdot (4m) \cdot m'}{(r-x)^2}$

For the net gravitational force on the third particle to be zero, these two opposing forces must be equal in magnitude:

$F_1 = F_2$

Substitute the force equations into the condition:

$\frac{G \cdot m \cdot m'}{x^2} = \frac{G \cdot (4m) \cdot m'}{(r-x)^2}$

Cancel the common terms G, m, and m' from both sides:

$\frac{1}{x^2} = \frac{4}{(r-x)^2}$

Take the square root of both sides to simplify the equation:

$\frac{1}{x} = \frac{2}{r-x}$

Cross-multiply to solve for x:

$r - x = 2x$

Add x to both sides:

$r = 3x$

Divide by 3:

$x = \frac{r}{3}$

2. **Answer :** B) 2

The acceleration due to gravity on the surface of a planet is given by the formula:

$g = \frac{G \cdot M}{R^2}$

The mass M of a planet can be expressed in terms of its density ρ and volume V:

$M = \text{Volume} \times \text{Density} = \frac{4}{3}\pi \cdot R^3 \cdot \rho$

Substitute this mass into the gravity formula:

$g = \frac{G \cdot \left(\frac{4}{3}\pi \cdot R^3 \cdot \rho\right)}{R^2}$

Simplify the expression:

$g = \frac{4}{3}\pi \cdot G \cdot \rho \cdot R$

This shows that acceleration due to gravity is directly proportional to both density and radius:

$g \propto \rho \cdot R$

Let g_p be the gravity on the planet and g_e be the gravity on Earth.

The problem states that the density is the same ($\rho_p = \rho_e$) and the radius is twice that of Earth ($R_p = 2R_e$).

Set up the ratio of gravity on the planet to that on Earth:

$\frac{g_p}{g_e} = \frac{\rho_p \cdot R_p}{\rho_e \cdot R_e}$

Substitute the given relations into the ratio:

$\frac{g_p}{g_e} = \frac{\rho_e \cdot (2R_e)}{\rho_e \cdot R_e}$

Cancel the common terms ρ_e and R_e:

$\frac{g_p}{g_e} = 2$

3. **Answer :** B) Linear

The acceleration due to gravity at a depth d inside a uniform Earth is given by the formula:

$g_d = g\left(1 - \frac{d}{R}\right)$

Expand the equation:

$g_d = g - \frac{g \cdot d}{R}$

In this equation, g and R are constants.

The relationship between g_d and depth d forms the equation of a straight line ($y = mx + c$).

Therefore, the variation of acceleration due to gravity with depth is linear.

4. **Answer :** C) $\frac{9F}{2}$

The initial gravitational force between two masses m_1 and m_2 separated by a distance r is:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

According to the problem, one mass becomes half ($m_1' = \frac{m_1}{2}$) and the distance becomes one-third ($r' = \frac{r}{3}$).

The new gravitational force F' is:

$F' = \frac{G \cdot m_1' \cdot m_2}{(r')^2}$

Substitute the new values into the formula:

$F' = \frac{G \cdot \left(\frac{m_1}{2}\right) \cdot m_2}{\left(\frac{r}{3}\right)^2}$

Square the new distance term:

$F' = \frac{G \cdot \left(\frac{m_1}{2}\right) \cdot m_2}{\frac{r^2}{9}}$

Rearrange the fractions:

$F' = \frac{1}{2} \cdot 9 \cdot \frac{G \cdot m_1 \cdot m_2}{r^2}$

Group the numerical coefficients:

$F' = \frac{9}{2} \cdot \left(\frac{G \cdot m_1 \cdot m_2}{r^2}\right)$

Substitute the original force F:

$F' = \frac{9F}{2}$

5. **Answer :** A) 2

The acceleration due to gravity at the surface of the Earth is g.

The acceleration due to gravity at a depth d is given by the formula:

$g_d = g\left(1 - \frac{d}{R}\right)$

The problem asks for the ratio at a depth of $d = \frac{R}{2}$.

Substitute this depth into the formula:

$g_d = g\left(1 - \frac{\frac{R}{2}}{R}\right)$

Simplify the fraction inside the parentheses:

$g_d = g\left(1 - \frac{1}{2}\right)$

Calculate the value:

$g_d = g\left(\frac{1}{2}\right)$

$g_d = \frac{g}{2}$

Find the ratio of gravity at the surface to gravity at this depth:

Ratio = $\frac{g}{g_d}$

Substitute the calculated value of g_d:

Ratio = $\frac{g}{\frac{g}{2}}$

Ratio = 2

6. **Answer :** D) Both B and C

The universal law of gravitation states that the force between two bodies is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

The formula is given by:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

This explicitly shows that the force depends on the masses (m_1, m_2) and the distance (r).

The constant G does not change, indicating the force is independent of the medium.

7. **Answer :** B) $g = 0$

The acceleration due to gravity at a distance r from the center of the Earth is given by:

$g = \frac{G \cdot M}{r^2}$

As the distance r approaches infinity ($r \to \infty$), the denominator becomes infinitely large.

$g = \lim_{r \to \infty} \frac{G \cdot M}{r^2}$

Any finite value divided by an infinitely large value approaches zero.

$g = 0$

8. **Answer :** A) Same

The acceleration due to gravity on a planet can be written in terms of its density ρ and radius R:

$g = \frac{4}{3}\pi \cdot G \cdot \rho \cdot R$

This indicates that gravity is directly proportional to the product of density and radius.

$g \propto \rho \cdot R$

The problem states that the new density is double ($\rho' = 2\rho$) and the new radius is half ($R' = \frac{R}{2}$).

The new acceleration due to gravity g' will be:

$g' \propto \rho' \cdot R'$

Substitute the new values:

$g' \propto (2\rho) \cdot \left(\frac{R}{2}\right)$

Multiply the terms together:

$g' \propto \rho \cdot R$

Since the proportional value remains exactly the same as the initial condition, the value of g becomes the same.

9. **Answer :** C) Medium

According to Newton's law of universal gravitation, the formula for gravitational force is:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

The terms in this equation are mass (m), distance (r), and the universal gravitational constant (G).

Because G is a universal constant, it does not change regardless of what material or vacuum separates the two masses.

Therefore, gravitational force is perfectly independent of the intervening medium.

10. **Answer :** A) g doubles

The formula for acceleration due to gravity on the surface of the Earth is:

$g = \frac{G \cdot M}{R^2}$

If the mass of the Earth M becomes double ($M' = 2M$) while the radius R remains the same.

The new acceleration due to gravity g' is:

$g' = \frac{G \cdot M\prime}{R^2}$

Substitute the new mass into the formula:

$g' = \frac{G \cdot (2M)}{R^2}$

Pull the constant factor to the front:

$g' = 2 \cdot \left(\frac{G \cdot M}{R^2}\right)$

Substitute the original gravity expression:

$g' = 2g$

11. **Answer :** D) $F = \frac{GMm}{R^2}$

According to Newton's law of universal gravitation, every particle attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

For Earth (mass M) and an object (mass m) separated by distance R, the formula is mathematically represented as:

$F = \frac{G \cdot M \cdot m}{R^2}$

12. **Answer :** C) Gravitational force

Water naturally flows from a higher elevation (hills) to a lower elevation (sea) due to the attractive force exerted by the Earth on the mass of the water.

This attractive force exerted by the Earth on any body is called gravity or gravitational force.

13. **Answer :** B) 6.6×10^{-8} dyne $cm^2 g^{-2}$

The value of the universal gravitational constant G in the SI system is:

$G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2 \cdot kg^{-2}}$

To convert to the CGS system, we must use the following conversions:

$1\ \mathrm{N} = 10^5$ dyne

$1\ \mathrm{m} = 10^2\ \mathrm{cm} \Longrightarrow 1\ \mathrm{m^2} = 10^4\ \mathrm{cm^2}$

$1\ \mathrm{kg} = 10^3\ \mathrm{g} \Longrightarrow 1\ \mathrm{kg^{-2}} = 10^{-6}\ \mathrm{g^{-2}}$

Substitute these conversions into the SI value of G:

$G = (6.67 \times 10^{-11}) \cdot (10^5\ \text{dyne}) \cdot \left(10^4\ \mathrm{cm^2}\right) \cdot \left(10^{-6}\ \mathrm{g^{-2}}\right)$

Combine the powers of 10:

$G = 6.67 \times 10^{-11+5+4-6}\ \text{dyne} \cdot \mathrm{cm^2} \cdot \mathrm{g^{-2}}$

$G = 6.67 \times 10^{-8}\ \text{dyne} \cdot \mathrm{cm^2} \cdot \mathrm{g^{-2}}$

Note: The source document options slightly round this to 6.6×10^{-8}.

14. **Answer :** B) Gravity

The Earth's atmosphere is composed of various gas molecules.

Because these gas molecules have mass, they are subjected to the attractive pull of the Earth.

This attractive force exerted by the Earth on any body is gravity.

It prevents the gas molecules from escaping into space, holding the atmosphere to the Earth.

15. **Answer :** C) $g = \frac{GM}{R^2}$

The weight of an object on the surface of the Earth is the gravitational force acting on it.

$W = m \cdot g$

The gravitational force according to Newton's law is:

$F = \frac{G \cdot M \cdot m}{R^2}$

Equate the two expressions for the force on the body:

$m \cdot g = \frac{G \cdot M \cdot m}{R^2}$

Cancel the mass of the object m from both sides:

$g = \frac{G \cdot M}{R^2}$

16. **Answer :** A) $g\left(1 - \frac{h}{R}\right)$

The value of acceleration due to gravity at the surface of the Earth is g.

As one moves towards the center of the Earth to a depth h, the effective mass of the Earth pulling the object inwards decreases.

The derivation using a uniform density model yields the standard formula for gravity at depth h:

$g_h = g\left(1 - \frac{h}{R}\right)$

17. **Answer :** (B) Decreases

The acceleration due to gravity at a distance r from the center of the Earth (where $r > R$) is:

$g = \frac{G \cdot M}{r^2}$

This shows an inverse square relationship with distance.

As we go away from the surface, the distance r increases.

Therefore, the acceleration due to gravity must decrease.

18. **Answer :** C) $g\left(1 - \frac{2h}{R}\right)$

The formula for acceleration due to gravity at a height h above the Earth's surface is:

$g_h = \frac{G \cdot M}{(R+h)^2}$

Factor out R from the denominator:

$g_h = \frac{G \cdot M}{R^2\left(1+\frac{h}{R}\right)^2}$

Substitute $g = \frac{G \cdot M}{R^2}$:

$g_h = g\left(1+\frac{h}{R}\right)^{-2}$

For $h < R$, we can use the binomial approximation $(1+x)^n \approx 1 + nx$:

$g_h = g\left(1-\frac{2h}{R}\right)$

Note: There is a typo in the source document's option letters where C comes before B. The correct mathematical expression matches the position labeled C.

19. **Answer :** A) $\frac{2}{3}m$

Let the mass of point A be $m_A = 4k$ and the mass of point B be $m_B = 3k$.

The distance between A and B is 1 m.

Let mass C (mass M) be placed at a distance x from mass A.

The distance of mass C from mass B will be $1 - x$.

The gravitational force between A and C is:

$F_{AC} = \frac{G \cdot m_A \cdot M}{x^2} = \frac{G \cdot (4k) \cdot M}{x^2}$

The gravitational force between B and C is:

$F_{BC} = \frac{G \cdot m_B \cdot M}{(1-x)^2} = \frac{G \cdot (3k) \cdot M}{(1-x)^2}$

The problem states that the force between A and C is $\frac{1}{3}$ of the force between B and C.

$F_{AC} = \frac{1}{3}F_{BC}$

Substitute the force expressions into the given relation:

$\frac{G \cdot (4k) \cdot M}{x^2} = \frac{1}{3} \cdot \frac{G \cdot (3k) \cdot M}{(1-x)^2}$

Cancel the common terms G, k, and M from both sides:

$\frac{4}{x^2} = \frac{1}{3} \cdot \frac{3}{(1-x)^2}$

Simplify the right side:

$\frac{4}{x^2} = \frac{1}{(1-x)^2}$

Take the square root of both sides:

$\frac{2}{x} = \frac{1}{1-x}$

Cross-multiply:

$2 \cdot (1-x) = x$

$2 - 2x = x$

$2 = 3x$

$x = \frac{2}{3}$ m

LEVEL-II

1. **Answer :** A) 4.17×10^{42}

Note: There is a typo in the source document for the charge of an electron; it states 1.6×10^{19} C but should be 1.6×10^{-19} C. The calculation below uses the correct physical value to arrive at the intended answer.

The electric force between two electrons separated by a distance r is:

$F_e = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2}$

Substitute the given constants:

$F_e = (9 \times 10^9) \cdot \frac{(1.6\times10^{-19})^2}{r^2}$

Calculate the squared charge:

$F_e = \frac{9\times10^9 \cdot 2.56\times10^{-38}}{r^2}$

$F_e = \frac{23.04\times10^{-29}}{r^2}$

The gravitational force between two electrons is:

$F_g = G \cdot \frac{m_e^2}{r^2}$

Substitute the given constants:

$F_g = (6.67 \times 10^{-11}) \cdot \frac{(9.1\times10^{-31})^2}{r^2}$

Calculate the squared mass:

$F_g = \frac{6.67\times10^{-11} \cdot 82.81\times10^{-62}}{r^2}$

$F_g = \frac{552.3427\times10^{-73}}{r^2}$

Now, calculate the ratio of the electric force to the gravitational force:

Ratio $= \frac{F_e}{F_g}$

Substitute the simplified force expressions (the r^2 terms cancel out):

Ratio $= \frac{23.04\times10^{-29}}{552.3427\times10^{-73}}$

Calculate the numerical division:

Ratio $\approx 0.04171 \times 10^{-29-(-73)}$

Ratio $\approx 0.04171 \times 10^{44}$

Adjust the decimal to standard scientific notation:

Ratio $\approx 4.17 \times 10^{42}$

2. **Answer :** C) $5 \times 10^{-8} N$

The gravitational force between two masses is given by the formula:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

Given values:

$m_1 = 50$ kg

$m_2 = 60$ kg

$r = 2$ m

$G = 6.67 \times 10^{-11}$ N $\cdot$ m^2/kg^2

Substitute the values into the formula:

$F = \frac{6.67 \times 10^{-11} \cdot 50 \cdot 60}{2^2}$

Calculate the numerator product:

$F = \frac{6.67 \times 10^{-11} \cdot 3000}{4}$

Divide 3000 by 4:

$F = 6.67 \times 10^{-11} \cdot 750$

Multiply the numbers:

$F = 5002.5 \times 10^{-11}$

Convert to standard scientific notation:

$F = 5.0025 \times 10^{-8}$ N

Rounding to a single digit as per the options:

$F \approx 5 \times 10^{-8}$ N

3. **Answer :** D) Anywhere in the universe

Newton's law of gravitation is a universal law.

The universal gravitational constant G does not change with location, time, or medium.

Therefore, the law applies to all objects with mass, anywhere in the universe.

4. **Answer :** D) Both (A) and (C)

According to Newton's law of gravitation, the force of attraction is given by:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

This formula shows that force is directly proportional to the product of the masses ($m_1 \cdot m_2$).

It also shows that force is inversely proportional to the square of the distance between them (r^2).

5. **Answer :** D) Gravitational force

Electrostatic and magnetic forces can be both attractive and repulsive depending on the charges or poles involved.

Frictional force is a resistive force that opposes relative motion.

Gravitational force, being based entirely on mass, is fundamentally always an attractive force.

6. **Answer :** D) $g_1 : g_2 = R_1\rho_1 : R_2\rho_2$

The acceleration due to gravity on the surface of a planet is given by:

$g = \frac{G \cdot M}{R^2}$

Substitute mass M = Volume × Density = $\frac{4}{3}\pi \cdot R^3 \cdot \rho$:

$g = \frac{G \cdot \left(\frac{4}{3}\pi \cdot R^3 \cdot \rho\right)}{R^2}$

Simplify the formula:

$g = \frac{4}{3}\pi \cdot G \cdot R \cdot \rho$

This establishes the direct proportionality $g \propto R \cdot \rho$.

For two planets, the ratio of their accelerations due to gravity is:

$\frac{g_1}{g_2} = \frac{R_1 \cdot \rho_1}{R_2 \cdot \rho_2}$

This matches the ratio $R_1\rho_1 : R_2\rho_2$.

7. **Answer :** A) Mass and acceleration due to gravity

The weight of a body is defined as the force with which it is attracted towards the center of a celestial body (like the Earth).

According to Newton's second law of motion ($F = m \cdot a$), in the context of gravity, the acceleration is g.

Therefore, Weight $W = m \cdot g$, which is the product of mass and acceleration due to gravity.

8. **Answer :** C) One-fourth of the initial force

The initial gravitational force is given by:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

When the distance r is doubled, the new distance is $r' = 2r$.

The new gravitational force F' becomes:

$F' = \frac{G \cdot m_1 \cdot m_2}{(2r)^2}$

Expand the squared term in the denominator:

$F' = \frac{G \cdot m_1 \cdot m_2}{4r^2}$

Factor out the $\frac{1}{4}$:

$F' = \frac{1}{4} \cdot \left(\frac{G \cdot m_1 \cdot m_2}{r^2}\right)$

Substitute the initial force F:

$F' = \frac{1}{4}F$

9. **Answer :** D) None of these

The value G is the universal gravitational constant.

By definition, a universal constant remains exactly the same everywhere and is entirely independent of the masses, sizes, or nature of the bodies involved.

10. **Answer :** B) Its weight will change.

Mass is the quantity of matter contained in a body.

The mass of an object is an intrinsic property and remains perfectly constant regardless of location.

Weight is the attractive force exerted by the earth (or any celestial body) on anybody.

Since the acceleration due to gravity on Earth is different from that on the Moon, the weight $W = m \cdot g$ will change.

11. **Answer :** A) $9.8\,m/s^2$

The acceleration due to gravity on Earth is:

$g_e = \frac{G \cdot M}{R^2} = 9.8\ \text{m/s}^2$

The planet has a mass four times that of Earth:

$M_p = 4M$

The planet has a diameter twice that of Earth, which means its radius is also twice:

$R_p = 2R$

Calculate the acceleration due to gravity on the planet g_p:

$g_p = \frac{G \cdot M_p}{R_p^2}$

Substitute the expressions for the planet's mass and radius:

$g_p = \frac{G \cdot (4M)}{(2R)^2}$

Expand the denominator:

$g_p = \frac{G \cdot 4M}{4R^2}$

Cancel the 4 from the numerator and denominator:

$g_p = \frac{G \cdot M}{R^2}$

This is exactly equal to the gravity on Earth:

$g_p = g_e = 9.8\ \text{m/s}^2$

12. **Answer :** A) Will increase

As the earth rotates, a body placed on its surface moves along the circular path and hence experiences a centrifugal force.

This reduces the apparent gravity. The formula for effective gravity at latitude λ due to Earth's rotation (angular velocity ω) is:

$g' = g - R \cdot \omega^2 \cdot \cos^2(\lambda)$

At the equator, latitude $\lambda = 0$, so $\cos(0) = 1$. The formula becomes:

$g'_{equator} = g - R \cdot \omega^2$

If the Earth stops rotating, its angular velocity ω becomes zero.

Substitute $\omega = 0$ into the formula:

$g'_{equator} = g - R \cdot (0)^2 = g$

Since the subtractive centrifugal term $R\omega^2$ disappears, the measured acceleration due to gravity g will increase.

13. **Answer :** D) Increase by 2%

The relationship between acceleration due to gravity and radius is:

$g = \frac{G \cdot M}{R^2}$

Take the natural logarithm of both sides:

$\ln(g) = \ln(G \cdot M) - 2 \cdot \ln(R)$

Differentiate both sides to find the fractional change, keeping mass M constant:

$\frac{\Delta g}{g} = 0 - 2 \cdot \frac{\Delta R}{R}$

The problem states the radius is decreased by 1%, so the fractional change is $\frac{\Delta R}{R} = -1\%$.

Substitute this into the derivative equation:

$\frac{\Delta g}{g} = -2 \cdot (-1\%)$

Calculate the result:

$\frac{\Delta g}{g} = +2\%$

A positive sign indicates an increase.

14. **Answer :** A) 3%

Using the same derived formula relating fractional changes in gravity and radius:

$\frac{\Delta g}{g} = -2 \cdot \frac{\Delta R}{R}$

The problem states the radius shrinks by 1.5%, which means $\frac{\Delta R}{R} = -1.5\%$.

Substitute this value into the equation:

$\frac{\Delta g}{g} = -2 \cdot (-1.5\%)$

Calculate the final percentage change:

$\frac{\Delta g}{g} = +3\%$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

15. **Answer :** A, B, C

The formula for acceleration due to gravity on the surface of a planet is:

$g = \frac{G \cdot M}{R^2}$

This explicitly shows that g depends on the mass of the planet (M) and the radius of the planet (R).

We can also express the mass in terms of density (ρ) and volume:

$g = \frac{G \cdot \left(\frac{4}{3}\pi \cdot R^3 \cdot \rho\right)}{R^2} = \frac{4}{3}\pi \cdot G \cdot R \cdot \rho$

This formulation shows that g also depends on the density of the planet.

Therefore, factors A, B, and C are all correct. It does not depend on the mass of the object.

16. **Answer :** A, B, D

Gravitational force is a mutually attractive force between any two masses, making statement A correct.

The work done by the gravitational force in moving an object between two points is independent of the path taken, making it a conservative force, so statement B is correct.

Gravitational force acts along the straight line joining the centers of the two masses, making statement D correct.

It is independent of the intervening medium, so statement C is false.

17. **Answer :** A, B, C

The universal gravitational constant G has an experimentally measured value of approximately $6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$, making statement A correct.

By definition, a universal constant does not depend on the nature or size of the interacting bodies, making statement B correct.

It also acts entirely independent of the surrounding medium, making statement C correct.

Because it is universally constant, it does not vary with position, rendering statement D false.

18. **Answer :** A, B, C

According to Newton's Third Law, the gravitational force exerted by mass 1 on mass 2 is equal and opposite to the force exerted by mass 2 on mass 1, meaning they form an action-reaction pair. Statement A is correct.

The gravitational constant G is a scalar magnitude, whereas the acceleration due to gravity g indicates a direction towards the center of mass, making it a vector. Statement B is correct.

Both G (via Cavendish experiment) and g (via pendulum or freefall) are determined experimentally. Statement C is correct.

While G is universal and constant, g varies depending on depth, height, and planetary mass/radius. Statement D is false.

ASSERTION AND REASON TYPE

19. **Answer :** A) Both A and R are true, R is correct explanation of A

At the exact center of the Earth, a body is surrounded uniformly by the Earth's mass in all directions.

The gravitational pull from any arbitrary small mass element of the Earth is perfectly balanced by an equal mass element situated exactly diametrically opposite to it.

Consequently, all these individual gravitational forces vectorially cancel each other out, resulting in a net gravitational force of zero.

Since Weight $W = F_g$, the weight of the body becomes zero. Thus, both statements are true and the reason correctly explains the assertion.

20. **Answer :** A) Both A and R are true, R is correct explanation of A

The Earth is not a perfect sphere; it is an oblate spheroid, meaning it is flattened at the poles and bulging at the equator.

Therefore, the radius R is minimum at the poles, making the base gravitational pull $g = \frac{GM}{R^2}$ mathematically maximum.

Additionally, as the earth rotates, a body placed on its surface moves along the circular path and hence experiences a centrifugal force.

The centrifugal force formula is $F_c = m \cdot R \cdot \omega^2 \cdot \cos(\lambda)$.

At the poles, the latitude $\lambda = 90°$, making $\cos(90°) = 0$, which results in a zero centrifugal force opposing gravity. Both statements are correct, and the reason is the valid explanation.

MATRIX MATCHING TYPE

21. **Answer :** (A) - (P), (B) - (Q), (C) - (R), (D) - (S)

The formula for acceleration due to gravity at a height h is $g_h = \frac{GM}{(R+h)^2}$, which clearly decreases proportional to $\frac{1}{(R+h)^2}$.

The formula for acceleration due to gravity at depth is $g_d = g\left(1 - \frac{d}{R}\right)$, which describes a strictly linear decrease with depth d.

At the exact center of a planet, the perfectly symmetrical distribution of mass causes all gravitational forces to cancel, rendering gravity to be identically zero.

For any given planet, the acceleration due to gravity reaches its absolute maximum magnitude exactly at the surface (before depth reduces it and height reduces it).

22. **Answer :** (A) - (P), (B) - (Q), (C) - (R), (D) - (S)

The mathematical expression for gravity at a small height h (where $h \ll R$) is derived as $g = g_0\left(1 - \frac{2h}{R}\right)$.

The mathematical expression for gravity at a depth d inside the planet is formally defined as $g = g_0\left(1 - \frac{d}{R}\right)$.

At the planet's geometric center, where $d = R$, substituting into the depth formula yields $g = g_0(1 - 1) = 0$.

Because the polar radius is the smallest distance to the Earth's center and lacks opposing centrifugal force, the gravitational acceleration is strictly maximum at the poles compared to the equator.

COMPREHENSION TYPE

23. **Answer :** B) Zero

Let the two equal masses be m and they be situated at coordinates $-x$ and $+x$.

The third body of mass M is placed exactly midway between them, which is at the origin (coordinate 0).

The gravitational force exerted by the left mass on M pulls it to the left with magnitude $F_{left} = \frac{G \cdot m \cdot M}{x^2}$.

The gravitational force exerted by the right mass on M pulls it to the right with magnitude $F_{right} = \frac{G \cdot m \cdot M}{x^2}$.

Because these two forces are equal in magnitude but perfectly opposite in direction, their vector sum is exactly zero.

24. **Answer :** B) Towards larger mass

If one of the masses is doubled, it becomes $2m$.

Assume the right mass is doubled. The force pulling to the right becomes $F_{right} = \frac{G \cdot (2m) \cdot M}{x^2}$.

The force pulling to the left remains $F_{left} = \frac{G \cdot m \cdot M}{x^2}$.

The net force is $F_{net} = F_{right} - F_{left}$.

Substitute the expressions:

$F_{net} = \frac{G \cdot (2m) \cdot M}{x^2} - \frac{G \cdot m \cdot M}{x^2} = \frac{G \cdot m \cdot M}{x^2}$

The net force is distinctly non-zero and points in the direction of the stronger pull, which corresponds to the larger mass.

25. **Answer :** B) One-fourth

The gravitational force between two point masses is governed by the inverse square law:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

When the distance between the acting masses is doubled, the new distance becomes $r' = 2r$.

Substitute this into the force equation:

$F' = \frac{G \cdot m_1 \cdot m_2}{(2r)^2}$

Expand the denominator:

$F' = \frac{G \cdot m_1 \cdot m_2}{4r^2}$

Factor out the constant:

$F' = \frac{1}{4} \cdot F$

26. **Answer :** B) Poles

Note: The source document contains a typo by listing the set of options A, B, C, D twice for this single question.

True gravitational acceleration is purely defined by Newton's law:

$g = \frac{G \cdot M}{R^2}$

Because the Earth is oblate, its polar radius (R_{pole}) is significantly smaller than its equatorial radius ($R_{equator}$).

Since gravity is inversely proportional to the square of the radius, a smaller radius yields a larger gravitational pull.

Consequently, true gravitational acceleration is maximum where the radius is minimum, which is at the poles.

27. **Answer :** B) Equator

Apparent weight reflects the net force experienced by a body, accounting for Earth's rotation.

As the earth rotates, a body placed on its surface moves along the circular path and hence experiences a centrifugal force. Due to it, the apparent weight of the body decreases.

The effective gravity formula is $g' = g - R \cdot \omega^2 \cdot \cos^2(\lambda)$.

At the equator ($\lambda = 0°$), $\cos(0°) = 1$, making the centrifugal subtraction term $R \cdot \omega^2$ maximum.

Additionally, the equatorial radius is maximum, making the true base gravity g minimum there.

Combining both a minimal base pull and a maximal centrifugal reduction makes the apparent weight strictly minimum at the equator.

INTEGER TYPE

28. Answer : 9

The initial gravitational force between two masses separated by distance r is:

$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$

When the distance becomes 3 times the original, the new distance is $r' = 3r$.

The new force F' is given by:

$F' = \frac{G \cdot m_1 \cdot m_2}{(3r)^2}$

Expand the denominator term:

$F' = \frac{G \cdot m_1 \cdot m_2}{9r^2}$

Factor out the scalar fraction:

$F' = \frac{1}{9} \cdot \left(\frac{G \cdot m_1 \cdot m_2}{r^2}\right)$

Substitute the initial force F:

$F' = \frac{1}{9} \cdot F$

The problem specifies the new force is $\frac{1}{x}$ times the original force.

Equating the fractions, we conclude $x = 9$.

29. Answer : 3

The acceleration due to gravity on a uniform spherical planet can be calculated using its density:

$g = \frac{G \cdot (\text{Volume} \cdot \text{Density})}{R^2}$

$g = \frac{G \cdot \left(\frac{4}{3}\pi \cdot R^3 \cdot \rho\right)}{R^2}$

Simplify the expression to show the dependencies:

$g = \frac{4}{3}\pi \cdot G \cdot R \cdot \rho$

This shows that g is directly proportional to both radius and density.

The problem states that both planets have the same radius, so we can ignore R when finding ratios.

The ratio of their gravities purely depends on the ratio of their densities:

$\frac{g_B}{g_A} = \frac{\rho_B}{\rho_A}$

We are given that the density of planet B is 3 times that of planet A ($\rho_B = 3\rho_A$).

Substitute this into the ratio:

$\frac{g_B}{g_A} = \frac{3\rho_A}{\rho_A}$

Cancel the common term to find the integer answer:

$\frac{g_B}{g_A} = 3$

30. **Answer :** 4

The formula relating acceleration due to gravity, mass, and radius is:

$g = \frac{G \cdot M}{R^2}$

Using the method of small fractional approximations from calculus, we take the natural logarithm of the equation:

$\ln(g) = \ln(G \cdot M) - 2\ln(R)$

Differentiate both sides. Since mass M and constant G do not change, their derivative is zero:

$\frac{\Delta g}{g} = -2 \cdot \frac{\Delta R}{R}$

Multiply both sides by 100 to convert to percentage changes:

% change in $g = -2 \cdot$ (% change in R)

The problem specifies the radius decreases by 2%, so the percentage change in R is -2%.

Substitute this value into the differential equation:

% change in $g = -2 \cdot (-2\%)$

Calculate the multiplication:

% change in $g = +4\%$

The positive sign indicates an increase. The approximate percentage increase is 4.

Escape Velocity, Satellites and Kepler's Laws

1. **Answer :** C) $6.4 \times 10^6 m$

The standard accepted value for the radius of the Earth is approximately 6400 km.

Convert this distance into meters to match the options.

$R = 6400$ km

$R = 6400 \times 10^3$ m

$R = 6.4 \times 10^6$ m

2. **Answer :** A) It sweeps equal areas in equal intervals of time.

This directly describes Kepler's second law of planetary motion, also known as the Law of Equal Areas.

The law states that a line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

3. **Answer :** B) All planets move in elliptical orbits, with the sun situated at one of the foci of the ellipse.

This is the formal definition of Kepler's first law of planetary motion, often called the Law of Orbits.

The Sun rests at one of the two focal points of the elliptical path, not the Earth.

4. **Answer :** B) 15.8 km/s

The formula for the escape velocity of a planet is:

$v_e = \sqrt{\frac{2GM}{R}}$

If the Earth's radius becomes half, the new radius is $R' = \frac{R}{2}$.

Substitute the new radius into the formula to find the new escape velocity v_e':

$v_e' = \sqrt{\frac{2GM}{\frac{R}{2}}}$

$v_e' = \sqrt{2 \cdot \frac{2GM}{R}}$

$v_e' = \sqrt{2} \cdot \sqrt{\frac{2GM}{R}}$

$v_e' = \sqrt{2} \cdot v_e$

Substitute the known original escape velocity $v_e = 11.2$ km/s:

$v_e' = 1.414 \cdot 11.2$

$v_e' = 15.8368$ km/s

Rounding to one decimal place gives 15.8 km/s.

5. **Answer :** B) $\sqrt{2}$

The escape velocity from the Earth's surface is given by the formula:

$v_e = \sqrt{2gR}$

The orbital velocity of a satellite very close to the Earth's surface is given by:

$v_o = \sqrt{gR}$

Calculate the ratio of the escape velocity to the orbital velocity:

Ratio = $\frac{v_e}{v_o}$

Ratio = $\frac{\sqrt{2gR}}{\sqrt{gR}}$

Ratio = $\sqrt{2}$

6. **Answer :** C) C

According to Kepler's second law, a planet's angular momentum is conserved.

$m \cdot v \cdot r =$ constant

This means velocity v is inversely proportional to distance r.

The kinetic energy is $\frac{1}{2}mv^2$, which means kinetic energy is minimum when velocity is minimum.

Velocity is minimum when the distance r from the Sun is maximum.

In the given figure, point C represents the aphelion, the point farthest from the Sun.

7. **Answer :** B) $-\frac{L^2}{2mr^2}$

The angular momentum L of a satellite of mass m in a circular orbit of radius r is:

$L = m \cdot v \cdot r$

Solve for velocity v:

$v = \frac{L}{m \cdot r}$

The kinetic energy K of the satellite is:

$K = \frac{1}{2} \cdot m \cdot v^2$

Substitute the expression for velocity:

$K = \frac{1}{2} \cdot m \cdot \left(\frac{L}{m \cdot r}\right)^2$

$K = \frac{1}{2} \cdot m \cdot \frac{L^2}{m^2 \cdot r^2}$

$K = \frac{L^2}{2 \cdot m \cdot r^2}$

For a bound satellite in a circular orbit, the total mechanical energy E is the negative of the kinetic energy:

$E = -K$

$E = -\frac{L^2}{2mr^2}$

8. **Answer :** A) 1:2

The orbital velocity v of a satellite in a circular orbit is:

$v = \sqrt{\frac{GM}{r}}$

The angular momentum L is:

$L = m \cdot v \cdot r$

Substitute the velocity into the angular momentum formula:

$L = m \cdot \left(\sqrt{\frac{GM}{r}}\right) \cdot r$

$L = m \cdot \sqrt{GM \cdot r}$

This shows that angular momentum is directly proportional to the square root of the orbital radius:

$L \propto \sqrt{r}$

Set up the ratio for the two different radii $r_1 = r$ and $r_2 = 4r$:

$\frac{L_1}{L_2} = \frac{\sqrt{r_1}}{\sqrt{r_2}}$

$\frac{L_1}{L_2} = \frac{\sqrt{r}}{\sqrt{4r}}$

$\frac{L_1}{L_2} = \frac{1}{2}$

9. **Answer :** B) 6R

The orbital radius r is measured from the center of the Earth.

It is the sum of the Earth's radius R and the height h above the Earth's surface.

$r = R + h$

We are given that the orbital radius is $7R$.

$7R = R + h$

Solve for h:

$h = 7R - R$

$h = 6R$

10. **Answer :** B) T^2R^{-3} = constant

Kepler's third law of planetary motion relates the time period T and the mean distance R.

$T^2 \propto R^3$

Introduce a constant of proportionality:

$T^2 = \text{constant} \cdot R^3$

Rearrange the equation to isolate the constant:

$\frac{T^2}{R^3} = \text{constant}$

Write it with a negative exponent:

$T^2R^{-3} = \text{constant}$

11. **Answer :** C) Greatest when it is closest to the sun.

According to Kepler's second law (law of equal areas), the angular momentum of a planet is conserved.

This necessitates that the planet moves faster when it is near the sun (perihelion) and slower when it is farther away (aphelion).

Therefore, its speed is greatest when it is closest to the Sun.

12. **Answer :** B) $\sqrt{gR}$

The potential energy of a particle of mass m at a height $h = R$ from the surface (distance $2R$ from the center) is:

$U = -\frac{GMm}{R+R}$

$U = -\frac{GMm}{2R}$

To escape, the total mechanical energy must be at least zero. The required kinetic energy K is:

$K + U = 0$

$\frac{1}{2}mv_e^2 - \frac{GMm}{2R} = 0$

$\frac{1}{2}mv_e^2 = \frac{GMm}{2R}$

Cancel m and the factor of $\frac{1}{2}$ from both sides:

$v_e^2 = \frac{GM}{R}$

We know that acceleration due to gravity at the surface is $g = \frac{GM}{R^2}$, which gives $GM = gR^2$.

Substitute this into the velocity equation:

$v_e^2 = \frac{gR^2}{R}$

$v_e^2 = gR$

$v_e = \sqrt{gR}$

13. **Answer :** B) $F_c = \frac{mv^2}{R}$

Centripetal force is the net force required to keep an object of mass m moving in a circular path of radius R with a linear speed v.

The standard formula derived from Newton's second law is $F_c = m \cdot a_c$, where $a_c = \frac{v^2}{R}$.

$F_c = \frac{mv^2}{R}$

14. **Answer :** D) $5^{\frac{5}{2}}$ years

Note: There appears to be a slight typo in the question's provided options. The correct mathematical relationship derived below results in $5^{3/2}$ years, but we will demonstrate the standard calculation.

According to Kepler's third law:

$T^2 \propto R^3$

Let T_E and R_E be the time period and orbital radius of the Earth.

$T_E = 1$ year

Let T_P and R_P be the time period and orbital radius of the planet.

$R_P = 5 \cdot R_E$

Set up the ratio:

$\left(\frac{T_P}{T_E}\right)^2 = \left(\frac{R_P}{R_E}\right)^3$

Substitute the known values:

$\left(\frac{T_P}{1}\right)^2 = \left(\frac{5R_E}{R_E}\right)^3$

$T_P^2 = 5^3$

Take the square root of both sides:

$T_P = 5^{\frac{3}{2}}$ years

15. **Answer :** B) $\frac{v_e}{\sqrt{2}}$

The initial kinetic energy required is $\frac{1}{2}mv^2$, and initial potential energy on the surface is $-\frac{GMm}{R}$.

At the maximum height $h = R$, the velocity is zero. The distance from the center is $2R$.

The final potential energy is $-\frac{GMm}{2R}$.

By conservation of energy:

$\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{2R}$

$\frac{1}{2}mv^2 = \frac{GMm}{R} - \frac{GMm}{2R}$

$\frac{1}{2}mv^2 = \frac{GMm}{2R}$

Multiply both sides by 2 and divide by m:

$v^2 = \frac{GM}{R}$

The escape velocity formula squared is $v_e^2 = \frac{2GM}{R}$.

Therefore, $\frac{GM}{R} = \frac{v_e^2}{2}$.

Substitute this back into our equation:

$v^2 = \frac{v_e^2}{2}$

$v = \frac{v_e}{\sqrt{2}}$

16. **Answer :** B) Decreases

Perigee is the point in a satellite's elliptical orbit where it is closest to the Earth.

Apogee is the point where it is farthest from the Earth.

According to the conservation of angular momentum, the satellite moves fastest at perigee and slowest at apogee.

As it moves from perigee to apogee, its speed decreases, which causes its kinetic energy to decrease.

17. **Answer :** B) 2 times

The formula for escape velocity in terms of density ρ and radius R is derived from $v_e = \sqrt{\frac{2GM}{R}}$.

Substitute mass $M = \frac{4}{3}\pi R^3 \rho$:

$v_e = \sqrt{\frac{2G}{R} \cdot \left(\frac{4}{3}\pi R^3 \rho\right)}$

$v_e = \sqrt{\frac{8}{3}\pi G \rho R^2}$

$v_e = R\sqrt{\frac{8}{3}\pi G \rho}$

This shows that v_e is directly proportional to radius R when density ρ is constant.

Since the planet has the same density but twice the radius, its escape velocity is exactly 2 times that of Earth.

18. **Answer :** B) $\frac{r}{4}$

Note: Option B's text was partially omitted in the OCR, but logically and mathematically, the correct option must represent $\frac{r}{4}$.

The orbital velocity v of a satellite is given by:

$v = \sqrt{\frac{GM}{r}}$

This means v is inversely proportional to the square root of the radius:

$v \propto \frac{1}{\sqrt{r}}$

Square both sides:

$r \propto \frac{1}{v^2}$

If the velocity is doubled ($v' = 2v$), then:

$r' \propto \frac{1}{(2v)^2}$

$r' \propto \frac{1}{4v^2}$

$r' = \frac{r}{4}$

19. **Answer :** A) -K

For a satellite in a circular orbit, the kinetic energy is $K = \frac{GMm}{2r}$.

The potential energy is $U = -\frac{GMm}{r}$.

This relationship means $U = -2K$.

The total energy E is the sum of kinetic and potential energy:

$E = K + U$

$E = K + (-2K)$

$E = -K$

20. **Answer :** A) The earth's gravity at take-off

The Earth has a much larger mass and stronger gravitational pull compared to the Moon.

To escape the Earth's surface and reach orbit or travel to the Moon, a spacecraft must overcome this immense gravitational force, accelerating to roughly 11.2 km/s.

This initial launch phase requires massive booster rockets and consumes the vast majority of the fuel used in the entire mission.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** A) 8 km/sec

The orbital velocity of a satellite revolving very near the Earth's surface is given by:

$v = \sqrt{gR}$

Substitute the standard values $g = 9.8$ m/s^2 and $R = 6.4 \times 10^6$ m:

$v = \sqrt{9.8 \times 6.4 \times 10^6}$

$v = \sqrt{62.72 \times 10^6}$

$v \approx 7.919 \times 10^3$ m/s

$v \approx 7.9$ km/s

This is closest to 8 km/sec.

2. **Answer :** D) $\sqrt{gR}$

The orbital velocity of a satellite at a height h is:

$v = \sqrt{\frac{GM}{R+h}}$

When height h is negligible ($h \approx 0$), the formula simplifies to:

$v = \sqrt{\frac{GM}{R}}$

Since $g = \frac{GM}{R^2}$, we can write $GM = gR^2$.

Substitute this into the simplified formula:

$v = \sqrt{\frac{gR^2}{R}}$

$v = \sqrt{gR}$

3. **Answer :** (C) $r^{\frac{3}{2}}$

According to Kepler's third law of planetary motion, the square of the time period T is directly proportional to the cube of the orbital radius r.

$T^2 \propto r^3$

Take the square root of both sides to find how T varies with r:

$T \propto \sqrt{r^3}$

$T \propto r^{\frac{3}{2}}$

4. **Answer :** B) Mass of the satellite

The mathematical expression for the orbital velocity of a satellite is:

$v = \sqrt{\frac{GM}{r}}$

In this equation, G is the universal gravitational constant, M is the mass of the Earth, and r is the radius of the orbit.

The mass of the satellite itself does not appear in the equation, meaning orbital velocity is entirely independent of it.

5. **Answer :** A) 3.5 km/s

The orbital velocity is inversely proportional to the square root of the orbital radius:

$v \propto \frac{1}{\sqrt{r}}$

Let $v_1 = 7$ km/s at radius $r_1 = R$.

We want to find v_2 at radius $r_2 = 4R$.

Set up the proportion:

$\frac{v_1}{v_2} = \sqrt{\frac{r_2}{r_1}}$

Substitute the known values:

$\frac{7}{v_2} = \sqrt{\frac{4R}{R}}$

$\frac{7}{v_2} = \sqrt{4}$

$\frac{7}{v_2} = 2$

$v_2 = \frac{7}{2}$

$v_2 = 3.5$ km/s

6. **Answer :** D) Ratio of total energy will be 4 but ratio of potential and kinetic energies will be 2

First, find the orbital radii from the center of the Earth.

$r_1 = R + R = 2R$

$r_2 = R + 7R = 8R$

Total energy is $E = -\frac{GMm}{2r}$. The ratio is $\frac{E_1}{E_2} = \frac{r_2}{r_1} = \frac{8R}{2R} = 4$.

Kinetic energy is $K = \frac{GMm}{2r}$. The ratio is $\frac{K_1}{K_2} = \frac{r_2}{r_1} = 4$.

Potential energy is $U = -\frac{GMm}{r}$. The ratio is $\frac{U_1}{U_2} = \frac{r_2}{r_1} = 4$.

Since all three ratios correctly evaluate to 4, statement D is the only wrong statement because it claims the potential and kinetic energy ratios will be 2.

7. **Answer :** D) 24 hours

A communication satellite must appear stationary over a specific point on the Earth's equator to maintain a continuous communication link.

To achieve this, it is placed in a geostationary orbit.

A geostationary orbit requires the satellite's period of revolution to perfectly match the Earth's rotational period, which is exactly 24 hours.

8. **Answer :** B) 7R

The actual calculated distance of a geostationary satellite from the center of the Earth is approximately 42,240 km.

The radius of the Earth R is 6400 km.

Divide the orbital distance by the Earth's radius to find the ratio:

Ratio = $\frac{42240}{6400}$

Ratio ≈ 6.6

This value is nearest to $7R$.

9. **Answer :** (B) The total angular momentum remains constant.

Planetary motion is dictated by gravity, which is a central force acting strictly along the line joining the planet and the sun.

Because the force is central, there is no torque applied to the planet ($\tau = 0$).

According to the law of conservation of angular momentum, when net torque is zero, the total angular momentum of the system remains perfectly constant.

10. **Answer :** (A) Continuously change in velocity

In a uniform circular motion, the object moves at a constant speed along a circular path.

However, because the path curves, the direction of the object's motion changes at every single instant.

Since velocity is a vector quantity defined by both speed and direction, a change in direction constitutes a continuous change in velocity.

11. **Answer :** (B) $\frac{v^2}{r}$

Centripetal acceleration is the acceleration directed towards the center of a circular path that keeps an object in uniform circular motion.

By definition in circular kinematics, it depends on the square of the linear velocity v and is inversely proportional to the radius r.

$a_c = \frac{v^2}{r}$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

12. **Answer :** A, C

Kepler's Third Law (Law of Periods) states that the square of the time period of revolution T of a planet around the sun is directly proportional to the cube of the semi-major axis R of its elliptical orbit.

$T^2 \propto R^3$

This proves option A is correct.

By converting the proportionality into a ratio for two different planets, we get:

$\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$

Which can be algebraically rewritten as:

$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{R_1}{R_2}\right)^3$

This proves option C is also correct.

13. **Answer :** A, C

The orbital speed is v_o and the escape speed is $v_e = \sqrt{2} \cdot v_o$.

If its speed increases to v_e, the required fractional increase is $\frac{\sqrt{2}v_o - v_o}{v_o} = \sqrt{2} - 1 \approx 0.414$.

This translates to a 41.4% increase, validating option A.

If the kinetic energy is doubled, the new kinetic energy $K_{new} = 2 \cdot \left(\frac{1}{2}mv_o^2\right) = mv_o^2$.

The potential energy is $U = -mv_o^2$. The total energy becomes $E = K_{new} + U = mv_o^2 - mv_o^2 = 0$, which is the condition for escape, validating option C.

14. **Answer :** D) Time period would increase

If the distance from the Earth (radius r) is increased, we can evaluate each parameter:

Linear velocity is $v \propto \frac{1}{\sqrt{r}}$, so it would decrease.

Angular velocity is $\omega \propto \frac{1}{r^{3/2}}$, so it would decrease.

Time period is governed by Kepler's third law $T \propto r^{3/2}$.

As r increases, the time period T will definitively increase.

15. **Answer :** B, C

If the radius of the orbit r decreases:

Kinetic energy $K = \frac{GMm}{2r}$ will increase because r is in the denominator.

Potential energy $U = -\frac{GMm}{r}$ has a larger magnitude but is negative, meaning it becomes a smaller (more negative) value, so it decreases.

Mechanical energy $E = -\frac{GMm}{2r}$ behaves similarly to potential energy and decreases.

Speed $v = \sqrt{\frac{GM}{r}}$ will increase.

ASSERTION AND REASON TYPE

16. **Answer :** A) Both A and R are true, R is correct explanation of A

Escape velocity is the minimum initial velocity required to overcome gravitational potential energy.

The necessary kinetic energy $\frac{1}{2}mv_e^2$ exactly equals the magnitude of the gravitational potential energy $\frac{GMm}{r}$ at that specific location.

Since potential energy is dependent on the distance from the center of the Earth, the required kinetic energy, and thus escape velocity, directly depends on the location of projection.

17. **Answer :** A) Both A and R are true, R is correct explanation of A

The magnitude of the potential energy is $|U| = \frac{GMm}{r}$.

The kinetic energy is $K = \frac{1}{2}mv^2$.

Substitute the expression for orbital velocity $v = \sqrt{\frac{GM}{r}}$:

$K = \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2$

$K = \frac{GMm}{2r}$

This demonstrates perfectly that the kinetic energy is exactly half the magnitude of the potential energy, and this conclusion relies directly on the given velocity formula.

MATRIX MATCHING TYPE

18. **Answer :** (A) - (P), (B) - (Q), (C) - (R), (D) - (S)

Escape velocity is governed by $v_e = \sqrt{\frac{2GM}{R}}$, which mathematically depends entirely on the planet's mass M, making it totally independent of the mass of the escaping particle itself.

Areal velocity represents Kepler's second law, which mandates that a planet sweeps equal areas in equal times, meaning it is strictly constant in orbit.

A geostationary satellite is designed to match Earth's rotation, possessing a specific time period of 24 hours.

A polar satellite orbits in a plane perpendicular to the equator, ensuring it passes over the Earth's poles.

19. **Answer :** (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

The mathematical relationship between escape velocity and orbital velocity near the Earth's surface equates to $v_e = \sqrt{2} \cdot v_o$.

The standard formula determining the orbital velocity at any given radius r is derived as $v = \sqrt{\frac{GM}{r}}$.

Kepler's third law dictates the time period of any satellite, showing $T \propto r^{\frac{3}{2}}$.

An object in orbit is perpetually in freefall toward Earth, yielding an environment where the normal force is zero, meaning apparent weight = 0.

COMPREHENSION TYPE

PASSAGE-I

20. **Answer :** A) $-\frac{E}{2}$

The total energy of a satellite in a circular orbit of radius r is inversely proportional to the radius.

$E_{total} = -\frac{GMm}{2r}$

This establishes the relation $E_{total} \propto \frac{1}{r}$.

At radius R, the total energy is $-E$.

When the radius is doubled to $2R$, the denominator doubles, halving the magnitude of the total energy.

$E_{new} = -\frac{E}{2}$

21. **Answer :** B) 84 minutes

The time period of a satellite revolving extremely close to the Earth's surface is calculated using:

$T = 2\pi\sqrt{\frac{R}{g}}$

Substitute the standard Earth values given:

$T = 2 \cdot 3.14159 \cdot \sqrt{\frac{6.4\times10^6}{9.8}}$

$T = 6.283 \cdot \sqrt{653061}$

$T = 6.283 \cdot 808.12$

$T \approx 5077$ seconds

Convert the time from seconds to minutes by dividing by 60:

$T \approx \frac{5077}{60}$

$T \approx 84.6$ minutes

22. **Answer :** A) $\frac{1}{2}$

The orbital speed v is inversely proportional to the square root of the orbital radius r:

$v \propto \frac{1}{\sqrt{r}}$

Let the initial radius be $r_1 = r$ and the new radius be $r_2 = 4r$.

Create a ratio for the new speed to the initial speed:

$\frac{v_2}{v_1} = \sqrt{\frac{r_1}{r_2}}$

Substitute the radii:

$\frac{v_2}{v_1} = \sqrt{\frac{r}{4r}}$

$\frac{v_2}{v_1} = \sqrt{\frac{1}{4}}$

$\frac{v_2}{v_1} = \frac{1}{2}$

PASSAGE-II

23. **Answer :** B) 0.96

The radius for a satellite just above the surface is $r_1 = 6400$ km.

The radius for the polar satellite is $r_2 = 6400 \text{ km} + 600 \text{ km} = 7000$ km.

The orbital speed is inversely proportional to the square root of the radius ($v \propto \frac{1}{\sqrt{r}}$).

Set up the ratio:

Ratio = $\sqrt{\frac{r_1}{r_2}}$

Ratio = $\sqrt{\frac{6400}{7000}}$

Ratio = $\sqrt{\frac{64}{70}}$

Ratio ≈ $\sqrt{0.91428}$

Ratio ≈ 0.956

24. **Answer :** C) 0 N

An orbiting satellite is perpetually in a state of freefall towards the Earth's center, driven solely by gravity.

Because both the satellite and the astronaut are accelerating at the exact same rate due to gravity, the floor of the satellite exerts no upward normal force on the astronaut.

Apparent weight is defined precisely by this normal force.

Therefore, the apparent weight is absolutely 0 N.

25. **Answer :** Note: The mathematically derived ratio is $\sqrt{2}L$, which is not listed in the options provided in the source text. However, demonstrating the math:

The angular momentum L for a circular orbit is:

$L = m \cdot v \cdot r$

Substitute the orbital velocity $v = \sqrt{\frac{GM}{r}}$:

$L = m \cdot \left(\sqrt{\frac{GM}{r}}\right) \cdot r$

$L = m \cdot \sqrt{GM \cdot r}$

This shows that $L \propto \sqrt{r}$.

If the new radius is $2r$, the new angular momentum L' is:

$L' = m \cdot \sqrt{GM \cdot (2r)}$

$L' = \sqrt{2} \cdot \left(m \cdot \sqrt{GM \cdot r}\right)$

$L' = \sqrt{2} \cdot L$

INTEGER TYPE

26. **Answer :** 1032

According to Kepler's third law, $T^2 \propto R^3$.

Let the current duration of the year be $T_1 = 365$ days and current distance be $R_1 = R$.

The new distance is $R_2 = 2R$.

Set up the ratio to find the new duration T_2:

$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{R_2}{R_1}\right)^3$

$\left(\frac{T_2}{365}\right)^2 = \left(\frac{2R}{R}\right)^3$

$\left(\frac{T_2}{365}\right)^2 = 2^3$

$\left(\frac{T_2}{365}\right)^2 = 8$

Take the square root of both sides:

$\frac{T_2}{365} = \sqrt{8}$

$\frac{T_2}{365} \approx 2.8284$

$T_2 \approx 365 \cdot 2.8284$

$T_2 \approx 1032.36$ days

Rounding to the nearest integer yields 1032.

27. **Answer :** 8

By Kepler's Third Law, $T \propto r^{\frac{3}{2}}$.

For a satellite close to Earth, $r = R$, so $T \propto R^{\frac{3}{2}}$.

For the new satellite, $r = 4R$, so $T' \propto (4R)^{\frac{3}{2}}$.

Find the ratio:

$\frac{T'}{T} = \frac{(4R)^{\frac{3}{2}}}{R^{\frac{3}{2}}}$

$\frac{T'}{T} = 4^{\frac{3}{2}}$

$\frac{T'}{T} = \left(\sqrt{4}\right)^3$

$\frac{T'}{T} = 2^3$

$\frac{T'}{T} = 8$

28. **Answer :** 11

The relationship between escape velocity v_e and orbital speed just above the surface v_o is:

$v_e = \sqrt{2} \cdot v_o$

Substitute the given orbital speed $v_o = 8$ km/s:

$v_e = \sqrt{2} \cdot 8$

$v_e = 1.414 \cdot 8$

$v_e = 11.312$ km/s

Rounding to the nearest integer gives 11.

Previous Year Questions

1. **Answer :** A) R. P, Q

The gravitational force on $M$is the vector sum of the forces due to the two masses m.

The gravitational force between two masses is

$F = \frac{GMm}{r^2}$

Let

$F_0 = \frac{GMm}{a^2}$

Configuration P

One $m$is at distance a: force = F_0(right)

Other $m$is at distance $2a$: force

$\frac{GMm}{(2a)^2} = \frac{F_0}{4}$

Both forces act in the same direction.

$F_P = F_0 + \frac{F_0}{4} = \frac{5F_0}{4}$

Configuration Q

Two masses are each at distance $a$on opposite sides of M.

Each exerts force F_0in opposite directions.

$F_Q = 0$

Configuration R

One $m$is at distance $a$horizontally: F_0

One $m$is at distance $a$vertically: F_0

These forces are perpendicular.

$F_R = \sqrt{F_0^2 + F_0^2} = \sqrt{2}\,F_0$

Comparison

$$F_R = \sqrt{2}F_0 \approx 1.414F_0$$
$$F_P = \frac{5}{4}F_0 = 1.25F_0$$
$$F_Q = 0$$

Since

$\sqrt{2} > \frac{5}{4} > 0$,

$\boxed{R > P > Q}$

Therefore, the order from greatest to least is R, P, Q.

2. **Answer :** (D) $4a_1 = 3a_2 = a_3$

According to Kepler's second law, the area swept by a planet per unit time is strictly constant.

$\frac{\text{Area}}{\text{Time}}$ = constant

Therefore, the rates must be equal:

$\frac{a_1}{3} = \frac{a_2}{4} = \frac{a_3}{12}$

To remove fractions, multiply the entire equation by the least common multiple of the denominators (12):

$12 \cdot \left(\frac{a_1}{3}\right) = 12 \cdot \left(\frac{a_2}{4}\right) = 12 \cdot \left(\frac{a_3}{12}\right)$

$4a_1 = 3a_2 = a_3$

3. **Answer :** C) F F F T

Statement (i) is typically false because standard trick questions omit or alter the astronomical distance value.

Statement (ii) is false because Earth rotates West to East, making stars appear to move from East to West.

Statement (iii) is false because Aryabhata, launched in 1975, was the first Indian satellite, not INSAT.

Statement (iv) is true because meteor showers are caused by Earth passing through specific comet debris trails, which occurs predictably at regular intervals every year.

4. **Answer :** B) Inversely Square 9.8 23.5

Newton's universal law of gravitation dictates that force is inversely (i) proportional to the square (ii) of the distance.

For a 1 kg object on Earth's surface, the force is its weight $W = m \cdot g = 1 \cdot 9.8 = 9.8$ N (iii).

If distance decreases by 10%, the new distance is $0.9r$.

The new force F' is proportional to $\frac{1}{(0.9r)^2} = \frac{1}{0.81r^2} \approx 1.2345 \cdot \left(\frac{1}{r^2}\right)$.

This represents an approximate increase of 23.5% (iv).

5. **Answer :** (A) f N

The universal gravitational force formula between two objects is $F = \frac{G \cdot m_1 \cdot m_2}{r^2}$.

This formula entirely depends on mass and distance.

Because G is a universal constant, gravitational force is intrinsically independent of the intervening medium.

Thus, placing the masses under sea-water does not change the force at all.

6. **Answer :** B) Faster More difficult Decrease Increase

Acceleration due to gravity is $g = \frac{GM}{R^2}$.

If the universal gravitational constant G increases by ten times, the acceleration due to gravity g will also proportionately increase (iv).

A higher gravitational pull will cause freely falling objects, like rain drops, to accelerate more quickly, falling faster (i).

Walking involves lifting one's mass against gravity; increased gravity makes this significantly more difficult (ii).

The time period of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$; an increased g clearly forces the time period to decrease (iii).

7. **Answer :** D) Acceleration of the particle of mass m_1 is proportional to m_2.

The magnitude of the gravitational force F experienced by both particles is perfectly identical due to Newton's Third Law ($F = \frac{Gm_1m_2}{r^2}$).

The acceleration of the first particle (m_1) is determined by Newton's Second Law:

$a_1 = \frac{F}{m_1}$

Substitute the gravitational force equation:

$a_1 = \frac{\left(\frac{Gm_1m_2}{r^2}\right)}{m_1}$

$a_1 = \frac{Gm_2}{r^2}$

This definitively shows that the acceleration of mass m_1 is directly proportional to the mass of the other particle, m_2.

8. **Answer :** A) 2 29.4 19.6 4.9

Let upward be the positive direction. For the first ball thrown upward:

$y_1 = v_{i1}t - \frac{1}{2}gt^2$

$y_1 = 49t - 4.9t^2$

For the second ball dropped from height $H = 98$ m:

$y_2 = 98 - \frac{1}{2}gt^2$

$y_2 = 98 - 4.9t^2$

They collide when their heights are equal ($y_1 = y_2$):

$49t - 4.9t^2 = 98 - 4.9t^2$

$49t = 98$

$t = 2$ s (This matches blank i)

Calculate their velocities at $t = 2$ s:

$v_1 = 49 - 9.8(2) = 49 - 19.6 = 29.4$ m/s (directed upwards, matches blank ii)

$v_2 = 0 - 9.8(2) = -19.6$ m/s (directed downwards with magnitude 19.6, matches blank iii)

Calculate perfectly inelastic collision velocity using conservation of momentum ($m_1 = m_2 = 0.1$ kg):

$m_1v_1 + m_2v_2 = (m_1 + m_2)v_{final}$

$(0.1)(29.4) + (0.1)(-19.6) = (0.2)v_{final}$

$2.94 - 1.96 = 0.2v_{final}$

$0.98 = 0.2v_{final}$

$v_{final} = 4.9$ m/s (This matches blank iv)

7. Heat

Temperature Scales and Thermal Expansion

LEVEL-I

1. **Answer :** B) Temperature

The direction of heat flow between two bodies in thermal contact is always determined by their temperature.

Heat naturally flows from a body at a higher temperature to a body at a lower temperature until thermal equilibrium is reached.

2. **Answer :** D) Both A & C

The formula relating density, mass, and volume is:

$\rho = \frac{m}{V}$

When a substance is heated, its mass (m) remains constant.

Due to thermal expansion, its volume (V) increases.

Because the volume increases while mass is constant, the density (ρ) decreases.

Thus, both density and volume change upon heating.

3. **Answer :** A) $160°C$

The relationship between the Fahrenheit and Celsius scales is given by the formula:

$F = \frac{9}{5}C + 32$

We are given that the Fahrenheit reading is twice the Celsius reading:

$F = 2C$

Substitute this into the temperature conversion formula:

$2C = \frac{9}{5}C + 32$

Subtract $\frac{9}{5}C$ from both sides:

$2C - 1.8C = 32$

$0.2C = 32$

$C = \frac{32}{0.2}$

$C = 160°C$

4. **Answer :** C) $90°C$

The formula for converting a reading from a faulty thermometer to a correct Celsius reading is:

$\frac{C-0}{100-0} = \frac{R-LFP}{UFP-LFP}$

Given values for the faulty thermometer:

Lower Fixed Point (LFP) = 6°

Upper Fixed Point (UFP) = 96°

Recorded Reading (R) = 87°

Substitute the given values into the formula:

$\frac{C}{100} = \frac{87-6}{96-6}$

$\frac{C}{100} = \frac{81}{90}$

$\frac{C}{100} = 0.9$

$C = 100 \times 0.9$

$C = 90°C$

5. **Answer :** C) a and b
Solids possess a definite shape, length, surface area, and volume, so they undergo linear, superficial, and cubical expansions.
Liquids and gases do not have a definite shape or surface area, but they occupy volume, so they only undergo cubical (volume) expansion.
Therefore, both statements a and b are correct.

6. **Answer :** A) 1:2:3
The relation between the coefficients of linear (α), superficial (β), and cubical (γ) expansion for an isotropic solid is:
$\beta = 2\alpha$
$\gamma = 3\alpha$
Therefore, the ratio $\alpha : \beta : \gamma$ is:
$\alpha : 2\alpha : 3\alpha$
$1 : 2 : 3$

7. **Answer :** A) $L_1\alpha_1 = L_2\alpha_2$
Let the initial lengths be L_1 and L_2.
For the difference in their lengths to be constant at all temperatures, their change in length for a given temperature change ΔT must be equal:
$\Delta L_1 = \Delta L_2$
The formula for change in length is $\Delta L = L\alpha\Delta T$. Substitute this into the equation:
$L_1\alpha_1\Delta T = L_2\alpha_2\Delta T$
Cancel ΔT from both sides:
$L_1\alpha_1 = L_2\alpha_2$

8. **Answer :** B) $5°C$
The formula for change in length is:
$\Delta L = L\alpha\Delta T$
Given:
$L = 1$ mm
$\Delta L = 5 \times 10^{-5}$ mm
$\alpha = 10 \times 10^{-6}$ K^{-1}
Substitute the values into the formula:
$5 \times 10^{-5} = 1 \times (10 \times 10^{-6}) \times \Delta T$
$5 \times 10^{-5} = 10^{-5} \times \Delta T$
Divide both sides by 10^{-5}:
$\Delta T = \frac{5 \times 10^{-5}}{10^{-5}}$
$\Delta T = 5°C$

9. **Answer :** C) 11×10^{-5} cm shorter
The formula for the change in length is:
$\Delta L = L\alpha\Delta T$
Given:
$L = 10$ cm
$\alpha = 11 \times 10^{-6}/°C$
Initial temperature = $20°C$, Final temperature = $19°C$. The change in temperature is:
$\Delta T = 19°C - 20°C = -1°C$
Substitute the values into the formula:
$\Delta L = 10 \times (11 \times 10^{-6}) \times (-1)$
$\Delta L = -110 \times 10^{-6}$ cm

$\Delta L = -11 \times 10^{-5}$ cm
The negative sign indicates a decrease in length. The bar will be 11×10^{-5} cm shorter.

10. **Answer :** B) $6.45 \times 10^{-6} k^{-1}$
The formula for the coefficient of linear expansion is:
$\alpha = \frac{\Delta L}{L_1 \Delta T}$
Given:
Initial length $L_1 = 3.521$ m
Final length $L_2 = 3.523$ m
Initial temperature $T_1 = 285$ K
Final temperature $T_2 = 373$ K
Calculate the change in length and temperature:
$\Delta L = 3.523 - 3.521 = 0.002$ m
$\Delta T = 373 - 285 = 88$ K
Substitute the values into the formula:
$\alpha = \frac{0.002}{3.521 \times 88}$
$\alpha = \frac{0.002}{309.848}$
$\alpha \approx 6.45 \times 10^{-6}$ K^{-1}

11. **Answer :** A) 0.03 cm
Let the standard correct length of the meter bar be $L = 1$ m.
The length of the first bar at $20°C$ (which was correct at $0°C$) is:
$L_1 = L(1 + \alpha \Delta T_1)$
$L_1 = 1[1 + (12 \times 10^{-6}) \times (20 - 0)]$
$L_1 = 1 + 240 \times 10^{-6}$ m
The length of the second bar at $20°C$ (which was correct at $25°C$) is:
$L_2 = L(1 + \alpha \Delta T_2)$
$L_2 = 1[1 + (12 \times 10^{-6}) \times (20 - 25)]$
$L_2 = 1 + (12 \times 10^{-6}) \times (-5)$
$L_2 = 1 - 60 \times 10^{-6}$ m
The difference in their lengths at $20°C$ is:
$\Delta L = L_1 - L_2$
$\Delta L = (1 + 240 \times 10^{-6}) - (1 - 60 \times 10^{-6})$
$\Delta L = 300 \times 10^{-6}$ m
Convert meters to centimeters:
$\Delta L = 300 \times 10^{-6} \times 100$ cm
$\Delta L = 30000 \times 10^{-6}$ cm
$\Delta L = 0.03$ cm

12. **Answer :** A) 0.0928 m
The formula for the increase in length is:
$\Delta L = L \alpha \Delta T$
Given:
Initial length $L = 50$ m
Initial temperature $T_1 = 16°C$
Final temperature $T_2 = 80°C$
$\alpha = 29 \times 10^{-6}$ K^{-1}
Calculate the change in temperature:
$\Delta T = 80 - 16 = 64°C$

Substitute the values into the formula:
$\Delta L = 50 \times (29 \times 10^{-6}) \times 64$
$\Delta L = 50 \times 1856 \times 10^{-6}$
$\Delta L = 92800 \times 10^{-6}$ m
$\Delta L = 0.0928$ m

13. **Answer :** A) $-3.75°C$
The formula for the area of a sheet at a temperature T is:
$A_T = A_0[1 + \beta(T - T_0)]$
Since $\beta = 2\alpha$, this becomes:
$A_T = A_0[1 + 2\alpha(T - T_0)]$
Let the initial common area be A.
For the brass sheet at common temperature T:
$A_{\text{brass}} = A[1 + 2\alpha_{\text{brass}}(T - 10)]$
For the steel sheet at common temperature T:
$A_{\text{steel}} = A[1 + 2\alpha_{\text{steel}}(T - 20)]$
Since their final areas are equal, equate the two expressions:
$1 + 2\alpha_{\text{brass}}(T - 10) = 1 + 2\alpha_{\text{steel}}(T - 20)$
$2(19 \times 10^{-6})(T - 10) = 2(11 \times 10^{-6})(T - 20)$
Cancel the common factor 2×10^{-6}:
$19(T - 10) = 11(T - 20)$
$19T - 190 = 11T - 220$
$19T - 11T = -220 + 190$
$8T = -30$
$T = -\frac{30}{8}$
$T = -3.75°C$

14. **Answer :** B) 2.584 mm^2
The formula for the increase in area is:
$\Delta A = A\beta\Delta T$
The coefficient of superficial expansion β is twice the linear expansion coefficient α:
$\beta = 2\alpha = 2 \times (19 \times 10^{-6}) = 38 \times 10^{-6}$ K^{-1}
Given:
Initial area $A = 800$ mm^2
Initial temperature $T_1 = 15°C$
Final temperature $T_2 = 100°C$
$\Delta T = 100 - 15 = 85°C$
Substitute the values into the formula:
$\Delta A = 800 \times (38 \times 10^{-6}) \times 85$
$\Delta A = 800 \times 3230 \times 10^{-6}$
$\Delta A = 2584000 \times 10^{-6}$ mm^2
$\Delta A = 2.584$ mm^2

15. **Answer :** B) Thrice the coefficient of linear expansion
The relation between cubical expansion coefficient (γ) and linear expansion coefficient (α) is derived from volume expanding in three dimensions.
$\gamma = 3\alpha$
Therefore, it is thrice the coefficient of linear expansion.

16. **Answer :** A) 358.8 k
The formula for volume expansion is:
$V_2 = V_1[1 + \gamma\Delta T]$

Given:
$V_1 = 440\ \text{mm}^3$
$V_2 = 480\ \text{mm}^3$
$\gamma = 12 \times 10^{-4}\ \text{K}^{-1}$
Initial temperature $T_1 = 283\ \text{K}$
Substitute the values into the formula:
$480 = 440[1 + (12 \times 10^{-4})\Delta T]$
$\frac{480}{440} = 1 + (12 \times 10^{-4})\Delta T$
$\frac{12}{11} = 1 + (12 \times 10^{-4})\Delta T$
$\frac{12}{11} - 1 = (12 \times 10^{-4})\Delta T$
$\frac{1}{11} = (12 \times 10^{-4})\Delta T$
$\Delta T = \frac{1}{11 \times 12 \times 10^{-4}}$
$\Delta T = \frac{10000}{132} \approx 75.75\ \text{K}$
Calculate the final temperature:
$T_2 = T_1 + \Delta T$
$T_2 = 283 + 75.75 = 358.75\ \text{K} \approx 358.8\ \text{K}$

17. **Answer :** B) $2.012\ l$
The formula for the final volume is:
$V_2 = V_1[1 + \gamma \Delta T]$
Given:
$V_1 = 2$ litres
$\gamma = 30 \times 10^{-5}\ \text{K}^{-1}$
Change in temperature $\Delta T = 40°C - 20°C = 20°C$
Substitute the values into the formula:
$V_2 = 2[1 + (30 \times 10^{-5}) \times 20]$
$V_2 = 2[1 + 600 \times 10^{-5}]$
$V_2 = 2[1 + 0.006]$
$V_2 = 2 \times 1.006$
$V_2 = 2.012\ l$

18. **Answer :** A) 1002.4
First, find the linear expansion coefficient (α) of the glass.
The formula for linear expansion is:
$\alpha = \frac{\Delta L}{L \Delta T}$
Given:
$L = 10$ cm
$\Delta L = 0.08$ mm $= 0.008$ cm
$\Delta T = 100°C - 0°C = 100°C$
Substitute the values:
$\alpha = \frac{0.008}{10 \times 100}$
$\alpha = 8 \times 10^{-6}/°\text{C}$
The coefficient of cubical expansion (γ) for the flask is:
$\gamma = 3\alpha = 3 \times (8 \times 10^{-6}) = 24 \times 10^{-6}/°\text{C}$
Now, find the final volume of the flask. The formula is:
$V_2 = V_1 + \Delta V = V_1 + V_1 \gamma \Delta T$
Given $V_1 = 1000$ cc and $\Delta T = 100°C$:

$\Delta V = 1000 \times (24 \times 10^{-6}) \times 100$
$\Delta V = 100000 \times 24 \times 10^{-6} = 2.4$ cc
Calculate the final volume:
$V_2 = 1000 + 2.4 = 1002.4$ cc

19. **Answer :** C) 3.6
First, determine the initial volume of the copper cube.
$V = \text{length}^3 = 10^3 = 1000$ c.c.
The coefficient of cubical expansion is:

$$\gamma = 3\alpha$$

$\gamma = 3 \times (12 \times 10^{-6}) = 36 \times 10^{-6}/°\text{C}$
The change in temperature is:
$\Delta T = 110°C - 10°C = 100°C$
The formula for the change in volume is:
$\Delta V = V\gamma\Delta T$
Substitute the values:
$\Delta V = 1000 \times (36 \times 10^{-6}) \times 100$
$\Delta V = 100000 \times 36 \times 10^{-6}$
$\Delta V = 3.6$ c.c.

20. **Answer :** C) $5 \times 10^{-5}/°C$
The fractional change in volume is given as a percentage:
$\frac{\Delta V}{V} = 0.3\% = \frac{0.3}{100} = 0.003$
The formula for cubical expansion is:
$\frac{\Delta V}{V} = \gamma\Delta T$
Given $\Delta T = 20°C$:
$0.003 = \gamma \times 20$
$\gamma = \frac{0.003}{20} = 0.00015 = 15 \times 10^{-5}/°\text{C}$
The coefficient of linear expansion (α) is one-third of the cubical expansion:
$\alpha = \frac{\gamma}{3}$
$\alpha = \frac{15 \times 10^{-5}}{3}$
$\alpha = 5 \times 10^{-5}/°\text{C}$

21. **Answer :** D) Will bend with steel on concave side
When a bimetallic strip is heated, the metal with the higher coefficient of linear expansion expands more.
Since $\alpha_{\text{Al}} > \alpha_{\text{steel}}$, aluminium expands more than steel for the same temperature rise.
To accommodate the longer length, the aluminium side forms the outer (convex) curve, forcing the steel side to become the inner (concave) curve.

22. **Answer :** A) Loses time
The time period of a simple pendulum is given by:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

When the temperature increases, the metallic pendulum undergoes linear expansion, causing its length (L) to increase.
Since the time period (T) is directly proportional to the square root of its length, an increase in length results in an increase in the time period.
Because each oscillation takes longer, the clock runs slower and thus loses time.

23. **Answer :** A) 474.5^0C

The formula for thermal expansion is:

$\Delta D = D_1 \alpha \Delta T$

Given:

Initial diameter $D_1 = 2$ m

Target diameter $D_2 = 2.01$ m

Change in diameter $\Delta D = 2.01 - 2 = 0.01$ m

$\alpha = 11 \times 10^{-6}/°\text{C}$

Initial temperature $T_1 = 20°C$

Substitute the values into the formula to find the change in temperature:

$0.01 = 2 \times (11 \times 10^{-6}) \times \Delta T$

$0.01 = 22 \times 10^{-6} \times \Delta T$

$\Delta T = \frac{0.01}{22 \times 10^{-6}}$

$\Delta T = \frac{10000}{22} \approx 454.5°\text{C}$

Calculate the final temperature:

$T_2 = T_1 + \Delta T$

$T_2 = 20 + 454.5 = 474.5°\text{C}$

24. **Answer :** A) 74 k

The formula for linear expansion is:

$\Delta D = D \alpha \Delta T$

Given:

Initial diameter $D = 15.02$ mm

Final diameter $D_f = 15.0$ mm

Change in diameter $\Delta D = 15.0 - 15.02 = -0.02$ mm

$\alpha = 18 \times 10^{-6}$ K^{-1}

Substitute the values into the formula to find the required temperature change:

$-0.02 = 15.02 \times (18 \times 10^{-6}) \times \Delta T$

$-0.02 = 270.36 \times 10^{-6} \times \Delta T$

$\Delta T = \frac{-0.02}{270.36 \times 10^{-6}}$

$\Delta T = \frac{-20000}{270.36}$

$\Delta T \approx -73.97$ K

The shaft must be cooled by approximately 74 K.

25. **Answer :** B) 15.55 s

The formula for the time gained or lost by a pendulum clock per day is:

$\Delta t = \frac{1}{2} \alpha \Delta T \times 86400$

Given:

$\alpha = 18 \times 10^{-6}/°\text{C}$

Initial temperature = $20°C$

Final temperature = $0°C$

Magnitude of temperature change $\Delta T = 20°C$

Time in one day = 86400 s

Substitute the values into the formula:

$\Delta t = \frac{1}{2} \times (18 \times 10^{-6}) \times 20 \times 86400$

$\Delta t = 9 \times 10^{-6} \times 20 \times 86400$

$\Delta t = 180 \times 10^{-6} \times 86400$

$\Delta t = 15.552$ s

LEVEL-II

1. **Answer :** A) 0.372 mm

The formula for linear expansion is:

$\Delta R = R\alpha\Delta T$

Given:

Initial radius $R = 30$ mm

$\alpha = 31 \times 10^{-6}$ K^{-1}

Change in temperature $\Delta T = 420°C - 20°C = 400°C$

Substitute the values into the formula:

$\Delta R = 30 \times (31 \times 10^{-6}) \times 400$

$\Delta R = 12000 \times 31 \times 10^{-6}$

$\Delta R = 372000 \times 10^{-6}$ mm

$\Delta R = 0.372$ mm

2. **Answer :** B) 280.5 mm^2

The formula for superficial expansion is:

$\Delta A = A\beta\Delta T$

First, calculate the initial surface area of the sphere ($A = 4\pi R^2$):

$A = 4 \times \pi \times (30)^2$

$A = 4 \times 3.14159 \times 900 \approx 11309.73$ mm^2

The coefficient of superficial expansion is $\beta = 2\alpha$:

$\beta = 2 \times 31 \times 10^{-6} = 62 \times 10^{-6}$ K^{-1}

The change in temperature is $\Delta T = 400°C$. Substitute the values into the formula:

$\Delta A = 11309.73 \times (62 \times 10^{-6}) \times 400$

$\Delta A = 11309.73 \times 0.0248$

$\Delta A \approx 280.48$ mm^2 ≈ 280.5 mm^2

3. **Answer :** C) 4207 mm^3

The formula for cubical expansion is:

$\Delta V = V\gamma\Delta T$

First, calculate the initial volume of the sphere ($V = \frac{4}{3}\pi R^3$):

$V = \frac{4}{3} \times \pi \times (30)^3$

$V = \frac{4}{3} \times 3.14159 \times 27000 \approx 113097.3$ mm^3

The coefficient of cubical expansion is $\gamma = 3\alpha$:

$\gamma = 3 \times 31 \times 10^{-6} = 93 \times 10^{-6}$ K^{-1}

The change in temperature is $\Delta T = 400°C$. Substitute the values into the formula:

$\Delta V = 113097.3 \times (93 \times 10^{-6}) \times 400$

$\Delta V = 113097.3 \times 0.0372$

$\Delta V \approx 4207.2$ mm^3 ≈ 4207 mm^3

4. **Answer :** B) 4:3

The formula for the expansion in length is:

$\Delta L = L\alpha\Delta T$

Since the expansions (ΔL) and the change in temperature (ΔT) are equal for both rods:

$L_1\alpha_1\Delta T = L_2\alpha_2\Delta T$

Divide both sides by ΔT:

$L_1\alpha_1 = L_2\alpha_2$

Rearrange to find the ratio of their initial lengths:

$\frac{L_1}{L_2} = \frac{\alpha_2}{\alpha_1}$

Given that the ratio of coefficients of linear expansion is 3:4 (so $\frac{\alpha_1}{\alpha_2} = \frac{3}{4}$):

$\frac{L_1}{L_2} = \frac{4}{3}$

5. **Answer :** A) $30 \times 10^{-6}/°C$

First, find the coefficient of linear expansion using the formula:

$\alpha = \frac{\Delta R}{R_1 \Delta T}$

Given:

$R_1 = 100$ cm

$R_2 = 100.1$ cm

$\Delta R = 100.1 - 100 = 0.1$ cm

$\Delta T = 100°C - 0°C = 100°C$

Substitute the values:

$\alpha = \frac{0.1}{100 \times 100}$

$\alpha = \frac{0.1}{10000} = 10^{-5} = 10 \times 10^{-6}/°\text{C}$

The coefficient of cubical expansion is $\gamma = 3\alpha$:

$\gamma = 3 \times (10 \times 10^{-6})$

$\gamma = 30 \times 10^{-6}/°\text{C}$

6. **Answer :** A) 2.59 sec

The formula for the error in time (seconds lost or gained per day) is:

$\Delta t = \frac{1}{2} \alpha \Delta T \times 86400$

Given:

$\alpha = 0.000012/°\text{C} = 12 \times 10^{-6}/°\text{C}$

$\Delta T = 20°C - 15°C = 5°C$

Substitute the values into the formula:

$\Delta t = \frac{1}{2} \times (12 \times 10^{-6}) \times 5 \times 86400$

$\Delta t = 6 \times 10^{-6} \times 432000$

$\Delta t = 2.592$ sec

7. **Answer :** A) $\frac{1}{86400}/°C$

The formula for time lost per day is:

$\Delta t = \frac{1}{2} \alpha \Delta T \times 86400$

Given:

Time lost $\Delta t = 12.5$ seconds

Change in temperature $\Delta T = 25°C - 0°C = 25°C$

Substitute the values into the formula:

$12.5 = \frac{1}{2} \alpha \times 25 \times 86400$

$12.5 = \alpha \times 12.5 \times 86400$

Divide both sides by 12.5:

$1 = \alpha \times 86400$

$\alpha = \frac{1}{86400}/°\text{C}$

8. **Answer :** A, D

The formula for linear expansion is:

$\Delta L = L_0 \alpha \Delta T$

From the formula, the change in length (ΔL) depends directly on:

The initial length (L_0)
The nature of the material (which gives α)
The temperature rise (ΔT)
Therefore, it depends on the initial length and the temperature rise.

9. **Answer :** A, C, D
The formula for the coefficient of linear expansion is:
$\alpha = \frac{\Delta L}{L \Delta T}$
Since both ΔL and L are measurements of length, their units cancel each other out in the ratio.
Thus, the unit of α only depends on the inverse of the temperature unit (K^{-1} or $^{\circ}C^{-1}$). It is completely independent of the units used for length, area, or mass.

10. **Answer :** B, D
When an isotropic material undergoes thermal expansion, every linear dimension expands in the same proportion, similar to a photographic enlargement.
Therefore, any length dimension L at temperature t relates to the original length by $L' = L(1 + \alpha t)$.
This rule applies equally to the radii of the holes (R_1, R_2), the separation between them (x), and the outer boundaries (a, b).
Thus, $\frac{x^1}{x} = (1 + \alpha t)$ and $\frac{b^1}{b} = (1 + \alpha t)$ are correct.

11. **Answer :** C) Different expansion
A thermostat utilizes a bimetallic strip, which consists of two different metals securely bonded together.
These metals have different coefficients of linear expansion.
When the temperature changes, one metal expands or contracts more than the other, causing the strip to bend. This mechanical bending is used to open or close an electrical circuit.

12. **Answer :** B, C
The formula for the initial circumference of a ring is $C = \pi D$.
When temperature increases by $\Delta T = 1^{\circ}C$, the increase in circumference is:
$\Delta C = C\alpha\Delta T = (\pi D)\alpha(1) = \pi D\alpha$
The formula for the initial area of the ring is $A = \frac{\pi D^2}{4}$.
The coefficient of superficial expansion is $\beta = 2\alpha$. The increase in area for $\Delta T = 1^{\circ}C$ is:$\Delta A = A\beta\Delta T = \left(\frac{\pi D^2}{4}\right)(2\alpha)(1) = \frac{\pi D^2 \alpha}{2}$

13. **Answer :** A) Both A and R are true, and R is the correct explanation of A
Expansion in one dimension is called linear expansion (change in length).
Expansion in two dimensions is called superficial expansion (change in surface area).
Expansion in three dimensions is called cubical expansion (change in volume).
Therefore, solids undergo these three classifications of expansion precisely because their dimensions (length, area, and volume) change upon heating.

14. **Answer :** A) Both A and R are true, and R is the correct explanation of A
Heat is a form of energy that transfers from one body to another due to a temperature difference.
When heat is added to or removed from a substance, its internal kinetic energy changes, which physically manifests as a change in temperature.
Thus, heat transfer is the causal action, and the resulting change in temperature is the observable effect.

15. **Answer :** A) 150.24 cm
The change in temperature is:
$\Delta T = 100°C - 20°C = 80°C$
First, find the change in length for the iron rod (ΔL_1):
$\Delta L_1 = L_1 \alpha_1 \Delta T$
$\Delta L_1 = 50 \times (12 \times 10^{-6}) \times 80 = 48000 \times 10^{-6} = 0.048$ cm
Next, find the change in length for the aluminium rod (ΔL_2):
$\Delta L_2 = L_2 \alpha_2 \Delta T$
$\Delta L_2 = 100 \times (24 \times 10^{-6}) \times 80 = 192000 \times 10^{-6} = 0.192$ cm
The total final length L' of the composite system is the sum of the initial lengths and their respective expansions:
$L' = (L_1 + L_2) + (\Delta L_1 + \Delta L_2)$
$L' = (50 + 100) + (0.048 + 0.192)$
$L' = 150 + 0.24 = 150.24$ cm

16. **Answer :** A) 0.24 cm
The total change in length (ΔL) of the composite system is the sum of the individual changes in length calculated in the previous step:
$\Delta L = \Delta L_1 + \Delta L_2$
$\Delta L = 0.048$ cm $+ 0.192$ cm
$\Delta L = 0.24$ cm

17. **Answer :** C) $20 \times 10^{-6}/^oC$
The effective coefficient of linear expansion for a composite bar is given by the formula:
$\alpha_{\text{avg}} = \frac{L_1\alpha_1 + L_2\alpha_2}{L_1 + L_2}$
Substitute the given values:
$\alpha_{\text{avg}} = \frac{50 \times (12 \times 10^{-6}) + 100 \times (24 \times 10^{-6})}{50 + 100}$
$\alpha_{\text{avg}} = \frac{600 \times 10^{-6} + 2400 \times 10^{-6}}{150}$
$\alpha_{\text{avg}} = \frac{3000 \times 10^{-6}}{150}$
$\alpha_{\text{avg}} = 20 \times 10^{-6}/°\text{C}$

18. **Answer :** B) $26 \times 10^{-6}/°C$
The relationship between the coefficient of linear expansion (α) and areal/superficial expansion (β) is:

$$\beta = 2\alpha$$

Given $\alpha = 13 \times 10^{-6}/°\text{C}$:
$\beta = 2 \times (13 \times 10^{-6})$
$\beta = 26 \times 10^{-6}/°\text{C}$

19. **Answer :** C) $57 \times 10^{-6}/^oC$
The relationship between the coefficient of linear expansion (α) and volume/cubical expansion (γ) is:
$\gamma = 3\alpha$
Given $\alpha = 19 \times 10^{-6}/°\text{C}$:
$\gamma = 3 \times (19 \times 10^{-6})$
$\gamma = 57 \times 10^{-6}/°\text{C}$

20. **Answer :** B) 0.0051 m
The formula for the increase in length is:
$\Delta L = L\alpha\Delta T$

Given:
$L = 10$ m
$\alpha = 17 \times 10^{-6}/°C$
$\Delta T = 60°C - 30°C = 30°C$
Substitute the values into the formula:
$\Delta L = 10 \times (17 \times 10^{-6}) \times 30$
$\Delta L = 5100 \times 10^{-6}$ m
$\Delta L = 0.0051$ m

21. **Answer :** a-S, b-R, c-P, T, d-Q
Evaluate each expression knowing that $\beta = 2\alpha$ and $\gamma = 3\alpha$:
For a):
$\frac{\alpha^2+\beta^2+\gamma^2}{\alpha^2} = \frac{\alpha^2+(2\alpha)^2+(3\alpha)^2}{\alpha^2} = \frac{\alpha^2+4\alpha^2+9\alpha^2}{\alpha^2} = \frac{14\alpha^2}{\alpha^2} = 14$
This matches S.
For b):
$\alpha = \frac{\beta}{2}$
This matches R.
For c):
$\gamma = 3\alpha = \frac{3\beta}{2}$
This matches P, T.
For d):
$\frac{\alpha\beta+\beta\gamma+\alpha\gamma}{\beta^2} = \frac{\alpha(2\alpha)+(2\alpha)(3\alpha)+\alpha(3\alpha)}{(2\alpha)^2} = \frac{2\alpha^2+6\alpha^2+3\alpha^2}{4\alpha^2} = \frac{11\alpha^2}{4\alpha^2} = \frac{11}{4}$
This matches Q.

22. **Answer :** a-Q, b-R, c-S, d-P
Evaluate each matching criteria conceptually:
a) Thermal expansion: Depends on initial dimensions, material properties (α), and temperature change. (Matches Q)
b) Expansion coefficients (α, β, γ): These are inherent material constants and depend only on the nature of the material itself. (Matches R)
c) Bimetallic strip: Used to compensate for temperature variations in delicate mechanisms, such as the balance wheel of a mechanical watch. (Matches S)
d) Invar steel: Has an extremely low coefficient of linear expansion, making it ideal for the rigid arms of a pendulum clock to ensure timekeeping accuracy doesn't drift with seasonal temperature changes. (Matches P)

23. **Answer :** 6
The condition for the difference in lengths to remain constant at all temperatures is:
$\Delta L_1 = \Delta L_2$
$L_1\alpha_1 = L_2\alpha_2$
Given:
$L_1 = 40$ cm
$\alpha_1 = 6 \times 10^{-6}/°C$
$\alpha_2 = 4 \times 10^{-6}/°C$
Substitute the values:
$40 \times (6 \times 10^{-6}) = L_2 \times (4 \times 10^{-6})$
$240 = 4 \times L_2$
$L_2 = \frac{240}{4} = 60$ cm

The length is expressed as $n \times 10^1$ cm:
$60 = n \times 10$
$n = 6$

24. **Answer :** 10
The relationship between the coefficient of linear expansion (α) and superficial expansion (β) is:
$\beta = 2\alpha$
Given $\alpha = 5 \times 10^{-6}$/°C:
$\beta = 2 \times (5 \times 10^{-6}) = 10 \times 10^{-6}/\circ C$
Therefore, the coefficient in terms of 10^{-6}/°C is 10.

25. **Answer :** 8
The relationship between the coefficient of cubical expansion (γ), superficial expansion (β), and linear expansion (α) is:
$\alpha = \gamma/3$
$\beta = 2\alpha$
Given $\gamma = 12 \times 10^{-6}$/°C:
First, find α:
$\alpha = 312 \times 10^{-6} = 4 \times 10^{-6}/\circ C$
Next, calculate β:
$\beta = 2 \times (4 \times 10^{-6}) = 8 \times 10^{-6}/\circ C$
Therefore, the coefficient in terms of 10^{-6}/°C is 8.

Expansion of Liquids and Gases

1. **Answer :** B) $1 \times 10^{-3}/^\circ C$
The apparent volume fraction of the liquid at temperature T is given by:
$f_2 = f_1(1 + \gamma_a \Delta T)$
Substitute the given values $f_1 = \frac{1}{6}$, $f_2 = \frac{1}{5}$, and $\Delta T = 200^\circ C$:
$\frac{1}{5} = \frac{1}{6}(1 + \gamma_a \times 200)$
Multiply both sides by 6:
$\frac{6}{5} = 1 + 200\gamma_a$
$1.2 = 1 + 200\gamma_a$
$0.2 = 200\gamma_a$
$\gamma_a = \frac{0.2}{200}$
$\gamma_a = 1 \times 10^{-3}/^\circ C$

2. **Answer :** C) 0.0005/°C
The formula for the coefficient of apparent expansion γ_a is:
$\gamma_a = \frac{m_{expelled}}{m_{remaining} \times \Delta T}$
Substitute the given ratio $\frac{m_{expelled}}{m_{remaining}} = \frac{1}{50}$ and $\Delta T = 40^\circ C$:
$\gamma_a = \frac{1}{50 \times 40}$
$\gamma_a = \frac{1}{2000}$
$\gamma_a = 0.0005/^\circ C$

3. **Answer :** A) 0.000155/°C
The relationship between real expansion, apparent expansion, and volume expansion of the vessel is:
$\gamma_r = \gamma_a + \gamma_v$

The volume expansion of the vessel γ_v is three times its linear expansion α:

$\gamma_v = 3\alpha$

$\gamma_v = 3 \times 0.000009/°C$

$\gamma_v = 0.000027/°C$

Now, calculate the apparent expansion γ_a:

$\gamma_a = \gamma_r - \gamma_v$

$\gamma_a = 0.000182 - 0.000027$

$\gamma_a = 0.000155/°C$

4. **Answer :** C) $10^{-3}/°C$

The change in temperature is:

$\Delta T = 100°C - 75°C$

$\Delta T = 25°C$

Using the apparent expansion formula:

$\gamma_a = \frac{m_{expelled}}{m_{remaining} \times \Delta T}$

Substitute the known values:

$\gamma_a = \frac{\frac{1}{40}}{25}$

$\gamma_a = \frac{1}{1000}$

$\gamma_a = 10^{-3}/°C$

5. **Answer :** D) 5.53 gm

The mass expelled is approximately given by:

$m_{expelled} = m_{initial} \times \gamma_a \times \Delta T$

Substitute $m_{initial} = 400$ gm, $\gamma_a = \frac{1}{6500}/°C$, and $\Delta T = 90°C$:

$m_{expelled} = 400 \times \frac{1}{6500} \times 90$

$m_{expelled} = \frac{36000}{6500}$

$m_{expelled} = \frac{360}{65}$

$m_{expelled} \approx 5.538$ gm

Option D (5.53 gm) is the closest matching value.

6. **Answer :** A) 15.2 ml

linear expansion of glass is $1.0 \times 10^{-5}/°C$.

First, find the volume expansion of the glass flask:

$\gamma_v = 3\alpha$

$\gamma_v = 3 \times (1.0 \times 10^{-5})$

$\gamma_v = 3.0 \times 10^{-5}/°C$

$\gamma_v = 0.3 \times 10^{-4}/°C$

Calculate the apparent expansion of mercury:

$\gamma_a = \gamma_r - \gamma_v$

$\gamma_a = 1.82 \times 10^{-4} - 0.3 \times 10^{-4}$

$\gamma_a = 1.52 \times 10^{-4}/°C$

Calculate the volume of mercury expelled:

$V_{expelled} = V_{initial} \times \gamma_a \times \Delta T$

$V_{expelled} = 1000 \text{ ml} \times (1.52 \times 10^{-4}) \times 100°C$

$V_{expelled} = 15.2$ ml

7. **Answer :** D) 0.000333/°C

Determine the change in temperature:

$\Delta T = 85°C - 10°C$

$\Delta T = 75°C$

Find the remaining mass of the liquid:

$m_{remaining} = m_{initial} - m_{expelled}$

$m_{remaining} = 82 \text{ gm} - 2 \text{ gm}$

$m_{remaining} = 80 \text{ gm}$

Use the apparent expansion formula:

$\gamma_a = \frac{m_{expelled}}{m_{remaining} \times \Delta T}$

$\gamma_a = \frac{2}{80 \times 75}$

$\gamma_a = \frac{2}{6000}$

$\gamma_a = \frac{1}{3000}$

$\gamma_a \approx 0.000333/°C$

8. **Answer :** B) $\frac{\gamma_R}{21}$

The relationship between the coefficients is:

$\gamma_r = \gamma_a + \gamma_v$

Substitute $\gamma_a = \frac{6}{7}\gamma_r$ and $\gamma_v = 3\alpha$:

$\gamma_r = \frac{6}{7}\gamma_r + 3\alpha$

$\gamma_r - \frac{6}{7}\gamma_r = 3\alpha$

$\frac{1}{7}\gamma_r = 3\alpha$

$\alpha = \frac{\gamma_r}{21}$

9. **Answer :** D) 0.0001/°C

Calculate the initial mass of the mercury:

$m_{initial} = 1040 \text{ gm} - 40 \text{ gm}$

$m_{initial} = 1000 \text{ gm}$

Calculate the remaining mass of the mercury after heating:

$m_{remaining} = 1000 \text{ gm} - 10 \text{ gm}$

$m_{remaining} = 990 \text{ gm}$

Find the apparent expansion of mercury ($\Delta T = 100°C$):

$\gamma_a = \frac{m_{expelled}}{m_{remaining} \times \Delta T}$

$\gamma_a = \frac{10}{990 \times 100}$

$\gamma_a = \frac{1}{9900}$

$\gamma_a \approx 0.000101/°C \approx 0.0001/°C$

Find the cubical expansion of glass (γ_v):

$\gamma_v = \gamma_r - \gamma_a$

$\gamma_v = 0.0002 - 0.0001$

$\gamma_v = 0.0001/°C$

10. **Answer :** $1.8 \times 10^{-4}/°C$

The relation for heights in a U-tube at different temperatures is:

$h_t = h_0(1 + \gamma_r \Delta T)$

Substitute the given values:

$101.8 = 100(1 + \gamma_r \times 100)$

$1.018 = 1 + 100\gamma_r$

$0.018 = 100\gamma_r$

$\gamma_r = \frac{0.018}{100}$

$\gamma_r = 1.8 \times 10^{-4}/°C$

11. **Answer :** B) 48 gm

Find the volume coefficient of expansion for glass:

$\gamma_g = 3\alpha$

$\gamma_g = 3 \times (8 \times 10^{-6})$

$\gamma_g = 24 \times 10^{-6}/°C$

Find the apparent expansion of the liquid:

$\gamma_a = \gamma_r - \gamma_g$

$\gamma_a = 500 \times 10^{-6} - 24 \times 10^{-6}$

$\gamma_a = 476 \times 10^{-6}/°C$

Calculate the mass of the liquid the vessel can hold at $80°C$:

$m_t = \frac{m_0}{1+\gamma_a \Delta T}$

$m_t = \frac{50}{1+(476\times10^{-6})\times80}$

$m_t = \frac{50}{1+0.03808}$

$m_t = \frac{50}{1.03808}$

$m_t \approx 48.16$ gm

12. **Answer :** D) $1 \times 10^{-4}/°C$

The real expansion of the liquid is the same in both vessels:

$\gamma_r = \gamma_{aA} + \gamma_{vA} = \gamma_{aB} + \gamma_{vB}$

Rearrange to find the difference in volume expansions of the vessels:

$\gamma_{vA} - \gamma_{vB} = \gamma_{aB} - \gamma_{aA}$

Substitute the given apparent expansions:

$\gamma_{vA} - \gamma_{vB} = (21 \times 10^{-4}) - (18 \times 10^{-4})$

$\gamma_{vA} - \gamma_{vB} = 3 \times 10^{-4}/°C$

Convert to linear expansion differences ($\gamma = 3\alpha$):

$3\alpha_A - 3\alpha_B = 3 \times 10^{-4}$

$3(\alpha_A - \alpha_B) = 3 \times 10^{-4}$

$\alpha_A - \alpha_B = 1 \times 10^{-4}/°C$

13. **Answer :** C) $4°C$

Due to the anomalous expansion of water, it contracts when heated from $0°C$ to $4°C$, reaching its minimum volume and maximum density precisely at $4°C$.

14. **Answer :** D) First decreases and then increases

Water exhibits anomalous expansion between $0°C$ and $4°C$. As it is heated from $0°C$ to $4°C$, its volume decreases. Heating it further from $4°C$ to $10°C$ causes its volume to increase.

15. **Answer :** B) More dense

As the lake cools, water at $4°C$ becomes the most dense. This denser water sinks to the bottom, allowing the surface water to continue cooling towards the freezing point.

16. **Answer :** D) Anomalous expansion of water

When cooled from $4°C$ to $0°C$, water undergoes anomalous expansion. The increase in volume exerts immense outward pressure on the rigid glass walls, causing the bottle to break.

17. **Answer :** B) Water expands from $4°C$ to $0°C$

A freezing mixture cools the jar rapidly. As the water's temperature drops from $4°C$ to $0°C$, it expands anomalously, breaking the confined jar.

18. **Answer :** B) Rises on heating

The apparent expansion is positive when $\gamma_r > \gamma_v$. Since the liquid expands more than the container, the liquid level rises on heating.

19. **Answer :** A) 13.13 gm/cc

First, calculate the temperature change in Celsius:

$T = 473\ \text{K} - 273 = 200°C$

Use the formula for variation of density with temperature:

$d_t = \frac{d_0}{1+\gamma\Delta T}$

Substitute the given values:

$d_t = \frac{13.6}{1+(0.18\times10^{-3})\times200}$

$d_t = \frac{13.6}{1+0.036}$

$d_t = \frac{13.6}{1.036}$

$d_t \approx 13.127\ \text{gm/cc}$

20. **Answer :** C) 1.0006

Find the volume expansion coefficient:

$\gamma = 3\alpha$

$\gamma = 3 \times (10 \times 10^{-6})$

$\gamma = 30 \times 10^{-6}/°C$

The ratio of densities is given by:

$\frac{d_{20}}{d_{40}} = \frac{1+\gamma T_{40}}{1+\gamma T_{20}}$

Since γ is very small, this can be approximated as:

$\frac{d_{20}}{d_{40}} \approx 1 + \gamma(T_{40} - T_{20})$

$\frac{d_{20}}{d_{40}} \approx 1 + (30 \times 10^{-6}) \times 20$

$\frac{d_{20}}{d_{40}} \approx 1 + 600 \times 10^{-6}$

$\frac{d_{20}}{d_{40}} \approx 1.0006$

21. **Answer :** B) 8.99 gm/cc

Find the volume expansion coefficient of copper:

$\gamma = 3\alpha$

$\gamma = 3 \times (17 \times 10^{-6})$

$\gamma = 51 \times 10^{-6}/°C$

Calculate the density at $20°C$:

$d_t = \frac{d_0}{1+\gamma\Delta T}$

$d_{20} \approx d_0(1 - \gamma\Delta T)$

$d_{20} = 9[1 - (51 \times 10^{-6}) \times 20]$

$d_{20} = 9[1 - 0.00102]$

$d_{20} = 9 \times 0.99898$

$d_{20} \approx 8.99\ \text{gm/cc}$

22. **Answer :** A) $2.12 \times 10^{-4}/°C$

Use the density formula:

$d_{100} = \frac{d_0}{1+\gamma\Delta T}$

Substitute the values:

$9.4 = \frac{10}{1+\gamma\times100}$

$1 + 100\gamma = \frac{10}{9.4}$

$1 + 100\gamma \approx 1.0638$

$100\gamma = 0.0638$

$\gamma = 6.38 \times 10^{-4}/°C$

Find the linear expansion coefficient:

$\alpha = \frac{\gamma}{3}$

$\alpha = \frac{6.38\times10^{-4}}{3}$

$\alpha \approx 2.12 \times 10^{-4}/°C$

23. **Answer :** C) $-123°C$

Convert the initial temperature to Kelvin:

$T_1 = 27°C + 273 = 300$ K

Apply the combined gas law:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Substitute $P_1 = 30$, $V_1 = V$, $P_2 = 1$, and $V_2 = 15V$:

$\frac{30\times V}{300} = \frac{1\times15V}{T_2}$

$\frac{1}{10} = \frac{15}{T_2}$

$T_2 = 150$ K

Convert back to Celsius:

$T_2 = 150 - 273$

$T_2 = -123°C$

24. **Answer :** D) $927°C$

Convert the initial temperature to Kelvin:

$T_1 = 27°C + 273 = 300$ K

Apply the combined gas law:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Substitute $P_2 = 2P_1$ and $V_2 = 2V_1$:

$T_2 = T_1\left(\frac{P_2}{P_1}\right)\left(\frac{V_2}{V_1}\right)$

$T_2 = 300 \times 2 \times 2$

$T_2 = 1200$ K

Convert to Celsius:

$T_2 = 1200 - 273$

$T_2 = 927°C$

25. **Answer :** B) $\frac{1}{320}/°C$

The volume coefficient of a gas based on an initial state V_1 at T_1 is given by:

$\gamma_v = \frac{V_2-V_1}{V_1(T_2-T_1)}$

Substitute $V_1 = 100$, $V_2 = 125$, and the temperatures:

$\gamma_v = \frac{125-100}{100(100-20)}$

$\gamma_v = \frac{25}{100\times80}$

$\gamma_v = \frac{25}{8000}$

$\gamma_v = \frac{1}{320}/°C$

26. **Answer :** B) $327°C$
Convert initial temperature to Kelvin:
$T_1 = 27°C + 273 = 300$ K
Apply Gay-Lussac's law at constant volume:
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Substitute $P_2 = 2P_1$:
$\frac{P_1}{300} = \frac{2P_1}{T_2}$
$T_2 = 600$ K
Convert to Celsius:
$T_2 = 600 - 273$
$T_2 = 327°C$

27. **Answer :** C) 25 cm^3
Apply Boyle's law at constant temperature:
$P_1V_1 = P_2V_2$
Substitute $P_1 = 1$ atm, $V_1 = 100$ cc, and $P_2 = 4$ atm:
$1 \times 100 = 4 \times V_2$
$V_2 = \frac{100}{4}$
$V_2 = 25$ cm^3

28. **Answer :** C) $\frac{1}{273}$
According to Charles's Law, the volume of an ideal gas at constant pressure expands by $\frac{1}{273}$ of its original volume at $0°C$ for every $1°C$ (or 1 K) rise in temperature.

29. **Answer :** B) 1000 cc
Apply Boyle's law for constant temperature:
$P_1V_1 = P_2V_2$
Substitute $P_1 = 760$ mm Hg, $V_1 = 500$ cc, and $P_2 = 380$ mm Hg:
$760 \times 500 = 380 \times V_2$
$V_2 = \frac{760 \times 500}{380}$
$V_2 = 2 \times 500$
$V_2 = 1000$ cc

LEVEL-II

1. **Answer :** B) $\gamma = 2\alpha$
The volume of the liquid column at any temperature is given by:$V_t = V_0(1 + \gamma\Delta t)$
The volume of the liquid can also be written in terms of the tube's cross-sectional area and length:
$V_t = A_tL_t$
Since the tube expands, its area at temperature t is:
$A_t = A_0(1 + 2\alpha\Delta t)$
The problem states the length of the liquid column does not change, meaning $L_t = L_0$:
$V_t = A_0(1 + 2\alpha\Delta t) \times L_0$
$V_t = V_0(1 + 2\alpha\Delta t)$
Equating the two expressions for V_t:
$V_0(1 + \gamma\Delta t) = V_0(1 + 2\alpha\Delta t)$
$\gamma = 2\alpha$

2. **Answer :** B) 1.016 m

Note: There is a typographical error in the problem statement. The cubical expansion of glass should be $0.000025°C^{-1}$ for the calculation to match the correct scientific answer choices.

Find the area expansion coefficient of the capillary bore:

$\beta_g = \frac{2}{3}\gamma_g$

$\beta_g = \frac{2}{3}(0.000025)$

$\beta_g \approx 0.00001667/°C$

The length of the mercury thread l_t is the volume of mercury divided by the cross-sectional area of the bore:

$l_t = \frac{V_t}{A_t}$

$l_t = \frac{V_0(1+\gamma_r \Delta t)}{A_0(1+\beta_g \Delta t)}$

$l_t \approx l_0[1 + (\gamma_r - \beta_g)\Delta t]$

Substitute the values:

$l_{100} = 1 \times [1 + (0.000182 - 0.00001667) \times 100]$

$l_{100} = 1 \times [1 + (0.00016533) \times 100]$

$l_{100} = 1 \times [1 + 0.016533]$

$l_{100} \approx 1.016$ m

3. **Answer :** A) $627°C$

Convert the initial temperature to Kelvin:

$T_1 = 27°C + 273 = 300$ K

Apply Charles's law at constant pressure:

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

Substitute $V_2 = 3V_1$:

$\frac{V_1}{300} = \frac{3V_1}{T_2}$

$T_2 = 900$ K

Convert to Celsius:

$T_2 = 900 - 273$

$T_2 = 627°C$

4. **Answer :** D) 100 k

Convert the initial temperature to Kelvin:

$T_1 = 27°C + 273 = 300$ K

Apply the combined gas law:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Substitute $P_1 = 30$, $V_1 = V$, $P_2 = 1$, and $V_2 = 10V$:

$\frac{30 \times V}{300} = \frac{1 \times 10V}{T_2}$

$\frac{1}{10} = \frac{10}{T_2}$

$T_2 = 100$ K

5. **Answer :** 250 K

For a closed vessel, volume is constant. Use Gay-Lussac's law:

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$

Substitute $P_2 = 1.04P_1$ (4% increase) and $T_2 = T_1 + 10$:

$\frac{P_1}{T_1} = \frac{1.04P_1}{T_1+10}$

$T_1 + 10 = 1.04T_1$
$0.04T_1 = 10$
$T_1 = \frac{10}{0.04}$
$T_1 = 250$ K

6. **Answer :** 400 K

For a closed vessel, the volume is constant.
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
$\frac{\Delta P}{P} = \frac{\Delta T}{T}$
Substitute $\frac{\Delta P}{P} = 0.005$ (0.5% increase) and $\Delta T = 2$:
$0.005 = \frac{2}{T}$
$T = \frac{2}{0.005}$
$T = 400$ K

7. **Answer :** B, D

Water undergoes anomalous expansion, contracting as it warms from $0°C$ to $4°C$. It reaches its minimum volume and maximum density exactly at $4°C$, which corresponds to 277 K (273 + 4). Thus, both options B and D are correct.

8. **Answer :** A, B, D

Normal liquids expand when heated, meaning their volume increases while their density decreases (Statement C is true). Statements A, B, and D describe behaviors contrary to this physical principle and are therefore false.

9. **Answer :** A) $\frac{1+60\gamma_{Fe}}{1+60\gamma_{Hg}}$

By Archimedes' principle, the weight of the floating block equals the weight of the displaced liquid. The submerged fraction k is the ratio of the solid's density to the liquid's density:
$k_1 = \frac{\rho_{Fe,0}}{\rho_{Hg,0}}$
At $60°C$, the densities change due to expansion:
$k_2 = \frac{\rho_{Fe,60}}{\rho_{Hg,60}}$
$k_2 = \frac{\frac{\rho_{Fe,0}}{1+60\gamma_{Fe}}}{\frac{\rho_{Hg,0}}{1+60\gamma_{Hg}}}$
$k_2 = k_1\left(\frac{1+60\gamma_{Hg}}{1+60\gamma_{Fe}}\right)$
Therefore, the ratio is:
$\frac{k_1}{k_2} = \frac{1+60\gamma_{Fe}}{1+60\gamma_{Hg}}$

10. **Answer :** B) $5 \times 10^{-3}/°C$

Calculate the buoyant forces (apparent loss of weight) at $0°C$ and $50°C$:
$F_{b0} = 55 - 30 = 25$ g
$F_{b50} = 55 - 35 = 20$ g
Assuming the expansion of the sinker is negligible compared to the liquid ($\gamma_S \approx 0$):
$F_{b50} = \frac{F_{b0}}{1+\gamma_L \Delta T}$
$20 = \frac{25}{1+\gamma_L \times 50}$
$1 + 50\gamma_L = \frac{25}{20}$
$1 + 50\gamma_L = 1.25$
$50\gamma_L = 0.25$
$\gamma_L = 0.005/°C = 5 \times 10^{-3}/°C$

11. **Answer :** B) $8.33 \times 10^{-4}/^{\circ}C$

The fraction immersed k is proportional to the density of the liquid (assuming the solid's expansion is negligible):

$k_1 \propto \frac{1}{\rho_{L20}}$

$k_2 \propto \frac{1}{\rho_{L100}}$

The ratio of these fractions gives:

$\frac{k_2}{k_1} = \frac{\rho_{L20}}{\rho_{L100}}$

Substitute density formula $\rho_{L20} = \rho_{L100}(1 + \gamma_L \Delta T)$:

$\frac{0.80}{0.75} = 1 + \gamma_L \times (100 - 20)$

$\frac{80}{75} = 1 + 80\gamma_L$

$\frac{16}{15} = 1 + 80\gamma_L$

$80\gamma_L = \frac{1}{15}$

$\gamma_L = \frac{1}{1200}$

$\gamma_L \approx 8.33 \times 10^{-4}/^{\circ}C$

12. **Answer :** A) 3.44 cc

Calculate the volume coefficient of expansion for glass:

$\gamma_g = 3\alpha_g$

$\gamma_g = 3 \times (9 \times 10^{-6})$

$\gamma_g = 27 \times 10^{-6}/^{\circ}C$

Calculate the apparent expansion of mercury:

$\gamma_a = \gamma_r - \gamma_g$

$\gamma_a = (180 \times 10^{-6}) - (27 \times 10^{-6})$

$\gamma_a = 153 \times 10^{-6}/^{\circ}C$

Calculate the volume of mercury that overflows:

$\Delta V = V_0 \times \gamma_a \times \Delta T$

$\Delta V = 250 \times (153 \times 10^{-6}) \times (110 - 20)$

$\Delta V = 250 \times 153 \times 10^{-6} \times 90$

$\Delta V = 22500 \times 153 \times 10^{-6}$

$\Delta V = 3.4425$ cc

13. **Answer :** C) 2 gm

First, find the apparent expansion from the first case:

$\gamma_a = \frac{m_{exp}}{m_{rem} \times \Delta T}$

$\gamma_a = \frac{2}{(52-2) \times 100}$

$\gamma_a = \frac{2}{5000} = 4 \times 10^{-4}/^{\circ}C$

Apply this to the second case to find the expelled mass x:

$x = (104 - x) \times \gamma_a \times \Delta T_2$

$x = (104 - x) \times (4 \times 10^{-4}) \times 50$

$x = (104 - x) \times 0.02$

$x = 2.08 - 0.02x$

$1.02x = 2.08$

$x = \frac{2.08}{1.02} \approx 2.03$ gm

14. **Answer :** B) $4 \times 10^{-4}/^{\circ}C$

Using the data from the previous calculation:

$\gamma_a = \frac{m_{exp}}{m_{rem} \times \Delta T}$

$\gamma_a = \frac{2}{(52-2) \times 100}$

$\gamma_a = \frac{2}{5000}$

$\gamma_a = 4 \times 10^{-4}/^{\circ}C$

15. **Answer :** a-R, b-S, c-R, d-S

Apparent expansion (a) and its coefficient γ_a (c) depend on the nature of the vessel, the liquid, and the temperature (R).

Real expansion (b) and its coefficient γ_r (d) depend only on the nature of the liquid and the temperature (S).

16. **Answer :** a-S, b-Q, c-P, d-R

If γ_g is positive and less than γ_r (a), dynamic heating causes the vessel to expand first (level decreases) before the liquid expands more (level increases). (S)

If γ_g is negative (b), the vessel strictly contracts, pushing the liquid level higher immediately. (Q)

If $\gamma_g = \gamma_r$ (c), the apparent expansion is zero, so the liquid level does not change. (P)

If $\gamma_g > \gamma_r$ (d), the vessel expands more than the liquid, so the liquid level decreases. (R)

17. **Answer :** 7

Given that the absolute coefficient of expansion is 7 times the volume coefficient of the vessel:

$\gamma_r = 7\gamma_v$

Calculate the apparent expansion coefficient:

$\gamma_a = \gamma_r - \gamma_v$

$\gamma_a = 7\gamma_v - \gamma_v$

$\gamma_a = 6\gamma_v$

Find the ratio of absolute to apparent expansion:

$\frac{\gamma_r}{\gamma_a} = \frac{7\gamma_v}{6\gamma_v}$

$\frac{\gamma_r}{\gamma_a} = \frac{7}{6}$

Equating this to the given ratio $\frac{x}{6}$:

$\frac{x}{6} = \frac{7}{6}$

$x = 7$

18. **Answer :** 1

Find the volume expansion coefficient for the glass flask:

$\gamma_g = 3\alpha_g$

$\gamma_g = 3 \times (0.1 \times 10^{-4})$

$\gamma_g = 0.3 \times 10^{-4}/^{\circ}C$

Calculate the apparent expansion coefficient of mercury:

$\gamma_a = \gamma_r - \gamma_g$

$\gamma_a = 1.30 \times 10^{-4} - 0.3 \times 10^{-4}$

$\gamma_a = 1.0 \times 10^{-4}/^{\circ}C$

Calculate the volume of mercury spilled out:

$\Delta V = V_0 \times \gamma_a \times \Delta T$

$\Delta V = 1000 \times (1.0 \times 10^{-4}) \times 100$

$\Delta V = 10$ c.c.

Equating this to the given expression $10 \times x$:

$10 \times x = 10$

$x = 1$

Heat Capacity & Calorimetry

1. **Answer :** B) Specific heat capacity
The definition provided in the question describes the specific heat capacity of a substance.

2. **Answer :** C) J/kg K
The S.I. unit for heat is the Joule (J), for mass is the kilogram (kg), and for temperature is the Kelvin (K).
The formula for specific heat is:
$s = \frac{Q}{m\Delta T}$
Substituting the units gives J/kg K.

3. **Answer :** C) Both A and B
Heat capacity depends on the mass of the body and its specific heat (which depends on the nature of the substance).
The formula is:
$C = m \times s$

4. **Answer :** B) High specific heat
Water can absorb a large amount of heat with only a small rise in temperature because of its very high specific heat capacity.

5. **Answer :** B) 1:16
The formula for thermal capacity is:
$C = m \times s$
Mass can be expressed as volume times density:
$m = V \times d$
$m = \frac{4}{3}\pi r^3 \times d$
Substitute this into the thermal capacity formula:
$C = \frac{4}{3}\pi r^3 \times d \times s$
Establish the ratio for the two spheres:
$\frac{C_1}{C_2} = \left(\frac{r_1}{r_2}\right)^3 \times \left(\frac{d_1}{d_2}\right) \times \left(\frac{s_1}{s_2}\right)$
Substitute the given ratios:
$\frac{C_1}{C_2} = \left(\frac{1}{2}\right)^3 \times \left(\frac{3}{4}\right) \times \left(\frac{2}{3}\right)$
$\frac{C_1}{C_2} = \frac{1}{8} \times \frac{3}{4} \times \frac{2}{3}$
$\frac{C_1}{C_2} = \frac{6}{96}$
$\frac{C_1}{C_2} = \frac{1}{16}$

6. **Answer :** A) 50 cal/°C
The formula for thermal capacity is:
$C = m \times s$
Substitute the given values for mass and specific heat:
$C = 500 \times 0.1$
$C = 50$

7. **Answer :** B) 3:2
The thermal capacities of the two liquids are equal:
$C_A = C_B$
$m_A \times s_A = m_B \times s_B$

Express mass as volume times density:
$(V_A \times d_A) \times s_A = (V_B \times d_B) \times s_B$
Substitute the given values:
$(8 \times 0.5) \times s_A = (10 \times 0.6) \times s_B$
$4 \times s_A = 6 \times s_B$
$\frac{s_A}{s_B} = \frac{6}{4}$
$\frac{s_A}{s_B} = \frac{3}{2}$

8. **Answer :** C) 0.05
A change of 1°C is equivalent to a change of 1.8°F.
The heat required per degree Fahrenheit is therefore scaled by a factor of 1/1.8 (or 5/9).
$s_F = s_C \times \frac{5}{9}$
Substitute the given specific heat:
$s_F = 0.09 \times \frac{5}{9}$
$s_F = 0.05$

9. **Answer :** B) Latent heat
The heat absorbed or released during a change of state at a constant temperature is known as latent heat.

10. **Answer :** C) L = Q/m
The formula for latent heat defines it as the total heat divided by the mass undergoing the phase change.

11. **Answer :** C) Remains same
The mass of the floating ice is equal to the mass of the water it displaces. When the ice melts, it turns into that exact volume of water, keeping the level unchanged.

12. **Answer :** B) Regelation
The phenomenon of ice melting under pressure and refreezing when the pressure is reduced is called regelation.

13. **Answer :** D) B & C
A rise in temperature increases the internal energy. This can occur either by supplying heat or by doing work on the object (as per the first law of thermodynamics).

14. **Answer :** B) Due to evaporation of sweat
A fan moves the air, which increases the rate of evaporation of sweat from the skin. Evaporation is a cooling process.

15. **Answer :** C) Increasing the average distance between the molecules
During melting, the temperature and kinetic energy remain constant. The absorbed heat increases the potential energy by increasing the distance between molecules.

16. **Answer :** C) Lowering of melting point of ice at the bottom due to high pressure
The enormous weight of the iceberg creates high pressure at the base, which lowers the melting point of ice, causing it to melt there first.

17. **Answer :** D) 30136 J
The total heat required is the sum of the heat to melt the ice, the heat to warm the water to 100°C, and the heat to vaporize it.
$Q_{total} = Q_{melt} + Q_{warm} + Q_{boil}$
$Q_{total} = mL_{ice} + mc\Delta T + mL_{steam}$
Substitute the given values (where m = 0.01 kg):
$Q_{total} = (0.01 \times 0.335 \times 10^6) + (0.01 \times 4186 \times 100) + (0.01 \times 2.26 \times 10^6)$

$Q_{total} = 3350 + 4186 + 22600$
$Q_{total} = 30136$

18. **Answer :** B) 0.813 g
The final temperature of the mixture will be 20°C + 5°C = 25°C.
According to the principle of calorimetry, the heat lost by the steam equals the heat gained by the water.
$Q_{lost} = Q_{gained}$
$m \times L_{steam} + m \times c_w \times (100 - 25) = M_w \times c_w \times \Delta T$
Substitute the given values:
$m \times 540 + m \times 1 \times 75 = 100 \times 1 \times 5$
$540m + 75m = 500$
$615m = 500$
$m = \frac{500}{615}$
$m \approx 0.813$

19. **Answer :** A) 2 gm
First, calculate the heat required to completely melt the 6 gm of ice at 0°C.
$Q_{req} = m \times L_{ice}$
$Q_{req} = 6 \times 80 = 480$ cal
Next, calculate the heat required to raise this melted water from 0°C to 100°C.
$Q_{warm} = m \times c_w \times \Delta T$
$Q_{warm} = 6 \times 1 \times 100 = 600$ cal
Total heat needed to bring the ice to water at 100°C is:
$Q_{total} = 480 + 600 = 1080$ cal
This heat is provided by the condensation of a mass m_s of steam at 100°C.
$Q_{released} = m_s \times L_{steam}$
$1080 = m_s \times 540$
$m_s = \frac{1080}{540}$
$m_s = 2$

20. **Answer :** A) $\frac{M_1S_1+M_2S_2}{M_1+M_2}$
The total heat capacity of the mixture is the sum of the individual heat capacities.
$C_{mix} = C_1 + C_2$
$(M_1 + M_2)S_{mix} = M_1S_1 + M_2S_2$
$S_{mix} = \frac{M_1S_1+M_2S_2}{M_1+M_2}$

21. **Answer :** A) 3:2
Let the mass of each liquid be m. By the principle of calorimetry:
$Heat\ lost = Heat\ gained$
$m \times S_1 \times (T_1 - T_{mix}) = m \times S_2 \times (T_{mix} - T_2)$
Substitute the given temperatures:
$S_1 \times (40 - 36) = S_2 \times (36 - 30)$
$S_1 \times 4 = S_2 \times 6$
$\frac{S_1}{S_2} = \frac{6}{4}$
$\frac{S_1}{S_2} = \frac{3}{2}$

22. **Answer :** C) 40°C
Heat released if all steam condenses and cools to a final temperature T:
$Q_{lost} = m_sL_s + m_sc_w(100 - T)$
$Q_{lost} = 10 \times 540 + 10 \times 1 \times (100 - T)$

$Q_{lost} = 5400 + 1000 - 10T = 6400 - 10T$
Heat gained by ice melting and warming to T:
$Q_{gained} = m_i L_i + m_i c_w (T - 0)$
$Q_{gained} = 50 \times 80 + 50 \times 1 \times T$
$Q_{gained} = 4000 + 50T$
Equate the heat lost and heat gained:
$6400 - 10T = 4000 + 50T$
$2400 = 60T$
$T = 40$

23. **Answer :** A) 45°C
Since the two liquids are the same substance and have equal mass/volume, the final temperature is the simple average of their initial temperatures.
$T = \frac{T_1 + T_2}{2}$
$T = \frac{70+20}{2}$
$T = 45$

LEVEL-II

1. **Answer :** A) 3360
The specific heat capacity is derived from thermal capacity:
$s = \frac{C}{m}$
$s = \frac{8}{10} = 0.8$ cal/g°C
Convert to S.I. units (J/kg K) using the conversion 1 cal/g°C = 4200 J/kg K:
$s_{SI} = 0.8 \times 4200$
$s_{SI} = 3360$

2. **Answer :** C) Zy:8
The ratio of thermal capacities is:
$\frac{C_1}{C_2} = \frac{m_1 s_1}{m_2 s_2}$
Express mass as volume times density:
$\frac{C_1}{C_2} = \left(\frac{r_1}{r_2}\right)^3 \left(\frac{d_1}{d_2}\right)\left(\frac{s_1}{s_2}\right)$
Substitute the provided ratios ($1:2$, $Z:X$, $x:y$). Assuming standard ratio formatting where density ratio is $Z:1$ (simplifying from $Z:X$ given likely typo for 1 or X):
$\frac{C_1}{C_2} = \left(\frac{1}{2}\right)^3 \times \left(\frac{Z}{1}\right) \times \left(\frac{y}{1}\right)$
$\frac{C_1}{C_2} = \frac{1}{8} \times Z \times y$
$\frac{C_1}{C_2} = \frac{Zy}{8}$

3. **Answer :** C) yt2/xt1
The heats for both substances are equal:
$Q_1 = Q_2$
$m_1 s_1 \Delta T_1 = m_2 s_2 \Delta T_2$
Substitute the values for the substance and water (specific heat of water $s_2 = 1$):
$x \times s_1 \times t_1 = y \times 1 \times t_2$
Solve for the specific heat ratio $s_1/1$:
$s_1 = \frac{y \times t_2}{x \times t_1}$

4. **Answer :** C) 40°C
Using the principle of calorimetry:

$m_1 s_1(T_1 - T) = m_2 s_2(T - T_2)$
Substitute the proportional values $m_1 = 3$, $m_2 = 4$, $s_1 = 4$, $s_2 = 5$:
$3 \times 4 \times (60 - T) = 4 \times 5 \times (T - 20)$
$12(60 - T) = 20(T - 20)$
$720 - 12T = 20T - 400$
$1120 = 32T$
$T = \frac{1120}{32}$
$T = 40$

5. **Answer :** A) 0°C
Heat required to melt all ice:
$Q_{melt} = 50 \times 80 = 4000$ cal
Heat available from water cooling to 0°C:
$Q_{avail} = 50 \times 1 \times 60 = 3000$ cal
Since $Q_{avail} < Q_{melt}$, only some ice melts. The final mixture will contain ice and water at 0°C. If 50 g of water cools to 0°C, it gives 3000 cal.
3000 / 80 = 37.5 g of ice melts. The remaining mixture is at 0°C.
The final temperature will indeed be 0°C since there is not enough heat to melt all the ice.

6. **Answer :** B) 0.1
Heat lost by the iron ball equals the heat gained by the melting ice:
$m_{iron} s_{iron} \Delta T = m_{ice} L_{ice}$
Substitute the values (convert iron mass to grams):
$200 \times s_{iron} \times 100 = 25 \times 80$
$20000 \times s_{iron} = 2000$
$s_{iron} = \frac{2000}{20000}$
$s_{iron} = 0.1$

7. **Answer :** A) 30 g
The heat provided by the water cooling to 0°C is:
$Q = m_w \times c \times \Delta T$
$Q = 80 \times 1 \times 30 = 2400$ cal
This heat is used to melt a mass m of ice:
$Q = m \times L_{ice}$
$2400 = m \times 80$
$m = \frac{2400}{80}$
$m = 30$

8. **Answer :** A) 10°C
Specific heat capacity of water (c): 1 cal/g°C
Latent heat of fusion of ice (L): 80 cal/g
First, we find out how much heat is needed just to turn the 100 g of ice at 0°C into 100 g of liquid water at 0°C:
$Q_{\text{melt}} = m_{\text{ice}} \times L$
$Q_{\text{melt}} = 100\ \text{g} \times 80\ \text{cal/g} = 8{,}000$ cal
Next, we see how much heat the hot water can potentially give off if it were to cool all the way down to 0°C:
$Q_{\text{available}} = m_{\text{water}} \times c \times \Delta T$
$Q_{\text{available}} = 100\ \text{g} \times 1\ \text{cal/g°C} \times (100°\text{C} - 0°\text{C}) = 10{,}000$ cal

Because the hot water has more heat available (10,000 cal) than what is required to melt the ice (8,000 cal), all of the ice will melt, and the remaining energy will warm the entire mixture above 0°C.
Using the principle of conservation of energy (Heat Lost = Heat Gained):
Heat lost by hot water = Heat to melt ice + Heat to warm the melted ice
$m_{\text{water}} \cdot c \cdot (100 - T_f) = Q_{\text{melt}} + m_{\text{ice}} \cdot c \cdot (T_f - 0)$
Plugging in our values:
$100 \cdot 1 \cdot (100 - T_f) = 8{,}000 + 100 \cdot 1 \cdot T_f$
$10{,}000 - 100T_f = 8{,}000 + 100T_f$
$2{,}000 = 200T_f$
$T_f = 10°\text{C}$

9. **Answer :** A)
The temperature increases as ice heats to 0°C, stays constant during melting, increases again as water heats to 100°C, and then stays constant during boiling. Curve A correctly depicts this stepwise phase change behavior.

10. **Answer :** B, D
Specific heat ratio:
$c_1 : c_2 = 1:2$
Temperatures:
$T_1 = 2\theta,\ T_2 = \theta$
Equal masses of the two liquids are mixed.
Using the principle of calorimetry,
$mc_1(2\theta - T) = mc_2(T - \theta)$

Substituting $c_1 = 1,\ c_2 = 2,$
$(2\theta - T) = 2(T - \theta)$
$2\theta - T = 2T - 2\theta$
$4\theta = 3T$
$T = \frac{4\theta}{3}$
Thus, Statement B is true and Statement A is false.
For equal masses,
Heat capacity,
$C = mc$
Since masses are equal,
$C_1 : C_2 = c_1 : c_2 = 1:2$

11. **Answer :** A, B, C
The latent heat of fusion of ice is 80 cal/g (A). The latent heat of vaporization of water is 540 cal/g (B). Its molar equivalent is roughly 40.8 kJ/mol (C). All these represent standard true values.

12. **Answer :** A, B, C
The principle of calorimetry states that heat lost equals heat gained in an isolated system (A), which is a statement of the law of conservation of energy (B). The fundamental equation for heat exchange is $Q = ms\Delta T$ (C).

13. **Answer :** A) Both A and R are true, and R is correct explanation of A
Water has its maximum density at 4°C due to anomalous expansion. This denser water sinks, keeping the bottom of lakes at 4°C, while ice forms on top. Ice acts as a thermal insulator, preventing further freezing and allowing aquatic life to survive.

14. **Answer :** A) Both A and R are true, and R is correct explanation of A
Upthrust equals the weight of the displaced liquid ($V\rho g$). As temperature increases, the solid expands (increasing V) and the liquid expands (decreasing ρ). Usually, the liquid's density decrease outweighs the solid's volume expansion, reducing the upthrust.

15. **Answer :** B) 2520 J/kg K
The formula for heat is:
$Q = m \times s \times \Delta T$
Convert 1200 calories to Joules (1 cal = 4.2 J):
$Q_J = 1200 \times 4.2 = 5040$ J
Substitute the mass in kg and the temperature change:
$5040 = 0.025 \times s \times (90 - 10)$
$5040 = 0.025 \times s \times 80$
$5040 = 2 \times s$
$s = 2520$

16. **Answer :** C) 63 J/K
Heat released,
$Q = 1200$ cal Temperature change,
$\Delta T = 80$ K Convert calories to joules:
1200 cal $= 1200 \times 4.2 = 5040$ J
$C = \frac{Q}{\Delta T} = \frac{5040}{80} = 63$ J/K

Answer : D) 0.015 kg
Water equivalent is the mass of water that has the same heat capacity as the substance.
$W = \frac{C}{s_w}$
$W = \frac{63 \text{ J/K}}{4200 \text{ J/kg K}}$
$W = 0.015$ kg

18. **Answer :** C) Between both
When two bodies at different temperatures exchange heat, the final equilibrium temperature will always be between the lower and higher initial temperatures.

19. **Answer :** B) Temperatures equal
Heat transfer is driven by a temperature difference and ceases when thermal equilibrium is reached, meaning the temperatures are equal.

20. **Answer :** B) Hot -> cold
Heat spontaneously flows from an object at a higher temperature to an object at a lower temperature.

21. **Answer :** a-R, b-Q, c-T, d-P
This is the heating curve of a pure substance.
OA: Solid is heated.
AB: Melting (solid → liquid) at constant temperature.
BC: Liquid is heated.
CD: Boiling (liquid → vapour) at constant temperature.
DE: Vapour is heated.
Therefore, the correct matching is:
a) AB represents → S) Change of state from solid to liquid (melting)
b) CD represents → Q) Change of state from liquid to vapour (boiling)
c) BC represents → T) Liquid state of substance
d) DE represents → P) Vapour state of substance

22. **Answer :** A-P, B-Q, C-R, D-S
Heat during a temperature change is calculated using $Q = mc\Delta T$ (P). Heat during a change of state is calculated using $Q = mL$ (Q). Latent heat of fusion occurs during the Solid -> Liquid transition at constant temperature (R). Latent heat of vaporization occurs during the Liquid -> Vapour transition at constant temperature (S).

23. **Answer :** 9
Heat lost by the water cooling from 80°C to 0°C:
$Q = m_w \times c \times \Delta T$
$Q = 9 \times 1 \times 80 = 720$ cal
This heat melts a mass m of ice:
$Q = m \times L_{ice}$
$720 = m \times 80$
$m = \frac{720}{80}$
$m = 9$

24. **Answer :** 672
The heat required to melt the ice is given by the latent heat formula:
$Q = m \times L$
Substitute the given values:
$Q = 2 \times 336$
$Q = 672$ kJ

25. **Answer :** 2260000
To express kJ/kg in J/kg, multiply by 1000.
2260 kJ/kg = 2260 × 1000 J/kg
= 2260000 J/kg

Previous Year Questions

1. **Answer :** B) +1°C
Use the temperature scale conversion formula:
$\frac{C}{5} = \frac{F-32}{9}$
Substitute the correct Fahrenheit reading:
$\frac{C}{5} = \frac{122-32}{9}$
$\frac{C}{5} = \frac{90}{9}$
$\frac{C}{5} = 10$
$C = 50$
The correct temperature is 50°C. The faulty thermometer reads 49°C, correction = true - measured = 50 - 49 = +1°C.

2. **Answer :** C) 608°F
The problem states that the Fahrenheit reading is 90% more than the Celsius reading:
$F = C + 0.90C$
$F = 1.9C$
Use the conversion formula:
$\frac{C}{5} = \frac{1.9C-32}{9}$
$9C = 5(1.9C - 32)$
$9C = 9.5C - 160$
$0.5C = 160$

$C = 320$

Calculate F:

$F = 1.9 \times 320$

$F = 608$

3. **Answer :** B) 0.9 kg

Calculate the rate of heat transfer through the walls of the box:

$\frac{dQ}{dt} = \frac{kA\Delta T}{d}$

Substitute the values: $k = 10^{-2}$, $A = 1000 \times 10^{-4} = 0.1$ m^2, $\Delta T = 42$, $d = 2 \times 10^{-3}$ m:

$\frac{dQ}{dt} = \frac{10^{-2} \times 0.1 \times 42}{2 \times 10^{-3}}$

$\frac{dQ}{dt} = \frac{0.042 \times 10^{-3}}{2 \times 10^{-3}}$

$\frac{dQ}{dt} = 21$ J/s

Total heat transferred in 160 minutes:

$Q = 21 \times (160 \times 60)$

$Q = 201600$ J

Mass of ice melted:

$m = \frac{Q}{L}$

$m = \frac{201600}{336000}$

$m = 0.6$ kg

Remaining ice:

$m_{remaining} = 1.5 - 0.6$

$m_{remaining} = 0.9$ kg

4. **Answer :** B) 1925 W

Total heat that must be removed consists of cooling the water and freezing it. Heat to cool water:

$Q_1 = mc\Delta T$

$Q_1 = 15 \times 4200 \times 30 = 1890000$ J

Heat to freeze water:

$Q_2 = mL$

$Q_2 = 15 \times 336000 = 5040000$ J

Total heat:

$Q = 1890000 + 5040000 = 6930000$ J

Power is work over time (1 hour = 3600 s):

$P = \frac{Q}{t}$

$P = \frac{6930000}{3600}$

$P = 1925$ W

5. **Answer :** C) 320°F

Given that the Fahrenheit temperature is twice the Celsius temperature:

$F = 2C$

Use the conversion formula:

$\frac{F-32}{9} = \frac{C}{5}$

$\frac{2C-32}{9} = \frac{C}{5}$

$10C - 160 = 9C$

$C = 160$

Calculate F:

$F = 2 \times 160$

$F = 320°F$

6. **Answer :** B) 8:10:5

When equal masses of A and B are mixed:

$s_A(20 - 15) = s_B(24 - 20)$

$5s_A = 4s_B$

$s_A = \frac{4}{5}s_B$

When equal masses of B and C are mixed:

$s_B(26 - 24) = s_C(30 - 26)$

$2s_B = 4s_C$

$s_C = \frac{1}{2}s_B$

The ratio of specific heats $s_A : s_B : s_C$:

$\frac{4}{5}s_B : s_B : \frac{1}{2}s_B$

Multiply by 10 to clear denominators:

$8 : 10 : 5$

7. **Answer :** D) 12 g

Heat gained by the water:

$Q_{gained} = m_w c\Delta T$

$Q_{gained} = 100 \times 1 \times (90 - 24) = 6600$ cal

Heat lost by the steam condensing and cooling to 90°C:

$Q_{lost} = m_s L + m_s c(100 - 90)$

$Q_{lost} = m_s \times 540 + m_s \times 1 \times 10$

$Q_{lost} = 550m_s$

Equate heat lost and heat gained:

$550m_s = 6600$

$m_s = \frac{6600}{550}$

$m_s = 12$ g

8. **Answer :** C) 20°C

Heat required to melt all the ice:

$Q_{melt} = 37 \times 80 = 2960$ cal

Heat released if the water cools to 0°C:

$Q_{release} = 74 \times 1 \times 70 = 5180$ cal

Since $Q_{release} > Q_{melt}$, all ice melts and the mixture will rise to a final temperature T.

Heat balance equation:

$m_{ice}L + m_{ice}c(T - 0) = m_w c(70 - T)$

$37 \times 80 + 37 \times 1 \times T = 74 \times 1 \times (70 - T)$

Divide by 37:

$80 + T = 2 \times (70 - T)$

$80 + T = 140 - 2T$

$3T = 60$

$T = 20$